

# DATA REGISTRY REQUIREMENTS MANUAL

FOR THE ~~2022~~ 2025 BUILDING ENERGY EFFICIENCY  
STANDARDS

TITLE 24, PART 6, AND ASSOCIATED  
ADMINISTRATIVE  
REGULATIONS IN PART 1



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Text is subject to internal review and revision, and that it  
is not official guidance until approved by the Commission



**CALIFORNIA  
ENERGY COMMISSION**



**CALIFORNIA  
NATURAL  
RESOURCES  
AGENCY**

California Energy Commission

## **STAFF REPORT**

# **Data Registry Requirements Manual**

**~~2022~~ 2025 Building Energy Efficiency  
Standards Title 24, Part 6, and Associated  
Administrative Regulations in Part 1**

**Gavin Newsom, Governor**

**June 202~~2~~5 | CEC-400-202~~2~~5-~~011~~\*\*\* CMF**

## 4 Electronic Signatures

### 4.1 Overview

*Reference Joint Appendix JA7.6* describes the distinction between electronic and digital signature requirements. Digital signatures and digital certificate authorities are described in Section 5 below. From Section JA7.2 definitions: electronic signature is a "computer data compilation of any symbol or series of symbols executed, adopted, or authorized by an individual to be the legally binding equivalent of the individual's handwritten signature" (U.S. 21 Code of Federal Regulations (CFR) Section 11.3). The electronic signature process for Energy Standards-Code compliance documents is described in *Reference Joint Appendix JA7.6.3.2.1* (Electronic Signature Capability) and *JA7.6.3.2.3* (Signer Review and Signature Actions). Additional guidance for use of electronic signatures for registering Energy Standards-Code compliance documents follows.

### 4.2 Establishing a User Account and Signature Authority

Anyone who wishes to sign Energy Standards-Code compliance documents electronically using a HERSECC-Provider's data registry will be required to establish a user account ~~with a data registry~~ secured by a unique username and password. Creating a user account requires the applicant to provide sufficient information to establish their identity with the ~~registry~~ registration provider. Users who wish to sign compliance documents electronically must provide, at a minimum, the data that would have to be entered into each of the fields on the signature block for the compliance document. To be granted signature authority for certain document types, the applicant must also provide proof of possession of all requisite professional qualifications applicable to the scope contained on the compliance document to be signed. For example, if an HVAC contractor wants to register refrigerant charge verification documents, the HVAC contractor's C-20 license number would be required to be disclosed to the data registry, and this license number will be reported in the signature block when the contractor signs the document as the responsible person. The applicant may also provide a wet signature on the user account application that may be scanned electronically by the data registry provider and added to the user's profile. The scanned image of the wet signature may be overlaid on the completed electronic compliance document in the signature field after the user performs a signing action while logged into the data registry to generate an electronic signature for the document.

Once the user has established a username and password to access a personal account with a data registry, signing actions will be attributed to the user's account ~~as identified by the unique username and password of the account.~~

When an authorized user elects to sign a document, a signing action should be required, such as clicking on a control (such as a button or check box icon) in the data registry user interface, which then results in the data registry populating the signature

block with the user's professional information and an image or other approved representation of the user's signature.

~~Separate user accounts must be established with each data registry when more than one data registry is used by a user for registration of compliance documents.~~

## 4.3 Data Field Definitions for Signature Blocks

### 4.3.1 Electronic Images and Fonts Used for Electronic Signatures

*Reference Joint Appendix JA7.6.3.2.1.3* requires authorized users to provide an electronic image of their handwritten signature to the data registry for use in displaying their electronic signature and allows the registration provider to make available alternative methods for creating an electronic image for displaying electronic signatures. At a minimum, the data registry must make available functionality for use of an electronic (scanned) image of the user's wet signature. Other methods are allowed to be used if the data registry user interface provides the alternative signature creation functionality. Such alternative methods may allow use of a mouse or other pointing device to draw an electronic signature, or the data registry may make available the option for a user to select a font representation of their handwritten signature.

### 4.3.2 Signature Block Data Validation

Data fields in signature blocks are not required to have values assigned to them when submitting XML to the RG for publishing a PDF compliance document. However, including data values in signature block fields in the XML is allowable. Therefore, if values are assigned to the signature block data fields in the XML transmitted to the RG, the data must be valid in accordance with the schema for that signature block. Otherwise, the XML for that compliance document submitted to the RG will fail validation.

### 4.3.3 Appending the Signature Block Data

The requirements for a data registry provider to verify the unique professional information submitted by a user when applying for a signature authority is specified in JA7.6.3.2.1.2; additional guidance is described in Section 4.2 (above). Populating the signature block with a user's unique professional information is described in JA7.6.3.2.3; additional guidance for appending the signature block data is given in Sections 4.4 and 4.5 below.

## 4.4 Electronic Signature Process Flow

The process for electronically signing an Energy ~~Standards~~ Code compliance document should include attention to the following guidance points for the process flow:

- a. When data entry into the data registry has been completed for a compliance document, and the documentation author wishes to sign the document, the data registry must first call for and receive from the RG a ~~PDF~~ file report (PDF ~~printout~~ image) of the completed compliance document. This request makes

available to the document signer the ability to review the completed document before signing. As a prerequisite to making signing controls available to the documentation author signer, the data registry transmits a call to the RG that includes all required XML data to complete the document. The RG validates that data against the schema for that document and returns a PDF to the data registry.

- b. The RG does not register the document. The Data Registry conducts registration procedures by incorporating or appending signatures, registration numbers, date/time stamps, and so forth to the completed and validated PDF received from the RG.
- c. The signing actions of responsible persons, documentation authors, or field technicians are not required to occur at the same time as the data input.
- d. The identity of a responsible person signer is not required to be known to the documentation author.
- e. As required by JA7.6.3.2.3.4 when an authorized user signs a document, the registry must provide a display of the compliance document that allows the user to access any part of the compliance document information for review, including a display of the document declaration statement.
- f. The documentation author must review the PDF file received from the RG and then, by performing a signing action, certify compliance with the declaration statements on the document.
- g. When the documentation author signs the document, the data registry automatically appends the documentation author's signature block information into the document XML data portion of the registration package received from the RG. Then the data registry overlays the documentation author's signature image and the documentation author's signature block information onto the documentation author portion of the signature block in the PDF.
- h. The data registry makes available to the responsible person the ability to view, before signing, a copy of the completed PDF that was signed by the documentation author.
- i. The responsible person must review the PDF file received from the RG. By signing, the responsible person certifies compliance with the declaration statements on the document.
- j. When the responsible person (or authorized representative) signs the document, the data registry appends the responsible person's signature block information into the document XML data portion of the registration package received from the RG. Then the data registry overlays the responsible person's signature image, and signature block information onto the responsible person's portion of the signature block in the PDF. (See Section 4.5.2 for additional information on signatures given by authorized representatives of the responsible person on certificates of installation).



- k. The responsible person's signing should also trigger the digital signing (by the data registry) of the completed document PDF file and the digital signing of the XML data as described in Section 5 to complete the registration. The completed registration package should be a single file containing an embedded digitally signed PDF document file and digitally signed XML data.

This process flow for electronically signing compliance documents is recommended to avoid sending XML data to the RG multiple times for each document to populate the signature block fields for each signing.

Transporting bitmap or images of signatures to and from the RG is not supported. The user's signature image in the signature block on completed compliance documents is expected to be overlaid on the PDF document signature block by the data registry when the user provides the signing during PDF document registration.

## 4.5 Delegated Signatures and Authorized Representatives

### 4.5.1 Delegation of Signature Authority for Certificates of Installation

Section RA 2.5.1 of the ~~2022~~ Reference Residential Appendices states:

- The builder or subcontractor who is eligible under Division 3 of the Business and Professions Code to take responsibility for the construction or installation, or their authorized representative as specified in Standards Section 10-103(a)3A, shall provide an electronic signature to register the Certificate of Installation, to certify the information provided on the Certificate is true and correct, and confirm that the construction or installation complies with the requirements shown on the dwelling unit's Certificate of Compliance that was approved by the enforcement agency.

The phrase "builder or subcontractor" in this context means contractors and subcontractors that hold general or specialty classification licenses. Specialty classifications that typically install or alter ECC-verified measures include ~~including contractors holding various license classifications such as Class C-20 (-Warm-Air Heating, Ventilating, and Air-Conditioning), C-2 (Insulation), and C-36 (Plumbing)- (responsible person).~~

Section 10-103(a)3A of the ~~2022-2025~~ Energy ~~Standards Code~~ states:

- **Delegation of Signature Authority.** Except where prohibited by law, including but not limited to any requirements under Division 3 of the Business and Professions Code, the *Responsible Person* may delegate signature authority to third parties (*Authorized Representatives*) provided that there is a written agreement:
  - i. Between the Responsible Person and the person to be designated as the Authorized Representative.
  - ii. Specifying that the Authorized Representative may sign Certificates of Installation on behalf of the Responsible Person.

- iii. Specifying that the legal responsibility for construction or installation in the applicable classification for the scope of work specified on the Certificate of Installation document(s) remains with the Responsible Person.
- iv. That is signed by both the *Responsible Person* and the *Authorized Representative*.
- v. That is retained by the ~~HERSECC~~ Provider to which all compliance documents are submitted for the building to which the Certificate of Installation documentation pertains.
- vi. That is maintained in the ~~HERSECC~~ Provider Data Registry such that it is accessible for verification by, ~~included~~ including but not limited to, the Energy Commission and enforcement agencies.

When the Energy ~~Standards Code~~ requires the certificate of installation to be registered, the responsible person must become an authorized user of the applicable data registry and must establish the applicable required signature authority with the data registry. For a responsible person to delegate signature authority to another person, in accordance with Section 10- 103(a)3A, a document that identifies the responsible person's designated authorized representative(s) must be submitted to the data registry provider as prerequisite to granting authorized representatives signature authority. Authorized representatives must also be authorized users of the data registry.

Data registries must provide CEC staff and enforcement agencies a user interface capability to generate reports that disclose the status of authorized representatives of responsible person users as required by Energy ~~Standards Code~~ Section 10- 103(a)3A.

#### 4.5.2 Signature Block Data for Delegated Signatures

When a responsible person's signing action on a certificate of installation is performed by an authorized representative of the responsible person, the information appended to the signature block XML data portion of the registration package and overlaid on the responsible person's signature block on the PDF (Section 4.4 j) should be the responsible person's professional information — not the authorized representative's professional information.

Furthermore, the electronic signature appearance on the document should include a statement to the effect that the authorized representative's signature is provided on behalf of the responsible person.

The data registry should limit the availability of delegated signature functionality to only those users who have been designated as authorized representatives as described in Section 4.5.1.



## 5 Digital Signatures and ~~Certificate Authorities~~ Acceptable Technologies

### 5.1 Overview

Digital signatures are used to certify the authenticity of a document and to verify the authenticity and identity of document signers. Digital signatures also establish the state of the document at the time it was signed and make it possible to provide alerts to document recipients indicating whether the document has been changed since being signed. A digital signature is an electronic signature that incorporates cryptographic methods of originator authentication, allowing the identity of the signer and the integrity of the data to be verified. The regulations, adopted by the Secretary of State, that govern the use of digital signatures for use by public entities in California are found in the [California Code of Regulations, Title 2, Division 7, Chapter 10 Digital Signatures](#), accessible ~~using the following URL on the Secretary of State website:~~

~~<https://www.sos.ca.gov/administration/regulations/current-regulations/technology/digital-signatures>~~

As required by *Reference Joint Appendix* JA7.6.2.2.4, JA7.6.3.2.4, and JA7.7.1.4, when concluding the document registration procedure, the data registry must apply a digital signature to the electronically signed compliance document (PDF ~~from the RG~~) ~~using the data registry provider's digital certificate issued by a certificate authority approved by the California Secretary of State.~~

The registration package for the registered compliance document that contains the XML data corresponding to the information reported on the compliance document PDF must also be digitally signed. Use of a certificate authority ~~approved by the California Secretary of State~~ is not required for digitally signing the registration package data.

Digital signatures for Energy ~~Standards Code~~ electronic compliance documents and data should conform to the applicable conventions described in Sections 5.2 ~~and following~~, ~~5.3, 5.4, 5.5, and 5.6 below.~~

### 5.2 Digital Signature Certificates and Certificate Authorities

The California Secretary of State repealed the requirement to list its authorized certificate authorities. Reference Joint Appendix JA7.6.2.2.4 was revised to reflect this rulemaking change. "The Registration Provider's digital signature shall be created by a technology acceptable by the California Secretary of State." Definitions, criteria for acceptability, and additional requirements for Digital Signatures are available at the Secretary of State website.

Digital certificates protect against impersonation, certifying that a public key belongs to a specified entity. They are issued by a certificate authority (CA). A digital certificate binds a public key to the associated owner, whether that owner is an individual, a software

application, or some other entity. Digital certificates are also known as “public key certificates” because they give assurances about the ownership of a public key when an asymmetric key scheme is used. A digital certificate contains the public key for an entity and is a statement that the public key belongs to that entity.

If public keys are sent directly by the owner to another entity, there is a risk that the message could be intercepted, and the public key substituted with a different key. This action is known as a “Man in the Middle Attack.” The solution to this vulnerability is to exchange public keys through a trusted third party; thus, the user has a strong assurance that the public key is authentic. The ~~trusted third party, called a Certificate Authority~~ (CA) certificate authority incorporates the key into a digital certificate.

~~Data registry digital signatures for registering Energy Standards compliance document PDFs must use a CA approved by the State of California. A list of approved digital signature certification authorities can be accessed using the following URL:~~

~~<https://www.sos.ca.gov/administration/regulations/current-regulations/technology/digital-signatures/approved-certification-authorities>~~

~~The list of CAs approved by the State of California may change over time. Thus, if the CA used by a data registry for registering compliance documents is dropped from the list of CAs approved by the State of California, the data registry must revise the digital document signing process to use one of the CAs from the current list of approved CAs. If the data registry must change the CA used in the associated approval, it should notify the CEC of this change.~~

~~In 2020/2021, the Secretary of State suspended/repealed the requirement to list authorized CAs. This action does not suspend digital signatures requirements. Each active provider will continue to sign all compliance documents using digital signatures created using an acceptable technology's CA.~~

### 5.3 Document Components That Must Be Digitally Signed

At the conclusion of document registration, the compliance document PDF and the XML data corresponding to the information reported on the PDF are expected to be contained in a registration package as described in Section 4.4. At such time as the CEC may approve a document repository, then a copy of the completed registration package is expected to be exported to the CEC Document Repository as described in Section 3.1 and *Reference Joint Appendix JA7.7.1.4.*

- ~~• The completed registration package exported to the CEC Document Repository is referred to as a *transmission* package in Section JA7.7.1.4.~~

The data registry is expected to retain a copy of the completed registration/transmission package, in accordance with Title 24, section 10-103.3.

When exporting the registered compliance document PDF or data, there are two digital signature use cases to consider.

Use Case 1: An authorized user of the data registry downloads a copy of a registered compliance document PDF and must verify the provider's digital signature once the PDF is downloaded to the authorized user's computer. The user may also transmit the PDF file to others who will need to verify the data registry provider's digital signature. These users do not want or need the XML data corresponding to the information reported for that compliance document.

Use Case 2: The Data registry provider exports a completed transmission package to a document repository or to someone who requested both the registered compliance document PDF and the XML data corresponding to the information reported on the PDF. Once received, the requestor will need to verify the authenticity of both the registered compliance document PDF and the XML data contained in the completed transmission package.

### **5.3.1 Digital Signing Requirements for the Registered Compliance Document PDF**

- a. The registered compliance document PDF must be digitally signed at the time the document is registered using the data registry provider's digital certificate issued by a certificate authority approved by the California Secretary of State.
- b. Data registries should extract the digitally signed compliance document PDF from the transmission package for export to a user when only the PDF is required by the user. It must be possible for the user to verify the PDF digital signature by opening it with freeware such as Adobe Acrobat Reader. To ensure that PDF reader freeware can verify the authenticity of the registered PDF, data registries must provide a secure method for users to acquire the certificate issued by the approved CA of the data registry that will add the certificate to the user's local root certificate store, if necessary, as described in Section 5.5.
- c. If an export to a user of the registered compliance document PDF is contained in a transmission package, it must be possible for the user to extract the digitally signed PDF and verify the digital signature of the PDF by opening it with freeware such as Adobe Acrobat Reader.

### **5.3.2 Digital Signing Requirements for the Transmission Package**

- a. The transmission package must contain the compliance document PDF file that was digitally signed at the time the document was registered using the data registry provider's digital certificate issued by a CA approved by the California Secretary of State. If the transmission package is exported to a user, it must be possible for the user to extract the digitally signed PDF from the transmission package and verify the PDF digital signature by opening it with freeware such as Adobe Acrobat Reader. To ensure that PDF reader freeware can verify the authenticity of the registered PDF, data registries must provide a secure method for users to acquire the certificate issued by the approved CA of the data registry that will add the certificate to the user's local root certificate store, if necessary, as described in Section 5.5.
- b. The transmission package must contain the XML data corresponding to the

information reported on the registered compliance document PDF.

- c. For performance certificate of compliance documents, the transmission package must contain the data used to provide the input to the Title 24, Part 6 compliance software tool used to generate the registered performance certificate of compliance PDF contained in the transmission package.
- d. The entire contents of the registration/transmission package must be digitally signed at the conclusion of registration and signed independently of the digital signing of the compliance document PDF. The digital signing of the registration/transmission package is not required to use a digital certificate issued by a certificate authority approved by the California Secretary of State
- e. To keep the XML data and the PDF report linked, the data registries must retain the entire digitally signed registration/transmission package.

### 5.3.3 Digital Signature Appearance

The signature appearance is how the signature is displayed to the user on the completed compliance document PDF. When the compliance document PDF is signed, the signature appearance becomes part of the signed document. It is not part of the signature.

#### 5.3.3.1 The Content of the Digital Signature Appearance

Provide the following text in the digital signature appearance:

"Digitally signed by [data registry provider's name]. This digital signature is provided in order to secure the content of this registered document, and in no way implies Registration Provider responsibility for the accuracy of the information."

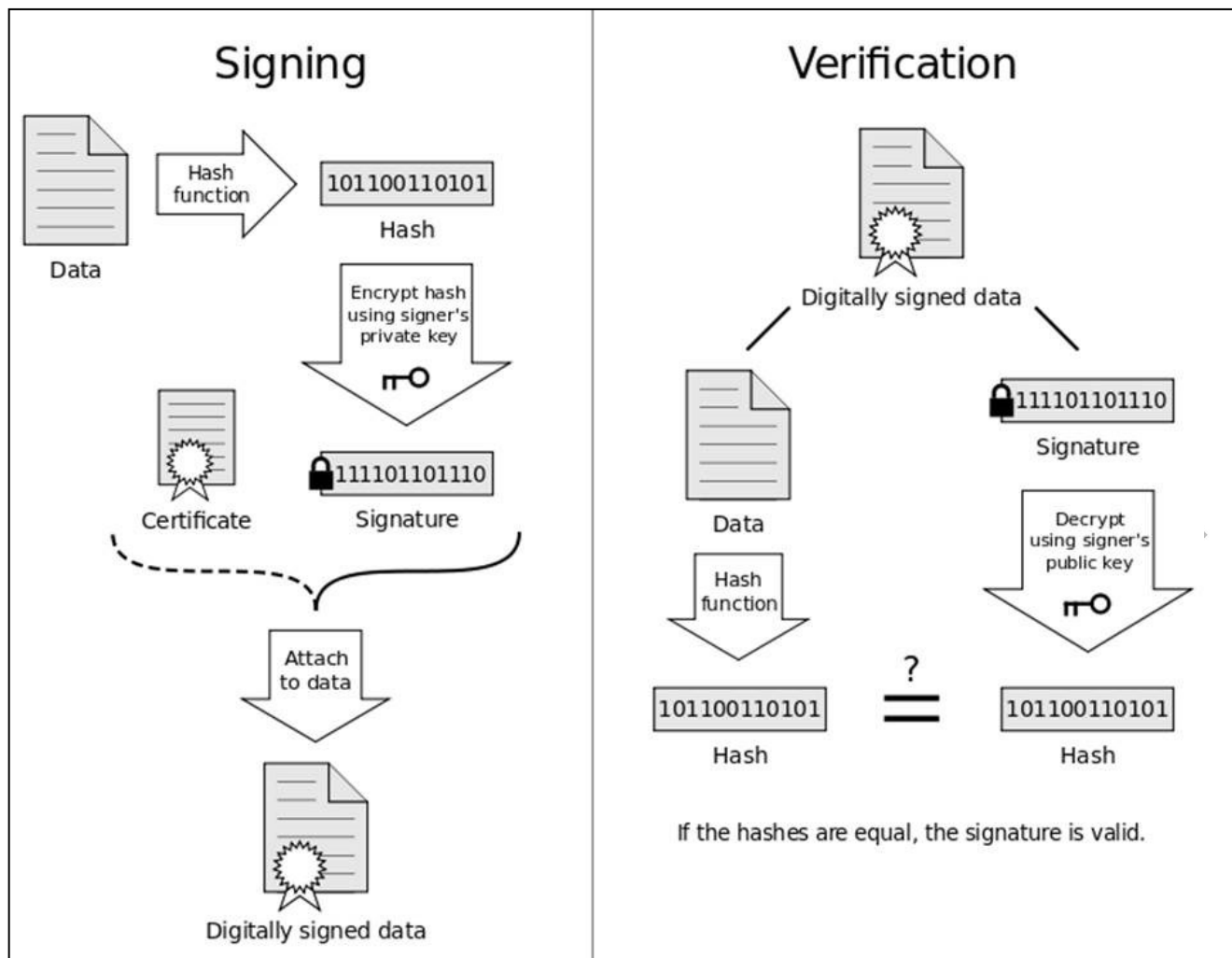
Do not include any other information such as graphic(s), watermark(s), date, or time stamps with the digital signature appearance.

#### 5.3.3.2 Placement of Digital Signature Appearance

Place the digital signature appearance once at the end of the compliance document, in a location that follows the responsible person's signature block.

Figure 5-1 describes the steps in the digital signing process and the digital verification process, which involves generating a hash and then applying the encryption key.

Figure 5-1: Process Flow Diagram for Digitally Signing a Document and Verifying the Digital Signature



Source: California Energy Commission

## 5.4 Considerations for Data Registry Digital Certificate Solutions

### 5.4.1 Approved Certificate Authorities

A digital certificate is attached to an electronic document to signify the sender's approval with their digital signature, a hash of the contents, and their public key for validation. A digital certificate also contains information such as the sender's name, the period for which the digital signature is valid, and the certificate authority (CA) that issued the digital certificate.

A CA issues digital certificates to a person or organization after verifying the information on their application. Since the initial development of digital signatures and digital certificates, companies and government agencies have implemented digital signature policies and published lists of approved CAs. California's Secretary of State publishes a [list of approved CAs](#) on its website accessible at:



<https://www.sos.ca.gov/administration/regulations/current-regulations/technology/digital-signatures/approved-certification-authorities>.

Data registry providers are required to assign their digital signature to registered compliance documents (PDF format documents) using a digital certificate issued by a CA approved by the Secretary of State. The digital certificate must be issued to the data registry organization, not to an individual. Digital signatures for registered compliance documents must clearly identify the data registry provider as the owner of the digital signature certificate.

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### **5.3.4 Digital Certificate Use Types**

There are three types of digital certificates that the provider may use.

#### **5.3.4.1 Digital Certificates for Document Signing**

- Advantages
  - Certificate satisfies the requirement to validate that the document hasn't changed since signing.
  - User doesn't have to perform additional actions to have the capability to verify authenticity of signed documents because the approved CA certificate comes with the operating system of their computer.
  - There are some automated solutions for organizations with high volume.
- Disadvantages
  - The organization is not listed as the owner in most offerings. The identity on the certificate is an employee name or a department in the organization.
  - Cost for high-volume solutions that require a hardware security module (HSM) is expensive compared to all other solutions.
  - Not offered by largest CAs.

#### **5.3.4.2 Digital Certificates for Code Signing**

- Advantages
  - Digital certificate authenticates the data registry provider as the owner and validates that the document ~~hasn't~~ had not changed since signing.
  - Timestamping avoids errors when a digital certificate expires after the document was signed by indicating the PDF was signed while the certificate was valid.
  - Price is reasonable. This solution must be incorporated into the data registry software development, which makes it automatic and not manual.
- Disadvantages
  - The CA digital certificate for this type is not automatically provided with the operating system of the computer, so the user has to perform actions to get the CA certificate added to their root certificates.

#### **5.3.4.3 SSL Certificates With Digital Signing of Documents**

- Advantages
  - An SSL certificate provides validation of the data registry provider and ~~their~~ its data registry website.
  - User rarely has to do anything because the root certificate of the CA is usually shipped with the user's operating system.

- Disadvantages
  - ~~May~~ Might not be available with SSL certificates ~~from most approved CAs.~~

### ~~5.4.3~~ 5.3.5 Digital Certificate Solution Options

Digital certificate solution options vary with the volume of digital signatures assigned annually. The following are examples of different digital certificate products, as surveyed in 2014 ~~from approved CAs.~~

~~Note: The price estimates provided in this section were acquired in 2014, thus, may not reflect current market pricing, and are provided for information only. However, the price estimates may still provide useful relative cost comparisons for these digital certificate solution options.~~

#### ~~5.4.3.1~~ 5.3.5.1 Manual Signing

Most CAs offer solutions that require manual signing of digital certificates. Some have limits on the number of signatures (up to 5,000 annually), ~~and prices range from \$400 to \$700 per year.~~ These systems come with a hardware key that contains the digital certificate and processing software that time stamps every digital signature, so it will be valid even after the digital certificate expires. Companies with a limited number of signatures per year may sign each certificate manually. However, for data registry providers with 20,000 or more signatures per year, manual signing is impractical.

#### ~~5.4.3.2~~ 5.3.5.2 Automatic Signing

CAs offer automated solutions, usually with unlimited signatures, for an annual fee plus the purchase of computer hardware that contains the digital certificate and processes digital signature requests, including the time stamping. CAs that are partners of Adobe's Certified Document Services (CDS) program are required to use hardware that complies with the National Institute of Standards and Technology (NIST) Federal Information Processing Standard, (FIPS) Publication 140-2 Security Requirements for Cryptographic Modules. This is usually referred to as the hardware security module (HSM) and comes in two forms. For systems that have a single server, an internal Peripheral Component Interconnect (PCI) card is the hardware solution. For systems that have multiple application servers, the HSM is an external appliance.

#### ~~5.4.3.3~~ Typical Vendor Pricing

~~Because the registered and signed compliance documents are in PDF format, and the digital signature must be verifiable using freeware such as Adobe Acrobat Reader, CAs that are on the Adobe Approved Trust List (AATL) are likely candidates for Data Registry Providers to consider. However, the candidate must also be on the Secretary of State's list of approved CAs. The CEC does not recommend or endorse any CA, but to demonstrate a pricing example for digital signatures, the following information is provided:~~

~~One CA offers an automated solution that provides unlimited digital signatures for \$15,000 per year. Its single server PCI card-based HSM ranges from \$5,000 to \$7,000. The multiple-server external appliance-based HSM ranges from \$12,000 to \$15,000. Applying this pricing~~

to a full-service data registry with roughly 200,000 signatures per year, the cost per registered compliance document would be \$0.15 the first year and \$0.075 each year after that.

#### 5.4.3.4 California Secretary of State Approved CAs and Adobe's Approved Trust List of CAs

CAs on the California Secretary of State list offer various services, and some specialize in providing services to government agencies. Refer to Table 5-1 below for a comparison of approved CAs. Those CAs that are also on Adobe's Approved Trust List (AATL) of organizations are identified in the second column. Adobe states the AATL members provide certificates that enable creation of trusted digital signatures whenever the signed document is opened in Adobe Reader or Acrobat.

The information in Table 5-1 is current as of September 2018 but is subject to change. Refer to the most current listings on the California Secretary of State list and Adobe's Approved Trust list when determining compliance with the data registry requirements.

Table 5-1: California Secretary of State List of Approved Certificate Authorities

| California Secretary of State list of Approved CAs<br><a href="https://www.sos.ca.gov/administration/regulations/current-regulations/technology/digital-signatures/approved-certification-authorities">https://www.sos.ca.gov/administration/regulations/current-regulations/technology/digital-signatures/approved-certification-authorities</a> | CAs also on Adobe Approved Trust List (AATL)<br><a href="https://helpx.adobe.com/acrobat/kb/approved-trust-list1.html">https://helpx.adobe.com/acrobat/kb/approved-trust-list1.html</a> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DigiCert                                                                                                                                                                                                                                                                                                                                          | DigiCert                                                                                                                                                                                |
| DocuSign, Corporation                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                         |
| Entrust Datacard Limited                                                                                                                                                                                                                                                                                                                          | Entrust Datacard Limited                                                                                                                                                                |
| GeoTrust, Inc.                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                         |
| GMO GlobalSign, Inc.                                                                                                                                                                                                                                                                                                                              | GMO GlobalSign, Inc                                                                                                                                                                     |
| Thawte, Inc.                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                         |
| Symantec Corporation                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                         |
| MaxMD/Park Avenue Capital, LLC                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                         |

Source: California Energy Commission

#### 5.5.4 Root Certificates for Validation of Document Authenticity

To ensure that PDF reader freeware can verify the authenticity of the registered PDF documents, data registries must provide a method for users to acquire the certificate issued by the approved CA of the data registry, which will add the certificate to the user's local root certificate store, if necessary. The suggested method is described in Subsection 5.5.1 below.

### **5.4.1 Data Registry Trusted-Source PDF Method**

- a. Each data registry must provide a link on one of the SSL-secured web pages that makes available a PDF file digitally signed by the data registry that the user can open (trusted-source PDF). It should not be necessary to log in to the data registry to navigate to this secure web page.
- b. The data registry must provide user instructions for viewing the status of the digital signature in this trusted-source PDF file.
- c. The data registry must provide user instructions to "trust" this trusted-source PDF file, which should result in the digital certificate of the data registry being added to the certificate store of the operating system of the user's local personal computing device.
- d. The data registry must provide user instructions to review the status of the digital signature in this trusted-source PDF file to confirm the status of the digital signature.

The status should indicate the signature is valid after adding the digital certificate of the registry to the certificate store of the operating system of the user's local personal computing device.

- e. If the digital signature in the trusted-source PDF file is not displayed as valid after performing the actions above, the data registry should provide further instructions for acquiring the certificate issued by the approved CA of the data registry that will add the certificate to the user's local root certificate store, if necessary.
- f. It may be necessary for the data registry to make available more than one version of a trusted-source PDF file type to make it possible for various types of PDF reader freeware applications to be used to validate registered compliance documents. The data registry must provide user instructions for selecting the correct trusted-source file for use with their PDF reader.

## **5.65.5 CBECC Software Output Data Security and Authentication for the Performance Certificate of Compliance**

The Residential and Nonresidential CBECC compliance manager-based software (compliance software) uses digital signing when generating analysis data for submission to the RG for creating the unregistered certificate of compliance. Each version of the compliance software employs a unique Rivest–Shamir–Adleman (RSA) algorithm-based public/private key pair to sign the data before sending it to the RG.

When transmitting data to the RG, the compliance software communicates using Secure Sockets Layer (SSL)-HTTPS (~~SSL~~) security technology to encrypt the communication. The RG will accept only HTTPS requests.

In addition to this basic authentication, the signed data are used to determine a watermarking status for compliance documents. Thus, in addition to the signed analysis output data, the compliance software sends parameters that the RG uses to determine how to process a request. These parameters include a "hash digest" or fixed length of arbitrary data that is based on the specific content of the analysis data, the public key for the signature, plus additional tokens. The RG reads the public key and compares the additional tokens to authenticate the connection made by the software. Once authenticated, the RG verifies the signature, processes the data, and may apply a watermark to the PDF report output, depending on the results of the verification.

## **5.75.6 RG Output Data Security and Authentication for the Performance Certificate of Compliance**

The RG employs XML digital signatures so that the certificate of compliance registration package produced by the RG can be verified by an approved data registry. Thus, the data registries must ensure that both the data and PDF documents used for registration have not been tampered with before submission to the Data Registry. This verification is accomplished using public key infrastructure (PKI) that employs a pair of public and /private keys.

The RG, using a secure private key, creates a hash, or fixed length of arbitrary data, that is based on the specific content of the data that has been processed. Any change made to the data being signed would result in a different hash value if that changed data were to be signed again using the same private key. This hash is the signature for that data.

The signed data can be verified by using the public key associated with the signature. Because the private key used to sign the data is secret, PKI uses the public key to verify that the data matches the signature provided. Data registry providers are given the public key that can be used to verify signed data.

The XML signing uses an "enveloped" signature, meaning that the signature is included in an XML element inside the data itself.

## **5.85.7 Data Exchanges Between External Digital Data**



## Sources (EDDS) and Data Registry

As an alternative to keyed-in data input for use in compliance document registration processes, the data registry may receive data transfers directly from external digital data sources as specified in Joint Appendix Section JA7.7.1.2. This receiving of data transfers may occur only if the working relationship between the data registry and the EDDS has been approved in accordance with the requirements in Joint Appendix Section JA7.8.

Section 5.8.1, 5.8.2, 5.8.3, and 5.8.4 below summarize the requirements for use of EDDS given in Joint Appendix JA7. Refer to JA7 for additional detail.

### ~~5.8.1~~ 5.7.1 EDDS Types

As specified in JA7.7.1.2.2, EDDS types ~~may~~ could include, ~~but are not limited to:~~

- a. Diagnostic instrument manufacturer services that incorporate wireless or web-based data-logging capabilities into the associated products, capture and store relevant information from field diagnostic testing procedures, and provide digital access to the stored data to the diagnostic tool owners and other parties to the field verification procedure.
- b. Third-party quality-control programs (TPQCP) that verify the work of participating installers, collect, and evaluate more detailed data than necessary for compliance, identify in real-time during the installation invalid and inaccurate installer testing and noncompliant installations, and enable corrected testing with the goal of bringing installations into compliance before the installer leaves the job site. TPQCP descriptions and requirements are specified in Appendix RA2.7.
- a. Internet-based datastore services that are administered by an EDDS provider who ensures the security and integrity of data input to the datastore service. Authorized users of Title 24, Part 6 Data Registries may elect to use EDDS datastore services for data input and subsequently transmit the stored data to a Title 24, Part 6 Data Registry while logged into the data registry during Title 24, Part 6 document registration.

### **5.7.2 EDDS Requirements**

As specified in JA7.7.1.2, the data upload to an EDDS. The data exchange between a data registry and an EDDS, including data upload and data exchange that is facilitated by an application programming interface (API), shall conform to the following:

- a. The data exchange from an EDDS to a data registry shall be initiated only by an authorized user of the data registry; only while the user is logged into his Title 24, Part 6 data registry user account; and only by use of a data exchange feature managed and made available to the user by the data registry user interface.
- b. The data exchange from an EDDS to a data registry shall not be an unattended automatic electronic data exchange transaction.
- c. The registration provider shall ensure the authorized user has the opportunity to review and revise the information transmitted to the data registry by use of an EDDS before making electronic signature controls available to the user.
- d. The registration provider shall be responsible for managing the security and integrity of the data exchange with the EDDS.
- e. The registration provider shall ensure that user data uploads to the EDDS and subsequent storage and maintenance of compliance data in the EDDS are done using best practices for secure data exchange and secure data storage.
- f. The registration provider shall ensure that the data exchange processes that import data into the data registry from the EDDS are performed using best practices for secure data exchange.
- g. The user's compliance data may be uploaded automatically to an EDDS datastore, such as by network-connected diagnostic field verification instruments, or it may be keyed in by the user using an EDDS services software user interface.
- h. The data transmitted from an EDDS to a data registry shall conform to the XML schema for each respective Title 24, Part 6 compliance document for which the data are to be used. All data provided to complete compliance documents shall be subjected to validation by the data registry software after the data is transmitted to the data registry.
- i. The current compliance document schemas approved by the CEC shall be made available to the EDDS services providers as needed to clarify the Title 24, Part 6 compliance document data requirements.

### **5.8.3.2 5.7.3 EDDS Approval**

#### **5.8.3.2 5.7.3.1 EDDS Application**

JA7.8 outlines the minimum content required for a registration provider EDDS application. The application must include detailed descriptions and specifications of the proposed EDDS. These descriptions and specifications will provide the CEC with the means to evaluate the EDDS application and the ability of the registration provider to show compliance with the regulatory requirements. The CEC will evaluate the applications based on the requirements in JA7.7.

To simplify the CEC review, the following four sections are recommended to be included in the application:

1. Introduction
2. Detailed description of the EDDS specifications and functionality. This description includes the full procedure for data collection, storage, maintenance, and data exchange and how requirements in JA7.7.1.2.1 will be met.
3. Demonstration of the EDDS service successfully and securely gather, storing, and transferring compliance data to the data registry.
4. Appendices, such as:
  - a. Self-test results (by the registration provider) of the transmission of compliance data between the EDDS and the data registry (JA7.8.2).
  - b. Details of contractual agreements with the EDDS service provider (JA7.8.2.6).
  - c. User manuals for the data registry and for the EDDS service software, features, and data registry process (JA7.8).
  - d. Application checklist of requirements from JA7.8.2 and the **HERS**ECC Regulations.
  - e. Other information at the discretion of the registration provider, such as a list of the compliance documents and data fields for which the EDDS will exchange compliance data.

Additional guidance on the application and approval process ~~may be~~ has been published by the CEC with respect to the 2022 Energy Code and the legacy HERS Program.<sup>1+</sup>

#### **5.8.3.2 5.7.3.2 EDDS Acceptance Testing**

Joint Appendix Section JA7.8 requires that when an application for a data registry approval includes use of EDDS, the CEC shall perform acceptance testing of the EDDS proposed to be used for data input by authorized users of the data registry.

Acceptance testing is necessary to verify that the proposed EDDS and data registry meet the requirements of JA7.7, conform to the specifications and features identified in the application, and are suitable for use for providing compliance document registration functionality required by the Energy Code. The recommended approach to satisfying the

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<sup>1</sup> Staff Paper: Approval Process for Use of an External Digital Data Source

acceptance testing requirements is for the registration provider to perform a live demonstration of the EDDS. The demonstration includes use of the EDDS and data registry to securely gather, store, and transfer compliance data to the data registry and produce a valid compliance document using the transferred compliance data. Each compliance document that receives data from the EDDS will be included in the demonstration. The CEC will observe the

<sup>1</sup>Footnotes:-

<https://efiling.energy.ca.gov/GetDocument.aspx?tn=234639-1&DocumentContentId=67486>

demonstration and evaluate EDDS and data registry functionality for conformance to the requirements of JA7.7 and specifications identified in the application.

### **5.8.3.3 5.7.3.3 Contractual Agreements Between Registry Provider and EDDS Services Provider**

JA7.8.2.6 requires applications for approval of data registry use of EDDS services to include documentation to disclose the details of the working agreement(s) or contract(s) between the registration provider and EDDS services entity. This documentation must include descriptions of the parties involved and the technologies used for the data exchanges between the EDDS and the data registry.

The contract executed between a registration provider and an ~~external digital data source (EDDS)~~ EDDS services provider is a prerequisite to approval of the EDDS for transmitting data to the data registry for Energy ~~Standards Code~~ document registration. The agreement is required to describe the specifications of any internet-based EDDS services or EDDS software used to store the compliance document data on behalf of authorized users of the data registry, including description of any internet-based data gateway interfaces used for sharing the compliance data with third parties.

A separate agreement is required for each working relationship between a data registry and an EDDS. EDDS services providers may be approved to provide services to any number of approved registration providers. Registration providers may be approved for use of any number of EDDS services providers.

### **5.8.3.4 5.7.3.4 EDDS Evaluation Criteria**

The CEC will evaluate any application or amendment submitted using the evaluation criteria included in the EDDS Service Application Checklist.<sup>2</sup> ~~This checklist is published by the CEC, and provides the specific evaluation criteria for each requirement, and includes suggestions of where in the application they should be addressed. The evaluation criteria are guided by the wording and intent of JA7, Title 24, Parts 1 and 6, and the HERSECHERS Regulations (Title 20, Section 1670). Any application for an EDDS to operate with a 2025 Energy Code Residential Data Registry must show how the EDDS will comply with the 2025 Energy Code and the ECC Program requirements of Title 24, Part 1, Section 10-103.3.~~

### **5.8.4 5.7.4 EDDS User Instructions**

<sup>2</sup> Checklist for 2022 EDDS Applicants

JA7.8.5.5 requires that when a data registry is approved to make available use of EDDS features to authorized users of the data registry for data input during document registration procedures, the data registry user manual shall include instructions for use of those features. The instructions shall describe use of the data registry user interface for EDDS data input procedures. Moreover, if the EDDS services provider has a user interface or software application that the user is expected to access and operate that is independent of the data registry user interface, a copy of the EDDS service or software user instructions shall be ~~2Footnotes: The checklist is located on Docket 19-HERS-01~~

~~<<https://efiling.energy.ca.gov/GetDocument.aspx?tn=234639>~~

~~1<https://efiling.energy.ca.gov/GetDocument.aspx?tn=234639-2&DocumentContentId=67487>>~~

included in the Data Registry User Manual. If the EDDS service or software user instructions contain proprietary information or intellectual property, the EDDS service or software user instructions do not need to be included in the Data Registry User Manual. However, the EDDS service or software user instructions must be made available to all authorized users that use the EDDS service or software.

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Text is subject to internal review and revision, and that it is not official guidance until approved by the Commission



## 6 Document Configuration Rules

### 6.1 Overview

High-level document configuration requirements are specified in *2025 Reference Joint Appendix JA7*. Section JA7.5.6 requires that data registries shall be capable of tracking all compliance documentation and maintaining the correct associations between related documents, including revisions and completion statuses for all documents within a building project. Section JA7.5.6.1 from *2025 Reference Joint Appendix JA7* is copied into Section 6.2.1 below. Additional guidance for completion of the project status report is given in Section 6.2.2.

### 6.2 Project Status Reports (PSR)

The PSR is a dynamic compliance document that is generated each time it is accessed so that it represents the latest status of a project's registration state and to display notes that could have been generated based on user input overrides on some document types. Prior versions of the PSR are not required to be retained.

#### ~~6.1.1~~ 6.2.1 Project Status Report Requirements Specified in ~~2022~~ Reference Joint Appendix JA7

##### ~~6.1.1.1~~ 6.2.1.1 Project Status Reports (quoted ~~F~~from JA7.5.6.1)

The status of completion of a project shall be reported by the data registry.

The data registry shall determine the documents required for a project based on the certificate of compliance and maintain a summary that reflects the current status of completion of the required documents and shall be readily accessible to authorized users of the data registry. Access to the report shall be facilitated by use of search parameters relevant to the project as listed in Sections JA7.5.6.1.1 and JA7.5.6.1.2.

Enforcement agencies may be authorized to enter notations into project records in data registries to communicate plan check and field inspection information to builders, designers, installers, and raters.

The project status report shall be made available in a printable format.

Minimum information requirements for the project status report shall include the following:

##### ~~6.1.1.2~~ 6.2.1.2 Project Status Report Information for Low-Rise Multifamily and Single-Family Residential Projects (From JA7.5.6.1.1)

- a. Project name.
- b. Project location (or address).
- c. Listing of the certificate of compliance documents required, date registered (or indicate not complete if the document record has been started but is not yet

registered), and registration number.

- d. Listing of the certificate of installation documents required, date registered (or indicate not complete if the document record has been started but is not yet registered), and registration number.
- e. Listing of the certificate of acceptance documents is not required, which must be communicated on the PSR.
- f. Listing of the certificate of verification documents required, date registered or indicate not complete if the document record has been started but is not yet registered, and registration number.
- g. Listing of the mandatory measure options required; options selected (refers to the certificate of installation and certificate of verification documentation).

### **6.1.1.3 6.2.1.3 Project Status Report Information for High-Rise Multifamily and Nonresidential Projects (From JA7.5.6.1.2)**

Note: Nonresidential document registration is contingent upon approval of a nonresidential data registry by the Commission.

- a. Project name.
- b. Project location (or address).
- c. Listing of the certificate of compliance documents required, date registered (or indicate not complete if the document record has been started but is not yet registered), and registration number.
- d. Listing of the certificate of installation documents required, date registered (or indicate not complete if the document record has been started but is not yet registered), and registration number.
- e. Listing of the certificate of acceptance documents required, date registered (or indicate not complete if the document record has been started but is not yet registered), and registration number.
- f. Listing of the certificate of verification documents required, date registered (or indicate not complete if the document record has been started but is not yet registered), and registration number.

### **6.1.2 6.2.2 Additional Guidance for Producing Residential Project Status Reports**

#### **6.1.2.1 6.2.2.1 Mandatory Measure Compliance Alternatives Determined at Installation Phase**

This section provides additional guidance for the topic identified in Section 6.2.1.2 f above.

When a mandatory requirement allows for compliance alternatives to be determined at the time of installation by the installing contractor, the project status report cannot determine

which compliance documents will be required for those features, if based only on the information given on the certificate of compliance.

Example Use Case — Compliance with Energy ~~Standards~~-Code Section 150.0(m)13:

The mandatory requirements in Section 150.0(m)13 for verification of ducted cooling system airflow rate and fan efficacy provides an exception which allows for use of an alternative compliance method when installation of a duct design that conforms to Table 150.0-B or Table 150.0-C was used. Thus, at the time of installation of the space conditioning system, the installer must select one of these options for compliance. Until the installing contractor selects an option for compliance and registers the applicable Installation Certificate (CF2R-MCH-01), the Data Registry will be unable to determine whether a MCH-22 and MCH-23 will be required to document the airflow rate and fan efficacy measurements, or otherwise whether a MCH-28 will be required to document the alternative duct design compliance option specified by Table 150.0-B or C.

Generally: When a certificate of compliance is registered for a project and the subsequent certificate of installation and certificate of verification compliance documents for a feature cannot be determined until after installation of the feature, the project status report should disclose the optional compliance choice(s) that must be determined by the installer at installation. After the installer determines the method of compliance and registers the applicable certificate of installation, the project status report should be updated to report the full set of compliance document requirements of the project consistent with the installer's compliance option choice.

Continuation of the 150.0(m)13 Example Use Case:

- a. After a Certificate of Compliance for the project is registered, but prior to registration of a CF2R-MCH-01 Certificate of Installation, the Project Status Report should indicate two options are available for compliance for the system; either a MCH-22 and MCH-23 will be required or, alternatively, a MCH-28 will be required.
- b. After a CF2R-MCH-01 for the dwelling is registered, if the MCH-01 determines airflow rate and fan efficacy verification will be documented for the system, the Project Status Report should indicate a MCH-22 and MCH-23 will be required. Otherwise, if the MCH-01 determines that the alternative compliance method using the Table 150.0-B or Table 150.0-C duct design was used, then the Project Status Report should indicate a MCH-28 will be required.

### ~~6.1.2.2~~ 6.2.2.2 Reporting Non-default Data Inputs

To streamline data entry for project compliance documents, many data fields have specified a default value that is expected to be correct for most project documentation situations. The user may leave the default value unchanged if it is correct, thus the user should not have to perform any data entry actions for these fields if the default value is correct.

When Compliance Document data field rules allow overriding a default value, and when the user elects to override that default value, the Project Status Report should include an

advisory message describing the:

- Data field name/identification for the overridden data.
- Default value for the data field.
- New value that was input as an alternative to the default value for the data field.

Allowing a user to override a default value may simplify the completion of project compliance documents by allowing the users the flexibility to avoid having to make revisions to parent documents when reporting features that are inconsistent with specifications on parent documents but are nonetheless still in compliance with the Energy ~~Standards~~Code.

Data Registry staff should not be required to determine whether the overridden/new data-input value complies with the Energy ~~Standards~~Code as part of the document registration process. The enforcement agency review of the project documentation at final inspection, or ~~HERSECC~~ quality assurance procedures, may ~~(or may not)~~ result in determinations/directives to revise one or more of the affected compliance documents or make changes to the feature(s) that were installed.

Example Use Case:

- The CF1R-PRF-01 specifies a single-zone ducted space cooling system with a single-speed condensing unit; the builder upgrades the dwelling to install a multi-zone ducted space cooling system with no bypass duct and a multi-speed compressor.
- On the CF2R-MCH-01 for the dwelling, the values in D09 and D10 must be overridden to accurately report the space conditioning system installed is Zonally Controlled and the Condenser is Multi-Speed.
- On the project Summary Report for the CF2R-MCH-01 the following messages must be reported:

Section D Field 09 Default Value Overridden:

- Default Value = Not Zonal
- Entered Value = Zonally Controlled

Section D Field 10 Default Value Overridden:

- Default Value = Single-Speed
- Entered Value = Multi-Speed

- An enforcement agency or ~~HERSECC~~ rater review of the overridden data finds that the compliance requirements for the Zonally Controlled system are the same as the requirements for the Not Zonal system. Since the compliance requirements are the same for the substitution, there is no need to revise the CF1R-PRF-01.

### ~~6.1.2.36.2.2.3~~ Delayed ~~HERSECC~~ Verification of Refrigerant Charge

- Excerpt from RA2.4.4:



"If necessary to avoid delay of approval of dwelling units completed when outside temperatures are below 55°F, the enforcement agency may approve compliance with the refrigerant charge verification requirements when installers have used the Weigh-in Charging Method described in Reference Residential Appendix RA3, Section RA3.2.3.1 and have not used the Section RA3.2.3.2 option for **HERSECC** verification compliance. This approval will be on the condition that installers submit to the enforcement agency a registered Certificate of Installation that includes a signed declaration indicating agreement to return to correct refrigerant charge if a **HERSECC** Rater determines at a later time when the outside temperature is 55°F or above, that correction is necessary. Installers must also notify homeowners that their systems have not had their charge verified. The **HERSECC** Provider shall track these projects to ensure a **HERSECC** Rater conducts the required refrigerant charge verification for all such systems."

- b. Excerpt from Certificate of Installation CF2R-MCH-25c for the Weigh-in Charging Procedure; Reference: Section F - Additional Requirements; field 04:

"When applicable and if necessary to avoid delay of approval of dwelling units completed when outside temperatures are below 55°F, the enforcement agency may approve compliance with the refrigerant charge verification requirements based on registration of this CF2R-MCH-25, documenting use of the RA3.2.3.1 HVAC Installer Weigh-In Charging Procedure when the optional Section RA3.2.3.2 **HERSECC** Rater Observation of Weigh-In Charging Procedure is not used. As condition for such enforcement agency approval, the responsible person's signature on this compliance document affirms the installer agrees to return to correct refrigerant charge if a **HERSECC** Rater determines at a later time, when the outside temperature is 55°F or greater, that refrigerant charge correction is necessary."

- c. Additional guidance for **HERSECC** Provider tracking and follow-up communications:

In order for the **HERSECC** Provider Data Registry to track these conditionally approved cold weather refrigerant charge verifications, and to ensure a **HERSECC** Rater conducts the required refrigerant charge verification when the outdoor air temperature is warmer, the Provider must be informed that the dwelling was approved at final inspection based on registration of only a CF2R-MCH-25, documenting use of the RA3.2.3.1 installer weigh-in charging procedure as allowed by RA2.4.4.

Note: There are no explicit requirements in the Energy Standards or in the Reference Appendices that direct the enforcement agency to disclose to the **HERSECC** Provider Data Registry any information about a project's building permit status, or the status of a required refrigerant charge verification. Therefore if the enforcement agency does not notify the Data Registry that a building was approved based on registration of only the CF2R-MCH-25, and if the required CF3R-MCH-25 has not been registered to complete the full set of required project documentation, it may be necessary for the Data Registry staff to follow up with the enforcement agency to determine the status of the building permit, and the status of the refrigerant charge verification for the building to determine whether refrigerant charge verification tracking is needed.

- d. Additional guidance for Data Registry Project Status Reporting for the necessary RA2.4.4 follow-up:



The Data Registry requirements in *Reference Joint Appendix JA7.5.6.1* state: "enforcement agencies may be authorized to enter notations into project records in data registries to communicate plan check and field inspection information to builders, designers, installers, and raters."

Thus, the Data Registry should make available data fields in the Project Status Report that enable enforcement agency persons, or **HERSECC** Raters, or Data Registry staff to flag a CF2R--MCH-25c for the project as requiring **HERS**field verification at a later time when the weather is warmer, when the enforcement agency has approved the dwelling based on registration of only a CF2R-MCH-25. The Data Registry should also make available the capability for users to enter notes that provide additional information useful for determining how and when the needed follow-up **HERS**field verification should be conducted, and if applicable, who should be contacted to perform the follow-up **HERS**field verification.

- e. Additional Guidance for Data Registry follow-up communications for prompting for **HERS**field verification according to RA2.4.4:

Once the Data Registry has set a flag to indicate that a follow-up **HERS**field verification is required, the Data Registry can be configured to automatically distribute reminder communications to the appropriate persons at predetermined time(s) or when predetermined conditions, such as warmer weather, are met.

The following table illustrates how the PSR could evolve during a compliance documentation workflow.

*Table 6-1: Contents of the Project Status Report per Project State*

| <b>Project State</b>                        | <b>PSR Contents (incremental)</b>                                                                                                              | <b>References</b>           | <b>Notes</b>                                                                                                                                                                                            |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>Compliance (no documents registered)</u> | <u>Optional at this stage.</u><br><u>Project site identification.</u><br><u>May allow AHJ user notations.</u>                                  | <u>JA7.5.6.1, 10-103(a)</u> | <u>If used, no documents are listed. PSR status is incomplete.</u>                                                                                                                                      |
| <u>Installation</u>                         | <u>Project site identification.</u><br><u>Registered certificates of compliance (CoC).</u><br><br><u>Mandatory features based on CoC data.</u> | <u>JA7.5.6.1, 10-103(a)</u> | <u>PSR status is incomplete.</u><br><u>CoC status is green/checked.</u><br><br><u>Other documents will be identified and given incomplete statuses.</u>                                                 |
| <u>Installation</u>                         | <u>Some certificates of installation (CoI) are ready for authoring.</u>                                                                        | <u>JA7.5.6.1, 10-103(a)</u> | <u>The registry will enable additional forms. These forms will be listed though the variants might not be known. Also, registration of something like the CF2RMCH01bE will enable additional forms.</u> |
| <u>Installation</u>                         | <u>Registered and enabled CoI forms.</u><br><u>Mandatory measures</u>                                                                          | <u>JA7.5.6.1, 10-103(a)</u> | <u>The registry should offer options like MCH-22, MCH-23, and MCH-28 then take away the mutually excluded</u>                                                                                           |

| <u>Project State</u>                 | <u>PSR Contents (incremental)</u>                                                                                                                                                                                                                                         | <u>References</u>                                                                            | <u>Notes</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>(updated).<br/>Content should be organized hierarchically by CoC, building, dwelling unit, system.<br/>Registry notes about changes to inputs that were defaulted based on CoC values.</p> <p>Indications of signature agreements (delegated signature authority).</p> | <p>DRRM section 6.2.2.2 covers "non-default" value overrides.</p> <p>10-103(a) item 3Avi</p> | <p>option.</p> <p>Forms like the MCH-25 have a registration prerequisite, but the registry could allow both to be authored and should list both on the PSR (-25 and -23).</p> <p>If the author cannot override any default inputs, then the CoC must be revised to reflect these changes, subject to 10-103(d)2.</p> <p>The use of delegated signature authority should be apparent for each document on the PSR.</p> <p>On the project page or elsewhere on the registry, the signature agreements must be accessible by AHJ and CEC users.</p> |
| <u>Installation and Verification</u> | <p>At least one certificate of verification (CoV) is enabled for authoring. Potentially all required forms are listed and have varying states of being required or registered.</p>                                                                                        | <p>DRRM Table 6-1</p>                                                                        | <p>As CoI forms are registered, the CoV counterparts should be made available on the registry and appear on the PSR.</p> <p>Registered forms are shown as complete; triggered forms are incomplete; overall project status is incomplete.</p>                                                                                                                                                                                                                                                                                                    |
| <u>Completed</u>                     | <p>All required forms are listed and registered.</p>                                                                                                                                                                                                                      | <p>10-103(d)2</p>                                                                            | <p>Project Status is complete when all listed forms have a complete status. All documents listed have registration numbers.</p> <p>AHJ must inspect the completed project to verify that indicated features are installed and that the required certificates were submitted. The PSR has an implied role at this stage.</p>                                                                                                                                                                                                                      |

Source: California Energy Commission

#### **6.2.2.4 Photographic Documentation**

Requirements for associated photographic documentation were added to the 2025 Energy Code update. JA7.5.6.3 states the conditions for these requirements. "If a registered document is associated with photographic evidence, it shall be stored [by the ECC-Provider] as a Joint Photographic Experts Group (JPEG) file." Per JA7.5.6.3(b), the ECC-Provider must make the files available to the CEC upon request. In practice, the photos may be linked from a project or site management page or elsewhere in the registry user environment normally accessible to CEC users to reduce manual requests for access.

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Text is subject to internal review and revision, and that it is not official guidance until approved by the Commission

### 6.3 General Configuration Rules

Table 6-2: Document Configuration Rules Applicable to ~~2022~~ 2025 Single-Family Compliance Documents ~~and HERS ECC Features~~

| Document Number | Document Type                          | Document Description                                                                      | Single-Family Document Configuration Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------|----------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CF1R-ADD-01-E   | Prescriptive Certificate of Compliance | Residential Building Additions Less Than or Equal To 1,000 ft <sup>2</sup> (Prescriptive) | <p>Prompt user to select compliance method and project scope.</p> <p>If compliance method is prescriptive and project scope is newly constructed addition less than or equal to 1,000 ft<sup>2</sup> (including ADUs), then require one CF1R-ADD-01 for the building.</p>                                                                                                                                                                                                                           |
| CF1R-ALT-01-E   | Prescriptive Certificate of Compliance | Residential Building Alterations (Prescriptive)                                           | <p>Prompt user to select compliance method and project scope.</p> <p>If compliance method is prescriptive and project scope is alteration to one or more building components, including an alteration to a space conditioning system<del>s</del>, then<br/>_require one CF1R-ALT-01 for the building.</p> <p>If the alteration is limited to only space conditioning system(s), then instead require use of the CF1R-ALT-02, which is applicable to only space conditioning system alterations.</p> |

| Document Number | Document Type                          | Document Description                                    | Single-Family Document Configuration Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|----------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CF1R-ALT-02-E   | Prescriptive Certificate of Compliance | Space Conditioning System(s) Alterations (Prescriptive) | <p>Prompt user to select compliance method and project scope.</p> <p>If compliance method is prescriptive and project scope is limited to alterations to space conditioning systems contained in a single family dwelling unit (heating systems, cooling systems, duct systems), then require one CF1R-ALT-02 for the single family dwelling unit.</p> <p>Else if compliance method is prescriptive, and a CF1R-ALT-01 or CF1R-ADD-01 are used for the project, and the building type is single family, then require one CF1R-ALT-02 for the single family dwelling unit identified on the CF1R that requires installation or alteration of a space conditioning system.</p> |
| CF1R-ENV-02-E   | Prescriptive Certificate of Compliance | Area Weighted Average Calculation Worksheet             | <p>Prompt user to declare whether they need to use an area weighted average to meet any of the prescriptive U-values or SHGC values.</p> <p>User elects to register a</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |



| Document Number | Document Type | Document Description | Single-Family Document Configuration Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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|                 |               |                      | <p>CF1R-ENV-02 when an area weighted average value is required to be calculated to demonstrate overall envelope compliance when a non-compliant value for an envelope feature is entered on the CF1R for the project. Require one CF1R-ENV-02 for each feature being area- weighted.</p> <p>When a CF1R-ENV-02 is required for compliance, registration of the CF1R-ENV-02 is a prerequisite to allowing registration of the parent CF1R (ALT--01, ADD-01, NCB-01).</p> <p>This worksheet is used to calculate the area-weighted average U-factors for building envelope features such as opaque exterior doors, walls, roofs, ceilings, floors, mass, and fenestration/glazing U-factors or Solar Heat Gain Coefficients (SHGC) values for prescriptive compliance.</p> |

| Document Number | Document Type                          | Document Description                                        | Single-Family Document Configuration Rules                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| CF1R-ENV-03-E   | Prescriptive Certificate of Compliance | Solar Heat Gain Coefficient (SHGC) Worksheet (Prescriptive) | <p>If the CF1R for the project is one of the following <del>prescriptive</del> CF1R types: CF1R-NCB-01, CF1R-ALT-01, or CF1R-ADD-01, and the value for Exterior Shading Device does not equal <del>"None"</del>, then require one CF1R-ENV-03 for the project.</p> <p>When a CF1R-ENV-03 is required for compliance, registration of the CF1R-ENV-03 document is a prerequisite to allowing registration of the parent CF1R-<del>(ALT-01, ADD-01, NCB-01)</del>.</p> |

| Document Number | Document Type                          | Document Description                                                 | Single-Family Document Configuration Rules                                                                                                                                                                                                                                                                                                                                                                                                      |
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| CF1R-ENV-04-E   | Prescriptive Certificate of Compliance | Cool Roof and Solar Reflectance Index (SRI) Worksheet (Prescriptive) | <p>If the CF1R for the project is one of the following prescriptive CF1R types:</p> <p>CF1R-NCB-01, CF1R-ALT-01, or CF1R-ADD-01, require one CF1R-ENV-04 for each roofing feature listed on the C1R that lists a value for Proposed SRI.</p> <p>When a CF1R-ENV-04 is required for compliance, registration of all applicable CF1R-ENV-04 documents is a prerequisite to allowing registration of the parent CF1R (ALT-01, ADD-01, NCB-01).</p> |
| CF1R-ENV-05-E   | Certificate of Compliance              | Alternative Default Fenestration Procedure (NA6) Worksheet           | <p>If the CF1R for the project is one of the following types:</p> <p>CF1R-PRF-01, CF1R-NCB-01, CF1R-ALT-01, or CF1R-ADD-01, and the value for Fenestration U-factor and/or SHGC Source is "ADFP" or "NA6 Equations," then require one CF1R-ENV-05 for the project.</p> <p>When a CF1R-ENV-05 is required for compliance, registration of the CF1R-</p>                                                                                          |

| Document Number | Document Type                          | Document Description                                                                                     | Single-Family Document Configuration Rules                                                                                                                                                                                                                                                                                                                                                                                                          |
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|                 |                                        |                                                                                                          | <p>ENV-05</p> <p>_document is a prerequisite to allowing registration of the parent CF1R (ALT-01, ADD-01, NCB-01, PRF-01).</p>                                                                                                                                                                                                                                                                                                                      |
| CF1R-ENV-06-E   | Prescriptive Certificate of Compliance | Interior and Exterior Insulation Layers Worksheet (Prescriptive)                                         | <p>If the CF1R for the project is one of the following prescriptive CF1R types:</p> <p>CF1R-NCB-01, CF1R-ALT-01, or CF1R-ADD-01, require one CF1R-ENV-06 for each mass wall listed on the CF1R that has a value for Interior and/or Exterior Insulation.</p> <p>When a CF1R-ENV-06 is required for compliance, registration of the CF1R-ENV-06 document is a prerequisite to allowing registration of the parent CF1R (ALT-01, ADD-01, NCB-01).</p> |
| CF1R-NCB-01-E   | Prescriptive Certificate of Compliance | Residential Newly Constructed Buildings, and Additions Greater Than 1,000 ft <sup>2</sup> (Prescriptive) | <p>Prompt user to select compliance method and project scope.</p> <p>If compliance method is prescriptive and project scope if newly</p>                                                                                                                                                                                                                                                                                                            |

| Document Number | Document Type                          | Document Description                                | Single-Family Document Configuration Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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|                 |                                        |                                                     | constructed building, or prescriptive newly constructed addition greater than 1,000 ft <sup>2</sup> , then require one CF1R-NCB-01 for the building.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| CF1R-PLB-01-E   | Prescriptive-Certificate-of-Compliance | Hydronic Heating System(s) Worksheet (Prescriptive) | <p>If the CF1R for the project is one of the following prescriptive CF1R types:</p> <p>CF1R-NCB-01, CF1R-ALT-01, CF1R-ADD-01, or CF2R-ALT-02, and the space conditioning heating system type identified on the CF1R is one of the following types: Hydronic, Combined Hydronic, Hydronic + Forced Air, or Combined Hydronic + Forced Air,</p> <p>then require one CF1R-PLB-01 for each of the hydronic heating systems listed on The CF1R</p> <p>When a CF1R-PLB-01 is required for compliance, registration of all applicable CF1R-PLB-01 documents is a prerequisite to allowing registration of the parent CF1R (ALT-01, ADD-01, NCB-01, ALT-02).</p> |



| Document Number | Document Type               | Document Description             | Single-Family Document Configuration Rules                                                                                                                                                                                                                                                                  |
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| CF2R-ELC-01-E   | Certificate of Installation | Electric Ready Requirements      | <p><del>If the CF1R is a CF1R-NCB, or if the CF1R is a CF1R-</del></p> <p><del>PRF and the Project Scope is a Newly Constructed Building, then require one CF2R-ELC-01 per dwelling unit.</del></p> <p>If the Project Scope is a Newly Constructed Building, require one CF2R-ELC-01 per dwelling unit.</p> |
| CF2R-ENV-01-E   | Certificate of Installation | Fenestration Installation        | <p>If the CF1R specifies fenestration features, require one CF2R-ENV-01 for each CF1R.</p> <p><del>Else if the CF1R does not specify fenestration features, a CF2R-ENV-01 is not required.</del></p>                                                                                                        |
| CF2R-ENV-03-E   | Certificate of Installation | Insulation Installation          | <p>If the CF1R specifies insulation features, require one CF2R-ENV-03 for each CF1R.</p> <p><del>Else if the CF1R does not specify insulation features, a CF2R-ENV-03 is not required.</del></p>                                                                                                            |
| CF2R-ENV-04-E   | Certificate of Installation | Roofing; Ventilation; Cool Roofs | <p>If the CF1R specifies Radiant Barrier or Cool Roof features, require one CF2R-ENV-04 for each CF1R.</p>                                                                                                                                                                                                  |

| Document Number                | Document Type                                        | Document Description                                                                                                                                                         | Single-Family Document Configuration Rules                                                                                                                                                                                                           |
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|                                |                                                      |                                                                                                                                                                              | <del>Else if the CF1R does not specify Radiant Barrier or Cool Roof features, a CF2R-ENV-04 is not required.</del>                                                                                                                                   |
| CF2R/CF3R-ENV-20a-H            | Certificate of Installation/ Verification            | Enclosure Air Leakage Diagnostic Test –<br>– Single Point Air Tightness Test with Manual Meter <u>or</u> <u>Automatic Meter,</u><br>– <u>or Multipoint Airtightness Test</u> | When the CF1R indicates requirement for Envelope Leakage <u>HERS field</u> verification for the dwelling, require one ENV-20 per building.                                                                                                           |
| <del>CF2R/CF3R-ENV-20b-H</del> | <del>Certificate of Installation/ Verification</del> | <del>Enclosure Air Leakage Diagnostic Test – Single Point Air Tightness Test with Automatic Meter</del>                                                                      | <del>When the CF1R indicates requirement for Envelope Leakage HERS verification for the dwelling, require one ENV-20 per building.</del>                                                                                                             |
| CF2R/CF3R-ENV-21-H             | Certificate of Installation/ Verification            | Quality Insulation Installation (QII) – Air Infiltration Sealing (Framing Stage)                                                                                             | When the CF1R indicates requirement for QII <u>HERS field</u> verification for the dwelling, compliance shall be demonstrated using all the QII field verification compliance documents (ENV- 21 & ENV-22).<br><br>Require one ENV- 21 per building. |
| CF2R/CF3R-ENV-22-H             | Certificate of Installation/ Verification            | Quality Insulation Installation (QII) – <u>(Insulating on</u>                                                                                                                | If the CF1R-PRF-01 indicates compliance credit for Nondefault SPF                                                                                                                                                                                    |

| Document Number | Document Type               | Document Description                                                                              | Single-Family Document Configuration Rules                                                                                                                                                                                                                                                                                                                                |
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|                 |                             | Installation Stage)                                                                               | R-value, then require one ENV-22 per building. Else, if the ENV-21 is required, require one ENV-22 per building.                                                                                                                                                                                                                                                          |
| CF3R-EXC-20-H   | Certificate of Verification | HERS-ECC-Rater Verification for of Existing Conditions for Performance Compliance for Alterations | <p>When credit for <u>verified</u> existing conditions is used on the CF1R, as a condition for CF1R registration, an EXC-20 that <del>verifies</del> <u>certifies</u> the existing conditions claimed on the CF1R shall first be registered.</p> <p>Required as a prerequisite to registration of a CF1R-PRF for an altered dwelling.</p>                                 |
| CF2R-LTG-01-E   | Certificate of Installation | Lighting – Single Family Dwellings                                                                | <p>If the building type on the CF1R is single family and the scope of the project is a Newly Constructed Building, then require one CF2R-LTG-01 per dwelling unit.</p> <p>Else prompt the user to declare whether the scope of the project includes lighting. If the scope includes lighting, then require one CF2R-LTG-01 per dwelling unit.</p> <p>Else CF2R-LTG-01</p> |

| Document Number | Document Type               | Document Description                                                                                    | Single-Family Document Configuration Rules                                                                                                                                                                                                                  |
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|                 |                             |                                                                                                         | is not required.                                                                                                                                                                                                                                            |
| CF2R-MCH-01a-E  | Certificate of Installation | Space Conditioning Systems, Ducts and Fans – for Performance Compliance for Newly Constructed Buildings | If the CF1R type is a CF1R-PRF-01 and the project scope is Newly Constructed Building, then require one CF2R- MCH-01a for each dwelling unit for which there are HVAC system compliance                                                                     |
| CF2R-MCH-01b-E  | Certificate of Installation | Space Conditioning Systems, Ducts and Fans – for Prescriptive Alterations                               | If the CF1R type is a CF1R-ALT-02, then require one CF2R-MCH-01b for each dwelling unit for which there are HVAC system compliance requirements given on the CF1R-ALT- 02.<br><br>Note: Refer also to the rules for configuration of the CF1R-ALT-02 above. |
| CF2R-MCH-01c-E  | Certificate of Installation | Space Conditioning Systems, Ducts and Fans – for Prescriptive Newly Constructed Buildings               | If the CF1R type is a CF1R-NCB-01, then require one CF2R-MCH-01c for each dwelling unit for which there are HVAC system compliance requirements given on the CF1R.                                                                                          |
| CF2R-MCH-01d-E  | Certificate of Installation | Space Conditioning Systems, Ducts and Fans – for Performance                                            | If the CF1R type is a CF1R-PRF-01 and the project scope is one                                                                                                                                                                                              |

| Document Number | Document Type               | Document Description                                        | Single-Family Document Configuration Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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|                 |                             | Compliance for Additions and Alterations, or Addition Alone | of the following two types: (1) Addition and/or alteration_ or (2) Newly Constructed – Addition Alone, then require one CF2R-MCH-01d for each dwelling unit for which there are HVAC compliance requirements given on the CF1R.                                                                                                                                                                                                                                                                                                                                                                       |
| CF2R-MCH-02-E   | Certificate of Installation | Whole House Fan                                             | <p>For single family projects in climate zones 8-14; if certificate of compliance type is prescriptive new construction (NCB), then require one CF2R-MCH-02 for the dwelling.</p> <p>Else, if the certificate of compliance type is prescriptive addition or alteration, then prompt user to declare whether a MCH-02 is needed.</p> <p>If the certificate of compliance type is performance (PRF), then if ventilation cooling system type from the CF1R is_ whole house fan_ and the field verification status is no, then require one CF2R-MCH-02_ for the dwelling.</p> <p>Else, CF2R-MCH-02_</p> |



| Document Number    | Document Type                            | Document Description                                                                                                                                                                                                                                                                            | Single-Family Document Configuration Rules                                                                                                                                                                                                                                                                                              |
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|                    |                                          |                                                                                                                                                                                                                                                                                                 | <p>is not required for the dwelling.</p> <p><del>Note: The whole house fan requirements are not applicable to multifamily buildings, thus CF2R-MCH-02 does</del></p> <p><del>are not applicable to multifamily dwellings.</del></p>                                                                                                     |
| CF2R-MCH-04-E      | Certificate of Installation              | Evaporative Coolers                                                                                                                                                                                                                                                                             | <p>Require one CF2R- MCH-04 for each evaporative cooling system installed in a dwelling unit when the cooling system type given on the CF2R-MCH-01 is one of the following types:</p> <ul style="list-style-type: none"> <li>*evaporative – direct,</li> <li>*evaporative – indirect</li> <li>*evaporative – indirect-direct</li> </ul> |
| CF2R/CF3R-MCH-20-H | Certificate of Installation/Verification | <p>Duct Leakage Measurement –</p> <ul style="list-style-type: none"> <li>– Completely New Duct System</li> <li>– <u>Low Leakage Ducts in Conditioned Space</u></li> <li>– <u>Low Leakage Air Handling Unit</u></li> <li>– <u>Complete Replacement</u></li> <li>– <u>Altered Duct</u></li> </ul> | <p>If the MCH-01 specifies a MCH-20 is required for a ducted indoor unit, then provide one MCH-20 for the ducted indoor unit.</p> <p>Each ducted indoor unit that requires verification must have its own MCH-20.</p>                                                                                                                   |

| Document Number                | Document Type                                       | Document Description                                                            | Single-Family Document Configuration Rules                                                                                                                                                                                                  |
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|                                |                                                     | <u>System and Sealing of All Accessible Leaks Using Smoke Test</u>              |                                                                                                                                                                                                                                             |
| <del>CF2R/CF3R-MCH-20b-H</del> | <del>Certificate of Installation/Verification</del> | <del>Duct Leakage Measurement—Low Leakage Ducts in Conditioned Space</del>      | <p><del>If the MCH-01 specifies a MCH-20 is required for a ducted indoor unit, then provide one MCH-20 for the ducted indoor unit.</del></p> <p><del>Each ducted indoor unit that requires verification must have its own MCH-20.</del></p> |
| <del>CF2R/CF3R-MCH-20c-H</del> | <del>Certificate of Installation/Verification</del> | <del>Duct Leakage Measurement—Low Leakage Air Handling Unit</del>               | <p><del>If the MCH-01 specifies a MCH-20 is required for a ducted indoor unit, then provide one MCH-20 for the ducted indoor unit.</del></p> <p><del>Each ducted indoor unit that requires verification must have its own MCH-20.</del></p> |
| <del>CF2R/CF3R-MCH-20d-H</del> | <del>Certificate of Installation/Verification</del> | <del>Duct Leakage Measurement—Complete Replacement or Altered Duct System</del> | <p><del>If the MCH-01 specifies a MCH-20 is required for a ducted indoor unit, then provide one MCH-20 for the ducted indoor unit.</del></p> <p><del>Each ducted indoor unit that requires verification must have its own MCH-</del></p>    |

| Document Number                | Document Type                                        | Document Description                                                                                                                                                                              | Single-Family Document Configuration Rules                                                                                                                                                                                                  |
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|                                |                                                      |                                                                                                                                                                                                   | <del>20.</del>                                                                                                                                                                                                                              |
| <del>CF2R/CF3R-MCH-20e-H</del> | <del>Certificate of Installation/ Verification</del> | <del>Duct Leakage Measurement — Sealing of All Accessible Leaks Using Smoke Test</del>                                                                                                            | <p><del>If the MCH-01 specifies a MCH-20 is required for a ducted indoor unit, then provide one MCH-20 for the ducted indoor unit.</del></p> <p><del>Each ducted indoor unit that requires verification must have its own MCH-20.</del></p> |
| CF2R/CF3R-MCH-21-H             | Certificate of Installation/ Verification            | Duct Location Verification                                                                                                                                                                        | <p>If the MCH-01 specifies a MCH-21 is required for a ducted indoor unit, then provide one CF2R-MCH-21 for the ducted indoor</p> <p>unit. Each ducted indoor unit that requires verification must have its own MCH- 21.</p>                 |
| CF2R/CF3R-MCH-22a-H            | Certificate of Installation/ Verification            | <p>Forced Air System Fan Efficacy (Watts/cfm)</p> <p>— Newly Installed Non-Zoned Systems or Zoned Multi-Speed Compressor Systems.</p> <p><del>Forced Air System Fan Efficacy (Watt/cfm)</del></p> | <p>If the MCH-01 specifies a MCH-22 is required for a ducted indoor unit, then provide one MCH-22 for each ducted indoor unit.</p> <p>Note: Each ducted indoor unit that requires verification must have its own MCH-22.</p>                |

| Document Number     | Document Type                            | Document Description                                                                                                                                                                                                                                                                                                                                                                                                                             | Single-Family Document Configuration Rules                                                                                                                                                                                   |
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|                     |                                          | <p><u>Newly Installed Zoned Single Speed Compressor System.</u></p> <p><del>Forced Air System Fan Efficacy (Watt/cfm) —</del></p> <p><del>n</del><u>Newly Installed Non-Zoned or Zoned Multi-Speed Compressor Systems with Central Fan Ventilation Cooling.</u></p> <p><del>Forced Air System Fan Efficacy (Watt/cfm) —</del></p> <p><del>— Newly Installed Zoned Single Speed Compressor Systems with Central Fan Ventilation Cooling</del></p> |                                                                                                                                                                                                                              |
| CF2R/CF3R-MCH-22b-H | Certificate of Installation/Verification | <p><del>Forced Air System Fan Efficacy (Watt/cfm) —</del></p> <p><del>Newly Installed Zoned Single Speed Compressor System</del></p>                                                                                                                                                                                                                                                                                                             | <p>If the MCH-01 specifies a MCH-22 is required for a ducted indoor unit, then provide one MCH-22 for each ducted indoor unit.</p> <p>Note: Each ducted indoor unit that requires verification must have its own MCH-22.</p> |
| CF2R/CF3R-MCH-22c-H | Certificate of Installation/Verification | <p><del>Forced Air System Fan Efficacy (Watt/cfm) —</del></p> <p><del>newly Installed</del></p>                                                                                                                                                                                                                                                                                                                                                  | <p>If the MCH-01 specifies a MCH-22 is required for a ducted indoor unit, then provide one MCH-22 for</p>                                                                                                                    |

| Document Number                | Document Type                                       | Document Description                                                                                                                                                                                                                                                                                                                                             | Single-Family Document Configuration Rules                                                                                                                                                                                                  |
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|                                |                                                     | <del>Non-Zoned or Zoned Multi-Speed Compressor Systems with Central Fan Ventilation Cooling</del>                                                                                                                                                                                                                                                                | <del>each ducted indoor unit.</del><br><br><del>Note: Each ducted indoor unit that requires verification must have its own MCH-22.</del>                                                                                                    |
| <del>CF2R/CF3R-MCH-22d-H</del> | <del>Certificate of Installation/Verification</del> | <del>Forced Air System Fan Efficacy (Watt/cfm) — Newly Installed Zoned Single-Speed Compressor Systems with Central Fan Ventilation Cooling</del>                                                                                                                                                                                                                | <del>If the MCH-01 specifies a MCH-22 is required for a ducted indoor unit, then provide one MCH-22 for each ducted indoor unit.</del><br><br><del>Note: Each ducted indoor unit that requires verification must have its own MCH-22.</del> |
| CF2R/CF3R-MCH-23a-H            | Certificate of Installation/Verification            | <p>Forced Air System Airflow Rate (cfm/ton)</p> <ul style="list-style-type: none"> <li>– Newly Installed non- Zoned Systems or Zoned Multi-Speed Compressor Systems,</li> <li>– <u>Newly Installed Zoned Single Speed Compressor Systems,</u></li> <li>– <u>Alternative to Compliance with Minimum System Airflow Requirement for Altered Systems</u></li> </ul> | <p>If the MCH-01 specifies a MCH-23 is required for a ducted indoor unit, then provide one MCH-23 for each ducted indoor unit.</p> <p>Note: Each ducted indoor unit that requires verification must have its own MCH-23.</p>                |



| Document Number     | Document Type                             | Document Description                                                                                                                                                                                                                                                                                                                                       | Single-Family Document Configuration Rules                                                                                                                                                                                   |
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|                     |                                           | <p><u>(Best That I Can Do),</u></p> <p><u>– Newly Installed Heating Only Non-Zoned Systems (Measurement Only),</u></p> <p><u>– Newly Installed Non- Zoned Systems or Zoned Multi-Speed Compressor with Central Fan Ventilation Cooling,</u></p> <p><u>– Newly Installed Zoned Single Speed Compressor Systems with Central Fan Ventilation Cooling</u></p> |                                                                                                                                                                                                                              |
| CF2R/CF3R MCH 23b H | Certificate of Installation/ Verification | Forced Air System Airflow Rate (cfm/ton) — Newly Installed Zoned Single Speed Compressor Systems                                                                                                                                                                                                                                                           | <p>If the MCH 01 specifies a MCH 23 is required for a ducted indoor unit, then provide one MCH 23 for each ducted indoor unit.</p> <p>Note: Each ducted indoor unit that requires verification must have its own MCH 23.</p> |
| CF2R/CF3R MCH 23c H | Certificate of Installation/ Verification | Forced Air System Airflow Rate (cfm/ton) — Alternative to Compliance with                                                                                                                                                                                                                                                                                  | If the MCH 01 specifies a MCH 23 is required for a ducted indoor unit, then provide one MCH 23 for each ducted                                                                                                               |

| Document Number                | Document Type                                       | Document Description                                                                                                                                         | Single-Family Document Configuration Rules                                                                                                                                                                                                  |
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|                                |                                                     | <del>Minimum System Airflow Requirement for Altered Systems (Best That I Can Do)</del>                                                                       | <del>indoor unit.</del><br><br><del>Note: Each ducted indoor unit that requires verification must have its own MCH-23.</del>                                                                                                                |
| <del>CF2R/CF3R-MCH-23d-H</del> | <del>Certificate of Installation/Verification</del> | <del>Forced Air System Airflow Rate (cfm/ton) — Newly Installed Heating Only Non-Zoned Systems (Measurement Only)</del>                                      | <del>If the MCH-01 specifies a MCH-23 is required for a ducted indoor unit, then provide one MCH-23 for each ducted indoor unit.</del><br><br><del>Note: Each ducted indoor unit that requires verification must have its own MCH-23.</del> |
| <del>CF2R/CF3R-MCH-23e-H</del> | <del>Certificate of Installation/Verification</del> | <del>Forced Air System Airflow Rate (cfm/ton) — Newly Installed Non-Zoned Systems or Zoned Multi-Speed Compressor with Central Fan Ventilation Cooling</del> | <del>If the MCH-01 specifies a MCH-23 is required for a ducted indoor unit, then provide one MCH-23 for each ducted indoor unit.</del><br><br><del>Note: Each ducted indoor unit that requires verification must have its own MCH-23.</del> |
| <del>CF2R/CF3R-MCH-23f-H</del> | <del>Certificate of Installation/Verification</del> | <del>Forced Air System Airflow Rate (cfm/ton) — Newly Installed Zoned Single-Speed Compressor Systems with Central Fan</del>                                 | <del>If the MCH-01 specifies a MCH-23 is required for a ducted indoor unit, then provide one MCH-23 for each ducted indoor unit.</del><br><br><del>Note: Each ducted indoor unit that requires</del>                                        |

| Document Number                | Document Type                                        | Document Description                                                                                                                                                                                                                                                                                                                    | Single-Family Document Configuration Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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|                                |                                                      | <del>Ventilation-Cooling</del>                                                                                                                                                                                                                                                                                                          | <del>verification must have its own MCH-23.</del>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| CF2R/CF3R-MCH-24a-H            | Certificate of Installation/ Verification            | <u>Enclosure Air Leakage Diagnostic Test Worksheet</u><br><u>– Single Point Air Tightness Test with Manual Meter or Automatic Meter,</u><br><u>– Multipoint Airtightness Test</u><br><del>Enclosure Air Leakage Worksheet – Single Point Air Tightness Test with Manual Meter or Automatic Meter,</del><br><del>– Multipoint Test</del> | <del>If the building type is multifamily, and the IAQ ventilation system type for the dwelling units in the building as specified on the CF1R-PRF-01 or CF1R-NCB-01 are one of the following types:</del><br><del>*Supply,</del><br><del>*Exhaust,</del><br><del>*Central Ventilation System – Supply, or</del><br><del>*Central Ventilation System – Exhaust,</del><br><del>then require one MCH-24 for each such dwelling unit in the building.</del><br><br><del>Else,</del> MCH-24 doc only used if called for by MCH-27 docs.<br><br><u>Require registration of the MCH-24 before registering the related MCH-27.</u> |
| <del>CF2R/CF3R-MCH-24b-H</del> | <del>Certificate of Installation/ Verification</del> | <del>Enclosure Air Leakage Worksheet – Single Point Air Tightness Test with Automatic Meter</del>                                                                                                                                                                                                                                       | <del>If the building type is multifamily, and the IAQ ventilation system type for the dwelling units in the building as specified on the CF1R-PRF-01 or CF1R-NCB-01 are one of the following types:</del><br><del>*Supply,</del>                                                                                                                                                                                                                                                                                                                                                                                           |

| Document Number     | Document Type                            | Document Description                                                                                                                                                                                                                                                                                                                                                                                      | Single-Family Document Configuration Rules                                                                                                                                                                                                                                            |
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|                     |                                          |                                                                                                                                                                                                                                                                                                                                                                                                           | <p><del>*Exhaust,</del><br/> <del>*Central Ventilation System — Supply, or</del><br/> <del>*Central Ventilation System — Exhaust,</del><br/> then require one MCH-24 for each such dwelling unit in the building.</p> <p>Else, MCH-24 doc only used if called for by MCH-27 docs.</p> |
| CF2R/CF3R-MCH-25a-H | Certificate of Installation/Verification | Refrigerant Charge Verification <ul style="list-style-type: none"> <li>– Superheat Method (Standard Charge Procedure),</li> <li>– <u>Sub-cooling Method (Standard Charge Procedure),</u></li> <li>– <u>Weigh-in Charging Procedure,</u></li> <li>– <u>Packaged System Manufacturer Refrigerant Charge Certification</u> <u>(CF2R-only for the <del>Y</del> (variant) does not require CF3R</u></li> </ul> | If the MCH-01 specifies a MCH-25 is required, then provide one MCH-25 for each space conditioning system.                                                                                                                                                                             |

| Document Number                | Document Type                                        | Document Description                                                                                                                      | Single-Family Document Configuration Rules                                                                           |
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|                                |                                                      | <u>verification)</u>                                                                                                                      |                                                                                                                      |
| <del>CF2R/CF3R-MCH-25b-H</del> | <del>Certificate of Installation/ Verification</del> | <del>Refrigerant Charge Verification— Sub-cooling Method (Standard Charge Procedure)</del>                                                | <del>If the MCH-01 specifies a MCH-25 is required, then provide one MCH-25 for each space-conditioning system.</del> |
| <del>CF2R/CF3R-MCH-25c-H</del> | <del>Certificate of Installation/ Verification</del> | <del>Refrigerant Charge Verification— Weigh-in-Charging Procedure</del>                                                                   | <del>If the MCH-01 specifies a MCH-25 is required, then provide one MCH-25 for each space-conditioning system.</del> |
| <del>CF3R-MCH-25d-H</del>      | <del>Certificate of Installation/ Verification</del> | <del>Refrigerant Charge Verification— Fault Indicator Display (FID) (embedded in the CF2R-MCH-25a,b,c and standalone CF3R-MCH-25d)</del>  | <del>If the MCH-01 specifies a MCH-25 is required, then provide one MCH-25 for each space-conditioning system.</del> |
| <del>CF2R-MCH-25f-E</del>      | <del>Certificate of Installation</del>               | <del>Refrigerant Charge Verification— Packaged System Manufacturer Refrigerant Charge Certification (CF2R only for the 'f' variant)</del> | <del>If the MCH-01 specifies a MCH-25 is required, then provide one MCH-25 for each space-conditioning system.</del> |

| Document Number     | Document Type                             | Document Description                                                                                                                                                                                                                   | Single-Family Document Configuration Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| CF2R/CF3R-MCH-26-H  | Certificate of Installation/ Verification | Rated Space Conditioning System Equipment Verification                                                                                                                                                                                 | When MCH-26 is required by the MCH-01, provide one MCH-26 for each space conditioning system according to the MCH-01.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| CF2R/CF3R-MCH-27a-H | Certificate of Installation/ Verification | <p>IAQ and Mechanical Ventilation</p> <p>– Single Family Attached/Detached Ventilation – HRV or ERV Fan Efficacy Performance Rating (W/CFM).</p> <p>– <u>Non- Dwelling Unit HRV or ERV Fan Efficacy Performance Rating (W/CFM)</u></p> | <p>following:</p> <p>*ADU Addition <math>\leq</math> 300 ft<sup>2</sup>,</p> <p>*ADU Addition <math>&gt;</math> 300 to <math>\leq</math> 400 ft<sup>2</sup>,</p> <p>*ADU Addition <math>&gt;</math> 400 to <math>\leq</math> 700 ft<sup>2</sup>, or</p> <p>*ADU Addition <math>&gt;</math> 700 to <math>\leq</math> 1,000 ft<sup>2</sup>, then require one MCH-27 for each dwelling unit.</p> <p>If the CF1R type is a CF1R-PRF-01 and indicates requirement for IAQ <u>HERSfield</u> Verification, then require one MCH- 27 for each dwelling unit.</p> <p>Else, the MCH-27 is not required for the dwelling.</p> <p>Note: Non-dwelling units are exempt from the indoor air quality ventilation requirements and shall use the MCH--27d to document.</p> <p><del>*ADU Addition <math>&gt;</math> 400 to <math>\leq</math> 700 ft<sup>2</sup>, or</del></p> <p><del>*ADU Addition <math>&gt;</math> 700 to <math>\leq</math> 1,000 ft<sup>2</sup>, then require</del></p> |



| Document Number    | Document Type                            | Document Description                                                                                           | Single-Family Document Configuration Rules                                                                                                                                                                                                                                                                                                                                                                                           |
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|                    |                                          |                                                                                                                | <p><del>one MCH-27 for each dwelling unit.</del></p> <p><del>If the CF1R type is a CF1R-PRF-01 and indicates requirement for IAQ-HERS Verification, then require one MCH-27 for each dwelling unit.</del></p> <p><del>Else, the MCH-27 is not required for the dwelling.</del></p> <p><del>Note: Non-dwelling units are exempt from the indoor air quality-ventilation requirements and shall use the MCH-27d to document.</del></p> |
| CF2R-MCH-27d-E     | Certificate of Installation              | <del>IAQ and Mechanical Ventilation—Non-Dwelling Unit—HRV or ERV Fan Efficacy Performance Rating (W/CFM)</del> | <p><del>If the CF1R type is a CF1R-NCB-01, then require one MCH-27 for each dwelling unit.</del></p> <p><del>If the CF1R type is a CF1R-ADD-01, and the project scope is one of the following:</del></p> <p><del>*ADU Addition <math>\leq</math> 300 ft<sup>2</sup>,</del></p> <p><del>*ADU Addition <math>&gt;</math> 300 to <math>\leq</math> 400 ft<sup>2</sup>,</del></p>                                                        |
| CF2R/CF3R-MCH-28-H | Certificate of Installation/Verification | Return Duct and Filter Grille Design According to Tables 150.0-B or C                                          | When MCH-28 is required by MCH-01, provide one MCH-28 for each space conditioning system according to the                                                                                                                                                                                                                                                                                                                            |

| Document Number                | Document Type                                        | Document Description                                                                                                                                                               | Single-Family Document Configuration Rules                                                                                                                    |
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|                                |                                                      |                                                                                                                                                                                    | MCH-01.                                                                                                                                                       |
| CF2R/CF3R-MCH-29-H             | Certificate of Installation/ Verification            | Supply Duct Surface Area and R-value; Buried Ducts; Deeply Buried Ducts                                                                                                            | When MCH-29 is required by MCH-01, provide one MCH-29 for each space conditioning system according to the MCH-01.                                             |
| CF2R/CF3R-MCH-30-H             | Certificate of Installation/ Verification            | Central Fan Ventilation Cooling Systems (CFVCS)                                                                                                                                    | When the CF1R- PRF-01 indicates Central Fan Ventilation Cooling System (VCS) was used, require one MCH-30 for each Central Fan VCS installed in the dwelling. |
| CF2R/CF3R-MCH-31a-H            | Certificate of Installation/ Verification            | Whole House Fan Airflow and Fan Efficacy<br>– Individual CFM and Watts Collection<br>– <u>Whole House Fan Airflow and Fan Efficacy – Individual CFM and Total Watts Collection</u> | When the CF1R- PRF-01 indicates compliance credit for <u>HERS-field</u> Cool Vent Verification, then require one MCH- 31 for each dwelling unit.              |
| <del>CF2R/CF3R-MCH-31b-H</del> | <del>Certificate of Installation/ Verification</del> | <del>Whole House Fan Airflow and Fan Efficacy – Individual CFM and Total Watts Collection</del>                                                                                    | <del>When the CF1R- PRF-01 indicates compliance credit for <u>HERS</u> Cool Vent Verification, then require one MCH- 31 for each dwelling unit.</del>         |
| CF2R/CF3R-MCH-32-H             | Certificate of Installation/ Verification            | Local Mechanical Exhaust – Kitchen Range Hood or                                                                                                                                   | When the CF1R indicates requirement for Kitchen Ventilation <u>HERS-field</u>                                                                                 |

| Document Number    | Document Type                             | Document Description                                                                         | Single-Family Document Configuration Rules                                                                                                                                                                                                                                                                                                             |
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|                    |                                           | Other Vented Fan – Airflow Rate or Capture Efficiency                                        | <p>verification for the dwelling, require one MCH- 32 per kitchen.</p> <p>Note: CBECC currently does not have the ability to model kitchens therefore <del>HERS-ECC</del> registry should query the user as to how many kitchens are contained within the dwelling. Value must be greater than or equal to 1.</p>                                      |
| CF2R/CF3R-MCH-33-H | Certificate of Installation/ Verification | VCHP Compliance Credit                                                                       | When the MCH-33 is required by the MCH-01, provide one MCH-33 for each VCHP system installed.                                                                                                                                                                                                                                                          |
| CF2R-MCH-34-E      | Certificate of Installation               | Pre-Cooling Compliance Credit                                                                | When the CF1R-PRF indicates compliance credit for Pre-Cooling, require one MCH-34 per dwelling unit.                                                                                                                                                                                                                                                   |
| CF2R-PLB-02-E      | Certificate of Installation               | Single Dwelling Unit Hot Water Distribution and Heat Pump Water Heater <del>(Non-HERS)</del> | <p><del>If the building type is single-family, r</del>Require one CF2R- PLB-02 for the dwelling unit if the dwelling unit contains <del>installed a</del> water heating system <del>with a non-HERS-verified</del> distribution <del>type</del>, and the installed heat pump water heater.</p> <p>Else, if the dwelling unit does not have a water</p> |

| Document Number    | Document Type                            | Document Description                                           | Single-Family Document Configuration Rules                                                                                                                                                                                                                                                                                                                                                                                 |
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|                    |                                          |                                                                | heating system, then a CF2R-PLB-02 is not required.                                                                                                                                                                                                                                                                                                                                                                        |
| CF2R-PLB-03-E      | Certificate of Installation              | Pool and Spa Systems                                           | <p>Prompt the user to declare whether the scope of the project includes installation of a new pool or spa, or installation of a replacement pool or spa component.</p> <p>If the response is yes, then require one CF2R-PLB-03.</p> <p>Else, if the response is no, then the CF2R-PLB-03 is not required.</p>                                                                                                              |
| CF2R/CF3R-PLB-22-H | Certificate of Installation/Verification | Single Dwelling Unit Hot Water System Distribution <b>HERS</b> | <p><del>If the building type is single family,</del> Require one PLB-22 for the dwelling unit if the dwelling unit contains one or more water heating systems with a <b>HERS</b> verified distribution type, drain water heat recovery, or expanded-credit compact design, and the installed heat pump water heater.</p> <p>Else, if the dwelling does not have a water heating system, then a PLB-22 is not required.</p> |

| Document Number | Document Type               | Document Description      | Single-Family Document Configuration Rules                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| CF2R-PVB-01-E   | Certificate of Installation | Photovoltaic (PV) Systems | <p>If the CF1R project scope is Newly Constructed Building, only if a PV system is in the proposed design or CF1R indicates a PV Exceptions require one PVB-01 per building.</p> <p><u>If the CF2R-PVB-01-E values for "Qualifying Exceptions" does not match certain exceptions reported by the CF1R-PRF-01, do not proceed with registration. The exceptions include Battery Storage and Community Solar. Other non-default exceptions should be flagged and noted on the PSR.</u></p> |
| CF2R-PVB-02-E   | Certificate of Installation | Battery Storage Systems   | <p>If the CF1R-PRF-01 indicates compliance credit for "Battery Storage System," require one PVB-02 per building.</p> <p><del>Otherwise, if the CF2R-PVB-01-E indicates exceptions does not state "Battery Storage," require one PVB-02 per building do not proceed with registration.</del></p>                                                                                                                                                                                          |

| Document Number | Document Type               | Document Description              | Single-Family Document Configuration Rules                                                                                                        |
|-----------------|-----------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| CF2R-SRA-01-E   | Certificate of Installation | Solar Ready Buildings             | When the CF2R-PVB-01 indicates exception "No PV – limited solar access," require one SRA-01 per building.                                         |
| CF2R-SRA-02-E   | Certificate of Installation | Minimum Solar Zone Area Worksheet | When CF2R-SRA-02 is required by CF2R-SRA-01, require one CF2R-SRA-02 per building.                                                                |
| CF2R-STH-01-E   | Certificate of Installation | Solar Water Heating Systems       | When the CF1R indicates use of a solar water system, for compliance, require one CF2R-STH-01 for each solar DHW system installed in the building. |

Source: California Energy Commission



Table 6-3: Document Configuration Rules Applicable to 2025<sup>2</sup> Low-Rise Multifamily Compliance Documents and HERS Features

| Document Number | Document Type             | Document Description                                                  | Low-Rise Multifamily Document Configuration Rules                                                                                                                                                             |
|-----------------|---------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LMCC-ELC-01-E   | Certificate of Compliance | Low Rise Multifamily Electrical Power Distribution and Electric Ready | <p>If the project includes electrical power distribution, then require completion of the LMCC- ELC.</p> <p>If the LMCC was completed through an EDDS, then require an upload of the XML.</p>                  |
| LMCC-ENV-01-E   | Certificate of Compliance | Low Rise Multifamily Envelope                                         | <p>If the project includes prescriptive compliance for envelope components, then require completion of the LMCC-ENV.</p> <p>If the LMCC was completed through an EDDS, then require an upload of the XML.</p> |
| LMCC-LTI-01-E   | Certificate of Compliance | Low Rise Multifamily Indoor Lighting                                  | <p>If the project includes prescriptive compliance for indoor lighting, then require completion of the LMCC-LTI.</p> <p>If the LMCC was completed through an EDDS, then require an upload of the XML.</p>     |
| LMCC-LTO-01-E   | Certificate of Compliance | Low Rise Multifamily Outdoor Lighting                                 | <p>If the project includes outdoor lighting, then require completion of the LMCC-LTO. If the LMCC was completed</p>                                                                                           |

| Document Number | Document Type             | Document Description                    | Low-Rise Multifamily Document Configuration Rules                                                                                                                                                                          |
|-----------------|---------------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 |                           |                                         | through an EDDS, then require an upload of the XML.                                                                                                                                                                        |
| LMCC-LTS-01-E   | Certificate of Compliance | Low Rise Multifamily Sign Lighting      | <p>If the project includes sign lighting, then require completion of the LMCC-LTS.</p> <p>If the LMCC was completed through an EDDS, then require an upload of the XML.</p>                                                |
| LMCC-MCH-01-E   | Certificate of Compliance | Low Rise Multifamily Mechanical Systems | <p>If the project includes prescriptive compliance for mechanical systems, then require completion of the LMCC-MCH.</p> <p>If the LMCC was completed through an EDDS, then require an upload of the XML.</p>               |
| LMCC-MCH-02-E   | Certificate of Compliance | Low Rise Multifamily Mechanical Systems | <p>If the project includes prescriptive Alteration compliance for mechanical systems, then require completion of the LMCC-MCH-02.</p> <p>If the LMCC was completed through an EDDS, then require an upload of the XML.</p> |
| LMCC-PLB-01-E   | Certificate of Compliance | Low Rise Multifamily Water Heating      | If the project includes prescriptive compliance for water heating, then require completion of the LMCC-PLB.                                                                                                                |

| Document Number    | Document Type                            | Document Description                                                                                 | Low-Rise Multifamily Document Configuration Rules                                                                                                                                                    |
|--------------------|------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    |                                          |                                                                                                      | If the LMCC was completed through an EDDS, then require an upload of the XML.                                                                                                                        |
| LMCC-PRC-01-E      | Certificate of Compliance                | Low Rise Multifamily Process Systems                                                                 | <p>If the project includes covered processes, then require completion of the LMCC-PRC.</p> <p>If the LMCC was completed through an EDDS, then require an upload of the XML.</p>                      |
| LMCC-SAB-01-E      | Certificate of Compliance                | Low Rise Multifamily Solar Ready, Photovoltaic, and Battery Storage                                  | <p>If the project includes solar ready, PV, and/or batter storage, then require completion of the LMCC-SAB.</p> <p>If the LMCC was completed through an EDDS, then require an upload of the XML.</p> |
| LMCC-CXR-01-E      | Certificate of Compliance                | Nonresidential Building Commissioning                                                                | If the project is low-rise mixed-use and therefore contains nonresidential spaces, then require completion of LMCC-CXR.                                                                              |
| LMCI-ELC-01-E      | Certificate of Installation              | Electric Ready                                                                                       | If the LMCC project scope is New Construction, then require one LMCI-ELC-01 per building.                                                                                                            |
| LMCI/LMCV-ENV-21-H | Certificate of Installation/Verification | <del>HERS-ECC</del> Quality Insulation Installation (QII) – Air Infiltration Sealing (Framing Stage) | When the LMCC indicates requirement for QII <del>HERS-ECC</del> verification for the dwelling, compliance shall be demonstrated                                                                      |

| Document Number    | Document Type                             | Document Description                                                                                         | Low-Rise Multifamily Document Configuration Rules                                                                                                                                                                                                                                             |
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|                    |                                           |                                                                                                              | <p>using all the QII <del>HERS</del><u>field</u> verification compliance documents (ENV-21 &amp; ENV-22).</p> <p>Require one ENV-21 per building.</p>                                                                                                                                         |
| LMCI/LMCV-ENV-22-H | Certificate of Installation/ Verification | <del>HERS</del> <u>ECC</u> Quality Insulation Installation (QII) – Insulation Installation                   | <p>When the LMCC indicates requirement for QII <del>HERS</del><u>field</u> verification for the dwelling, compliance shall be demonstrated using all the QII <del>HERS</del><u>field</u> verification compliance documents (ENV-21 &amp; ENV-22).</p> <p>Require one ENV-22 per building.</p> |
| LMCV-EXC-20-H      | Certificate of Verification               | <del>HERS</del> <u>field</u> Verification for Existing Conditions for Performance Compliance for Alterations | <p>When credit for existing conditions is used on the LMCC, as a condition for LMCC registration, an EXC-20 that verifies the existing condition claimed on the LMCC shall first be registered.</p> <p>Required as a prerequisite to registration of a LMCC-PRF for an alteration.</p>        |
| LMCI-MCH-01a-E     | Certificate of Installation               | Space Conditioning Systems, Ducts and Fans – for Performance                                                 | If the LMCC-PRF is completed, and the project scope is a Newly Constructed Building, then require one LMCI-                                                                                                                                                                                   |

| Document Number                  | Document Type                             | Document Description                                                                                                                             | Low-Rise Multifamily Document Configuration Rules                                                                                          |
|----------------------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
|                                  |                                           | Compliance for Newly Constructed Buildings                                                                                                       | MCH- 01a for the building.                                                                                                                 |
| LMCI-MCH-01b-E                   | Certificate of Installation               | Space Conditioning Systems, Ducts and Fans – for Prescriptive Alterations                                                                        | If the LMCC-MCH is completed, and the project scope is an Alteration, then require one LMCI-MCH-01b for the building.                      |
| LMCI-MCH-01c-E                   | Certificate of Installation               | Space Conditioning Systems, Ducts and Fans – for Prescriptive Newly Constructed Buildings                                                        | If the LMCC-MCH-01-E is completed, and the project scope is a Newly Constructed Building, then require one LMCI-MCH- 01c for the building. |
| LMCI-MCH-01d-E                   | Certificate of Installation               | Space Conditioning Systems, Ducts and Fans – for Performance Compliance for Additions and Alterations, or Addition Alone                         | If the LMCC-PRF is completed, and the project scope is an Alteration, then require one LMCI-MCH-01d for the building.                      |
| LMCI/LMCV-MCH-20 <sup>a</sup> -H | Certificate of Installation/ Verification | <p><del>HERS</del> Duct Leakage Measurement</p> <p>– Completely New Duct System<sub>✓</sub></p> <p><u>– Low Leakage Ducts in Conditioned</u></p> | If the LMCI-MCH-01 specifies a LMCI-MCH-20 is required, then provide one MCH-20 per ducted unit.                                           |

| Document Number                | Document Type                                       | Document Description                                                                                                                                                                      | Low-Rise Multifamily Document Configuration Rules |
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|                                |                                                     | <u>Space,</u><br><br><u>– Low Leakage Air Handling Unit,</u><br><u>– Complete Replacement or Altered Duct System, or</u><br><br><u>– Sealing of All Accessible Leaks Using Smoke Test</u> |                                                   |
| <del>LMCI/LMCV MCH 20b-H</del> | <del>Certificate of Installation/Verification</del> | <del>HERS Duct Leakage Measurement — Low Leakage Ducts in Conditioned Space</del>                                                                                                         | <del>See rules for the MCH—20a.</del>             |
| <del>LMCI/LMCV MCH 20c-H</del> | <del>Certificate of Installation/Verification</del> | <del>HERS Duct Leakage Measurement — Low Leakage Air Handling Unit</del>                                                                                                                  | <del>See rules for the MCH—20a.</del>             |
| <del>LMCI/LMCV MCH 20d-H</del> | <del>Certificate of Installation/Verification</del> | <del>HERS Duct Leakage Measurement — Complete Replacement or Altered Duct System</del>                                                                                                    | <del>See rules for the MCH—20a.</del>             |
| <del>LMCI/LMCV MCH 20e-H</del> | <del>Certificate of Installation/Verification</del> | <del>HERS Duct Leakage Measurement — Sealing of All Accessible Leaks Using Smoke</del>                                                                                                    | <del>See rules for the MCH—20a.</del>             |



| Document Number                  | Document Type                                        | Document Description                                                                                                                                                                                                                                | Low-Rise Multifamily Document Configuration Rules                                                |
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|                                  |                                                      | <del>Test</del>                                                                                                                                                                                                                                     |                                                                                                  |
| LMCI/LMCV-MCH-21-H               | Certificate of Installation/ Verification            | <del>HERS</del> -Duct Location                                                                                                                                                                                                                      | If the LMCI-MCH-01 specifies a LMCI-MCH-21 is required, then provide one MCH-21 per ducted unit. |
| LMCI/LMCV-MCH-22 <del>a</del> -H | Certificate of Installation/ Verification            | <del>HERS Fan-Efficiency</del><br><u>Forced Air System Fan Efficacy (Watts/cfm)</u><br><br><u>– Newly Installed Non-Zoned Systems or Zoned Multi-Speed Compressor Systems,</u><br><br><u>– Newly Installed Zoned Single Speed Compressor System</u> | If the LMCI-MCH-01 specifies a LMCI-MCH-22 is required, then provide one MCH-22 per ducted unit. |
| <del>LMCI/LMCV-MCH-22b-H</del>   | <del>Certificate of Installation/ Verification</del> | <del>HERS Fan-Efficiency</del>                                                                                                                                                                                                                      | <del>See rules for the MCH-22a.</del>                                                            |
| <del>LMCI/LMCV-MCH-22c-H</del>   | <del>Certificate of Installation/ Verification</del> | <del>HERS Fan-Efficiency</del>                                                                                                                                                                                                                      | <del>See rules for the MCH-22a.</del>                                                            |
| <del>LMCI/LMCV-MCH-22d-H</del>   | <del>Certificate of Installation/ Verification</del> | <del>HERS Fan-Efficiency</del>                                                                                                                                                                                                                      | <del>See rules for the MCH-22a.</del>                                                            |
| LMCI/LMCV-MCH-23 <del>a</del> -H | Certificate of Installation/ Verification            | <del>HERS</del> - <u>Forced Air System Airflow Rate (cfm/ton)</u><br><u>– Newly Installed non- Zoned</u>                                                                                                                                            | If the LMCI-MCH-01 specifies a LMCI-MCH-23 is required, then provide one MCH-23 per ducted unit. |

| Document Number                | Document Type                                       | Document Description                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Low-Rise Multifamily Document Configuration Rules |
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|                                |                                                     | <p><u>Systems or Zoned Multi-Speed Compressor Systems,</u></p> <p><u>– Newly Installed Zoned Single Speed Compressor Systems,</u></p> <p><u>– Alternative to Compliance with Minimum System Airflow Requirement for Altered Systems (Best That I Can Do),</u></p> <p><u>– Newly Installed Heating Only Non-Zoned Systems (Measurement Only),</u></p> <p><u>– Newly Installed Non- Zoned Systems or Zoned Multi-Speed Compressor with Central Fan Ventilation Cooling</u></p> |                                                   |
| <del>LMCI/LMCV MCH 23b-H</del> | <del>Certificate of Installation/Verification</del> | <del>HERS Airflow Rate</del>                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <del>See rules for the MCH-23a.</del>             |

| Document Number                | Document Type                                       | Document Description                                                                                                                                                              | Low-Rise Multifamily Document Configuration Rules                                                              |
|--------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <del>LMCI/LMCV-MCH-23c-H</del> | <del>Certificate of Installation/Verification</del> | <del>HERS Airflow Rate</del>                                                                                                                                                      | <del>See rules for the MCH-23a.</del>                                                                          |
| <del>LMCI/LMCV-MCH-23d-H</del> | <del>Certificate of Installation/Verification</del> | <del>HERS Airflow Rate</del>                                                                                                                                                      | <del>See rules for the MCH-23a.</del>                                                                          |
| <del>LMCI/LMCV-MCH-23f-H</del> | <del>Certificate of Installation/Verification</del> | <del>HERS Airflow Rate</del>                                                                                                                                                      | <del>See rules for the MCH-23a.</del>                                                                          |
| LMCI/LMCV-MCH-24a-H            | Certificate of Installation/Verification            | <del>HERS</del> Enclosure Air Leakage <u>Test Worksheet</u><br>– Single Point Air Tightness Test with Manual Meter <u>or</u> <u>Automatic Meter</u> ,<br>– <u>Multipoint Test</u> | If the LMCI-MCH-01 specifies a LMCI-MCH-24 is required, then provide one MCH-24 per dwelling unit.             |
| <del>LMCI/LMCV-MCH-24b-H</del> | <del>Certificate of Installation/Verification</del> | <del>HERS Enclosure Air Leakage Worksheet – Single Point Air Tightness Test with Automatic Meter</del>                                                                            | <del>See rules for the MCH-24a.</del>                                                                          |
| LMCI/LMCV-MCH-25a-H            | Certificate of Installation/Verification            | <del>HERS</del> <u>ECC</u> Refrigerant Charge Verification<br>– Superheat Method (Standard Charge Procedure),<br>– <u>Sub-cooling Method</u>                                      | If the LMCI-MCH-01 specifies a LMCI-MCH-25 is required, then provide one MCH-25 per space conditioning system. |

| Document Number                | Document Type                                        | Document Description                                                                                                                                                                                | Low-Rise Multifamily Document Configuration Rules |
|--------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
|                                |                                                      | <u>(Standard Charge Procedure),</u><br><br><u>– Weigh-in Charging Procedure,</u><br><br><u>– Packaged System Manufacturer Refrigerant Charge Certification (does not require CF3R verification)</u> |                                                   |
| <del>LMCI/LMCV MCH 25b H</del> | <del>Certificate of Installation/ Verification</del> | <del>HERS Refrigerant Charge Verification — Sub-cooling Method (Standard Charge Procedure)</del>                                                                                                    | <del>See rules for the MCH— 25a.</del>            |
| <del>LMCI/LMCV MCH 25c H</del> | <del>Certificate of Installation/ Verification</del> | <del>HERS Refrigerant Charge Verification — Weigh-in Charging Procedure</del>                                                                                                                       | <del>See rules for the MCH— 25a.</del>            |
| <del>LMCI/LMCV MCH 25d H</del> | <del>Certificate of Installation/ Verification</del> | <del>HERS Refrigerant Charge Verification — Fault Indicator Display (FID)</del>                                                                                                                     | <del>See rules for the MCH— 25a.</del>            |

| Document Number                  | Document Type                                       | Document Description                                                                                                                               | Low-Rise Multifamily Document Configuration Rules                                                                         |
|----------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|                                  |                                                     | <del>(embedded in the LMCI-MCH-25a,b,c and standalone LMCI-MCH-25d)</del>                                                                          |                                                                                                                           |
| <del>LMCI/LMCV-MCH-25e-H</del>   | <del>Certificate of Installation/Verification</del> | <del>HERS Refrigerant Charge Verification—Winter Setup</del>                                                                                       | <del>See rules for the MCH-25a.</del>                                                                                     |
| <del>LMCI-MCH-25f-E</del>        | <del>Certificate of Installation/Verification</del> | <del>Refrigerant Charge Verification—Packaged System Manufacturer Refrigerant Charge Certification (LMCI only for the 'f' variant)</del>           | <del>See rules for the MCH-25a. <u>No ECC verification is required.</u></del>                                             |
| LMCI/LMCV-MCH-26-H               | Certificate of Installation/Verification            | <del>HERS</del> Rated System Verification                                                                                                          | If the LMCI-MCH-01 specifies a LMCI-MCH-26 is required, then provide one MCH-26 per space conditioning system.            |
| LMCI/LMCV-MCH-27 <del>b</del> -H | Certificate of Installation/Verification            | <del>HERS</del> IAQ and Mechanical Ventilation Heat Recovery Ventilation (HRV) or Energy Recovery Ventilation (ERV) Rated Performance Verification | If the LMCC-MCH or LMCC-PRF specifies a LMCI-MCH-27 is required, then provide one MCH- 27 <del>b</del> per dwelling unit. |
| LMCI/LMCV-MCH-28-H               | Certificate of Installation/Verification            | <del>HERS</del> Return Duct and Filter Grille Design                                                                                               | If the LMCC-MCH-01 or LMCC-PRF specifies a LMCI-MCH-                                                                      |

| Document Number    | Document Type                             | Document Description                                                                            | Low-Rise Multifamily Document Configuration Rules                                                              |
|--------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
|                    |                                           |                                                                                                 | 28 is required, then provide one MCH-28 per space conditioning system.                                         |
| LMCI/LMCV-MCH-29-H | Certificate of Installation/ Verification | <del>HERS</del> Supply Duct Surface Area and R-value; Buried Ducts; Deeply Buried Ducts         | If the LMCC-MCH-01 specifies a LMCI-MCH-29 is required, then provide one MCH-29 per space conditioning system. |
| LMCI/LMCV-MCH-32-H | Certificate of Installation/ Verification | <del>HERS</del> Local Mechanical Exhaust                                                        | If the LMCC-MCH or LMCC-PRF specifies a LMCI-MCH-32 is required, then provide one MCH-32 per kitchen.          |
| LMCI/LMCV-MCH-33-H | Certificate of Installation/ Verification | <del>HERS</del> VCHP Compliance Credit                                                          | If the LMCC-PRF specifies a LMCI-MCH-33 is required, then provide one MCH-33 per VCHP system installed.        |
| LMCI-PLB-01-E      | Certificate of Installation               | Multifamily Central Hot Water System Distribution<br>(Non-HERS)                                 | If the LMCC-PRF is completed, then require one LMCI-PLB-01 for the building.                                   |
| LMCI-PLB-02-E      | Certificate of Installation               | Single Individual Dwelling Unit Hot Water Distribution and Heat Pump Water Heater<br>(Non-HERS) | If the LMCC-PRF is completed, then require one LMCI-PLB-02 per dwelling unit.                                  |
| LMCI/LMCV-PLB-21-H | Certificate of Installation/ Verification | <del>HERS</del> Central Hot Water System Distribution                                           | If the LMCC-PRF specifies a LMCI-PLB-21 is required, then provide one PLB-21 for the building.                 |



| Document Number    | Document Type                             | Document Description                                                   | Low-Rise Multifamily Document Configuration Rules                                               |
|--------------------|-------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| LMCI/LMCV-PLB-22-H | Certificate of Installation/ Verification | <del>HERS</del> Individual Dwelling Unit Hot Water System Distribution | If the LMCC-PRF specifies a LMCI-PLB-22 is required, then provide one PLB-22 per dwelling unit. |

Source: California Energy Commission

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Table 6-4: Document Configuration Rules Applicable to 2025<sup>2</sup> Nonresidential  
(Including High-Rise Multifamily) Compliance Documents and HERS Features

| Document Number           | Document Type                          | Document Description                                                                                                                                                                                                                                      | Nonresidential Document Configuration Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NRCV-MCH-04a-H            | Certificate of Verification            | <p>Nonresidential Duct Leakage Measurement – Completely New Duct System.</p> <p>– <u>Low Leakage Air Handling Unit.</u></p> <p>– <u>Complete Replacement or Altered Duct System.</u></p> <p>– <u>Sealing of All Accessible Leaks Using Smoke Test</u></p> | <p>One NRCV-MCH-04 shall be required for each duct system identified as requiring duct leakage testing on the Certificate of Compliance.</p> <p>Alternatively, an approved Acceptance Test Technician Certification Provider and certified Acceptance Test Technician may perform the duct leakage verification as provided in the Reference Appendices NA1.9. The results of the acceptance test may be used to satisfy the condition of compliance. Systems verified under this procedure are not eligible for sampling.</p> |
| <del>NRCV-MCH-04c-H</del> | <del>Certificate of Verification</del> | <del>Nonresidential Duct Leakage Measurement – Low Leakage Air Handling Unit</del>                                                                                                                                                                        | <del>Same as the rule for the NRCV-MCH-04a-H</del>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <del>NRCV-MCH-04d-H</del> | <del>Certificate of Verification</del> | <del>Nonresidential Duct Leakage Measurement – Complete Replacement or Altered Duct</del>                                                                                                                                                                 | <del>Same as the rule for the NRCV-MCH-04a-H</del>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Document Number           | Document Type                          | Document Description                                                                                                                                            | Nonresidential Document Configuration Rules                                                                                                                                 |
|---------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           |                                        | <del>System</del>                                                                                                                                               |                                                                                                                                                                             |
| <del>NRCV-MCH-04e-H</del> | <del>Certificate of Verification</del> | <del>Nonresidential Duct Leakage Measurement – Sealing of All Accessible Leaks Using Smoke Test</del>                                                           | <del>Same as the rule for the NRCV-MCH-04a-H</del>                                                                                                                          |
| NRCV-MCH-24a-H            | Certificate of Verification            | Enclosure Air leakage <u>Test</u><br><del>Worksheet –</del><br>Single Point Test with Manual Meter <u>or</u> <u>Automatic Meter</u><br><u>– Multipoint Test</u> | MCH-24 doc only used if called for by NRCV-MCH-27 docs.                                                                                                                     |
| <del>NRCV-MCH-24b-H</del> | <del>Certificate of Verification</del> | <del>Enclosure Air Leakage Worksheet – Single Point Test with Automatic Meter</del>                                                                             | <del>MCH-24 doc only used if called for by NRCV-MCH-27 docs.</del>                                                                                                          |
| NRCV-MCH-27b-H            | Certificate of Verification            | IAQ and Mechanical Ventilation – High- Rise Residential Multifamily Ventilation                                                                                 | When building type is high-rise residential, prompt the user to declare whether a MCH-27 is needed. <del>If needed,</del> <u>Require</u> one NRCV-MCH-27 per dwelling unit. |

| Document Number           | Document Type                          | Document Description                                                                                                   | Nonresidential Document Configuration Rules                                                                                                                                                            |
|---------------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <del>NRCV-MCH-27c-H</del> | <del>Certificate of Verification</del> | <del>IAQ and Mechanical Ventilation—High-Rise Residential Multifamily Ventilation—Scheduled or Real-Time Control</del> | <del>Same as the rule for the NRCV-MCH-27b-H</del>                                                                                                                                                     |
| NRCV-MCH-32-H             | Certificate of Verification            | Verification of vented range hood airflow rate or capture efficiency                                                   | If the building type specified on the NRCC is high-rise residential, and if the dwelling unit has a kitchen, then require one MCH-32 for the dwelling unit.                                            |
| NRCV-PLB-21-H             | Certificate of Verification            | Nonresidential High-Rise Multifamily Central Hot Water System Distribution ( <del>HERSECC</del> )                      | <u>If the NRCC-PRF specifies a NRCV-PLB-21 is required, then provide one PLB-21 for the building.</u> <del>When HERS verification is required by NRCC, provide one NRCV-PLB-21 for the building.</del> |
| NRCV-PLB-22-H             | Certificate of Verification            | Nonresidential High-Rise Individual Dwelling Unit Hot Water System Distribution                                        | <u>If the NRCC-PRF specifies a NRCV-PLB-21 is required, then provide one PLB-21 for the building.</u> <del>When HERS verification is required by NRCC provide one PLB-22 for each dwelling unit.</del> |

Source: California Energy Commission

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## 7 Group Sampling Rules for ~~HERS~~ECC Verification Compliance

### 7.1 Overview

Residential ~~HERS~~field Verification, Testing, and Documentation Procedures are given in ~~2022~~ 2025 Reference Residential Appendix sections RA2 and RA3.

For newly constructed buildings, at the builder's option, compliance with ~~HERS~~field verification and diagnostic testing (~~HERS~~verificationFV&DT) requirements may be demonstrated for a group of dwelling units by performing at least one ~~HERS~~ECC-Rater verification for every measure within a designated group of dwelling units in which the same measures requiring ~~HERS~~ECC verification have been installed in each dwelling unit in the group. If the builder elects to demonstrate compliance utilizing group sampling for single-family construction, all applicable procedures described in Reference Residential Appendix Sections RA2.6.2, RA2.6.3, and RA2.6.4 must be followed. If the builder elects to demonstrate compliance utilizing group sampling for ~~low~~high-rise multifamily construction (~~not~~ greater than three habitable stories), all applicable procedures described in Reference Nonresidential Appendix Sections NA1.6 must be followed.

For alterations to existing buildings, group sampling is no longer an option. Section RA2.8 was revised to require ECC-Rater verification of all measures under alterations. The group sampling rules tables in this section should reflect this update but is not always stated explicitly. ~~building owners or their agents may carry out the actions that are assigned to builders in Reference Residential Appendix Sections RA2.1 through RA2.6. Refer to Reference Residential Appendix Section RA2.8 for installer requirements and HERS procedures for alterations to existing buildings.~~

When a Third Party Quality Control Program is used, the sampling procedure utilized is limited to sampling of a "closed" group as described in Section RA2.6.3. However, the sample tested may be selected and field verified from within a group of up to thirty dwelling units.

Refer to Reference Residential Appendix Section RA2.7 for Third Party Quality Control Program requirements.

### 7.2 General Group Sampling Rules

Group sampling rules are specified in ~~2022~~ 2025 Reference Residential Appendix RA2. Relevant sections from ~~2022~~ RA2.6 are copied into Section 7.2.1 below for convenience. Additional guidance for administering group sampling processes and registering Certificate of Verification documentation is given in Section 7.2.2.

#### 7.2.1 Group Sampling Rules Specified in ~~2025~~2 Residential Appendix RA2



### 7.2.1.1 Designation of Groups (From RA2.6.3.1)

After the initial model field verification and diagnostic testing is completed as specified in RA2.6.2, the builder or the builder's authorized representative shall determine a sampling procedure to be used and shall designate the dwelling units to include in the group of dwellings that require **HERS-field** verification. The maximum number of dwelling units allowed in a sample group may range from five, to seven, to thirty as described in Sections RA2.6.3.3, RA2.6.3.4, and RA2.7 respectively.

If multiple measures requiring **HERS-field** verification are installed, each dwelling unit in a designated group shall have the same measures requiring **HERS-field** verification as the other dwelling units in the designated group. If some dwelling units have installed a different set of measures requiring **HERS-field** verification, those dwelling units shall be in a separate group.

If the dwelling units in a designated group have multiple measures that require **HERS-field** verification, sample testing for individual measures may be conducted in any of the dwelling units in the group - it is not required that all of the sample tests for all of the individual measures be completed in the same dwelling unit. Individual measures shall be allowed to be included in a group regardless of whether compliance requires one sample test, or if compliance requires more than one sample test (up to 100 percent sample test rate) be reported for such individual measures.

Dwelling units in a designated group shall all be located within the same enforcement agency jurisdiction and subdivision or multifamily housing development. ~~An exception to this rule is given in Section RA2.8 for sample groups applicable to alterations.~~

If dwelling units have central forced-air space conditioning equipment that introduces outside air into the conditioned space utilizing means that connect outside air ventilation ducts directly to the dwelling unit's central forced air duct system (Central Fan-Integrated Ventilation System or CFI Ventilation System), the CFI ventilation technology shall be considered a separate measure for **HERS-field** verification sampling purposes, and dwellings with CFI ventilation systems shall be placed in separate groups from other dwelling units that do not utilize CFI ventilation technology.

### 7.2.1.2 Group Status — "Open" Groups and "Closed" Groups (From RA2.6.3.2)

Registration of the first Certificate of Installation, for the first dwelling in a sample group shall be required to "open" a new group. The date of the responsible person's registration signature for the first Certificate of Installation for the group shall establish the start date for the group. Additional dwellings may be entered into the registry and included in an "open" group over a period of time subject to registration of the Certificate of Installation documents to the registry for each additional dwelling. However, the group shall not remain "open" to receive additional dwellings for a period longer than six months after the start date of the group. A group may be "closed" at any time after the group has been "opened" at the option of the builder or builder's authorized representative, thus the size of a "closed" group may range from a minimum of one dwelling to a maximum of seven dwellings. When a group becomes classified as "closed", no additional dwellings shall be added to the group.

**7.2.1.3 Sampling of a "Closed" Group of Up to Seven Dwellings (From RA2.6.3.3)**

The following criteria shall be met as prerequisite to attaining ~~HERS-field~~ verification compliance for the group:

- a. All of the dwelling units contained in the sample group have been identified. A maximum of seven dwellings are allowed to be included in a "closed" sample group for ~~HERS-ECC~~ compliance.
- b. Installation of all the measures that require ~~HERS-field~~ verification has been completed in all the dwellings that are entered in the group, and registration of the Certificates of Installation for all the dwellings entered in the group has been completed.
- c. The group has been classified as a "closed" group in the Provider Data Registry
- d. At the request of the builder or the builder's authorized representative, a ~~HERS-Rater~~~~ECC-Rater~~ shall randomly select one dwelling unit from the "closed" sample group for field verification and diagnostic testing. If the dwelling unit meets the compliance requirements, this "tested" dwelling and each of the other "not-tested" dwellings in the group shall receive a registered Certificate of Verification.

**7.2.1.4 Sampling of an "Open" Group of Up to Five Dwellings (from RA2.6.3.4)**

The following criteria shall be met as prerequisite to attaining ~~HERS-field~~ verification compliance for the group:

- a. At least one dwelling unit from the sample group has been identified. A maximum of five dwellings are allowed to be included in an "open" sample group for HERS compliance.
- b. Installation of all the measures that require ~~HERS-field~~ verification shall be completed in all the dwellings that are entered in the group, and registration of the Certificates of Installation for all the dwellings entered in the group has been completed.
- c. At the request of the builder or the builder's authorized representative, a ~~HERS-Rater~~~~ECC-Rater~~ shall randomly select one dwelling unit from those currently entered into the "open" sample group for field verification and diagnostic testing. If the dwelling unit meets the compliance requirements, the "tested" dwelling and each of the other "not tested" dwellings currently entered into the group shall receive a registered Certificate of Verification. If less than five dwelling units have been entered into the group, the group shall be allowed to remain "open" and eligible to receive additional dwelling units. Dwelling units entered into the "open" group subsequent to the compliant ~~HERS-field~~ verification of the "tested" dwelling shall also receive a registered Certificate of Verification as a "not tested" dwelling subject to receipt of the registered Certificate of Installation by the ~~HERS-Provider Data Registry~~~~data registry~~ for the dwelling. The group shall be "closed" when it reaches the limit of five dwellings or when the six-month limit for "open" groups has been exceeded, or when the builder requests that the group be closed.

### 7.2.1.5 Additional Requirements Applicable to Group Sampling Procedures (From RA2.6.3.5)

The builder or the ~~HERS-Rater~~ECC-Rater may request removal of untested dwelling units from a group by notifying the ~~HERS-ECC-Provider~~ prior to selection of the dwelling sample that will be tested from an ~~"open"~~ or ~~"closed"~~ group and shall provide justification for the change. Removed dwelling units shall be field verified and diagnostically tested individually or shall be included in a subsequent group for sampling.

There are exceptions to the requirement to have completed Certificate of Installation data entered into the ~~HERS-Provider Data Registry~~data registry prior to selection of the dwelling unit to be tested in a group. Some ~~HERS-ECC~~ measures require multiple verifications during the construction process. A sample group is not required to be closed before ~~HERS-ECC~~ field verification and diagnostic testing can begin for the following measures. For these measures, the ~~HERS-Rater~~ECC-Rater is allowed to randomly select the dwelling unit to be field verified from those that are at the proper stage of construction to enable the first of the multiple verifications to be completed.

- a. **Quality Installation of Insulation** measure requires inspection of the air barrier and inspection of the insulation behind tubs and showers at framing rough-in. Verification of the wall, floor, and ceiling insulation must be completed prior to drywall installation. Attic insulation installation may require follow-up verification.
- b. **Buried Ducts** measure requires verification of the duct design prior to verification of the attic insulation.
- c. **Duct Surface Area** requires verification of the duct design prior to installation of the attic insulation.

The ~~HERS-Rater~~ECC-Rater, with no direction from the installer or builder, shall randomly select one dwelling unit from a "closed" sample group for field verification and diagnostic testing upon receiving the builder's, or builder representative's, request for ~~HERS-field~~ verification of that group. Alternatively, the ~~HERS-Rater~~ECC-Rater shall randomly select one dwelling unit from the dwellings currently entered into an "open" sample group upon receiving the builder's, or builder representative's, request for ~~HERS-~~verification of that group. The ~~HERS-Rater~~ECC-Rater shall ~~diagnostically test, and field-~~verify the selected dwelling unit. The ~~HERS-Rater~~ECC-Rater shall enter ~~the the test and/or field verification-~~ results into the ~~HERS-Provider Data Registry~~data registry regardless of whether the results indicate a pass or fail. If the test fails, then the failure must be entered into the Provider's Data Registry even if the installer immediately corrects the problem. In addition, the procedures in Section RA2.6.4 shall be followed.

If field verification and diagnostic testing determines that the requirements for compliance are met, the ~~HERS-Rater~~ECC-Rater shall enter the test results into the ~~HERS-Provider Data Registry~~data registry. Whereupon the Provider shall make available to the ~~HERS-Rater~~ECC-Rater, the builder, the enforcement agency, and other approved users of the ~~HERS-Provider Data Registry~~data registry, a registered copy of the Certificate of Verification for the "tested" dwelling, and for all other "not tested" dwelling units entered in the group at the time of the

sample test. The registered Certificate of Verification shall report the successful diagnostic testing results and conclusions regarding compliance for the tested dwelling unit. The registered Certificate of Verification shall also provide:

1. Building permit number for the dwelling unit.
2. Registration Number that conforms to the numbering convention specified in Reference Joint Appendix JA7.
3. Group Number that conforms to the numbering convention specified in Reference Joint Appendix JA7.
4. Time and date stamp of the Provider's issuance of the registered Certificate of Verification.
5. Provider's logo, water mark, or official seal.
6. Indication that the dwelling was a "tested" dwelling or was a "not-tested" dwelling in a sample group.

Whenever the builder changes subcontractors who are responsible for a feature that is being diagnostically field verified and tested, the builder shall notify the ~~HERS-Rater~~ECC-Rater of the subcontractor change and terminate sampling for any affected groups. All dwelling units utilizing features that require ~~HERS~~-verification for compliance that were installed by previous subcontractors or were subject to verification and testing under the supervision of a previous ~~HERS~~ Provider or ECC-Provider, for which the builder does not have a completed Certificate of Verification, shall be individually tested or included in a separate group for sampling. Dwelling units with installations completed by new subcontractors shall be individually tested or shall be included in a new sampling group.

The ~~HERS-Rater~~ECC-Rater shall not notify the builder when sample testing will occur prior to the completion of the work that is to be tested, or prior to registration of the Certificate of Installation.

The data registry ~~HERS-Provider~~ shall "close" any "open" group within six months after the earliest signature date shown on any Certificate of Installation for a dwelling entered in the group. When such group closure occurs, the ~~HERS-Provider~~ ECC-Provider shall notify the builder that the group has been "closed", then require that a sample dwelling shall be selected for field verification and diagnostic testing by a ~~HERS-Rater~~ECC-Rater if field verification has not yet been conducted on a sample dwelling entered in the group.



## 7.2.2 Additional Guidance for Group Sampling Procedures and Documentation

### 7.2.2.1 Certificate of Verification Documentation for Not-Tested Measures in a Sample Group

When a measure complies with a [HERS-field](#) verification as one of the "not-tested" measures in a sample group, a Certificate of Verification document for that feature for that measure should be created that does not include actual verification results data, but only includes the following items:

- Certificate of Verification Header** for the applicable compliance document for the [HERS-field](#) verification protocol for the [HERSECC](#) feature that was verified in the "tested" measure by the [HERS-RaterECC-Rater](#).
- Certificate of Verification Footer for the applicable compliance document for the [HERS-field](#) verification protocol for the [HERSECC](#) feature that was verified in the "tested" measure by the [HERS-RaterECC-Rater](#).
- Certificate of Verification Signature block** for the applicable compliance document for the [HERS-field](#) verification protocol for the [HERS-field](#) feature that was verified in the "tested" measure by the [HERS-RaterECC-Rater](#).
- Water-mark** that indicates the measure passed as a "not-tested" measure in the sample group.

The process used by the Data Registry for generating the Certificate of Verification document for not-tested dwellings in a sample group should include the following:

- Use the same URI call to the RG that is used for the tested version of the CF3R compliance document for that feature in the sample group. ~~When the compliance document type is a variant series type, it is necessary to include the variant letter (e.g., the "a" in CF3RMCH20aH) used for the tested dwelling doc.~~ The RG uses the docToken (e.g., CF3RMCH20aH) to pick the document header info.
- Send XML to the RG that includes a value = "NotTested" in the signature block field named "SampleGroupTestStatus". The RG reads the value of `<comp:responsiblePerson5_SampleGroupTestStatus>NotTested</comp:responsiblePerson5_SampleGroupTestStatus>` and validates using the CF3RFeatureNotTested schema (CF3RNoTestH.xsd). There is no need to include in the XML, any data other than that needed for the specific project or dwelling unit name, location, enforcement agency, and permit information that should be displayed in the header of the completed compliance document. The remainder of the signature block and footer data is expected to be appended/overlaid after the PDF format document has been produced by the RG and transmitted to the Data Registry as described in Section 4.4, with the exception that the signatures provided for the "tested" dwelling may be automatically used for the "not-tested" dwelling(s) as well. Registration of not-tested dwelling documents may be performed automatically in conjunction with the registration of the tested dwelling document for the group. The documentation author

and responsible person are not required to provide additional signing actions for the not-tested dwelling unit documents. Additionally, the Data Registry should ensure that each dwelling document in the sample group is given a unique registration number.

#### **7.2.2.2 Group Sampling Rules Applicable to Specific ~~2022-2025~~ Compliance Document Types and HERS-Features**

Additional guidance for specific ~~2022-2025~~ compliance documentation for group sampling, and guidance for specific HERS-verification features for single-family group sampling is provided in Table 7-1. Guidance for group sampling in low-rise multifamily dwellings is provided in Table 7-2. Guidance for group sampling in nonresidential buildings is provided in Table 7-3.

Some documents have been excluded from the group sampling tables for brevity. Certificates of compliance do not represent testing or verification of energy features, and so are not applicable to group sampling. Certificates of installation ~~that are strictly non-HERSECC-~~ are verified by the enforcement agency are also not applicable.

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Table 7-1: Group Sampling Rules Applicable to ~~2022~~ 2025 Single-Family Compliance Documents ~~and HERSECC Features~~

| Document Number     | Document Type                             | Document Description                                                                                                                                | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CF2R/CF3R-ENV-20a-H | Certificate of Installation/ Verification | <p>Enclosure Air Leakage Single</p> <p>-Point Test with Manual Meter <u>or Automatic Meter,</u></p> <p><u>- or Multipoint Airtightness Test</u></p> | <p><b>Envelope Leakage Feature</b></p> <p>When the CF1R-PRF indicates requirement for Envelope Leakage <del>HERS</del> <u>field</u> verification for the dwelling, then compliance may be demonstrated using <del>either one of the applicable ENV-20 variants (a, b).</del></p> <p>The same "<u>Enclosure Air Leakage test</u>" <u>verification protocol</u> <del>variant</del> does not need to be used in all dwellings in a sample group, thus any combination of ENV-20 <u>verification</u> <del>variants</del> can be used to qualify to be in the same sample group for envelope leakage credit features.</p> |

| Document Number                | Document Type                                        | Document Description                                                    | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <del>CF2R/CF3R-ENV-20b-H</del> | <del>Certificate of Installation/ Verification</del> | <del>Enclosure Air Leakage Single-Point Test with Automatic Meter</del> | <p><b>Envelope Leakage Feature</b></p> <p>When the CF1R-PRF indicates requirement for Envelope Leakage HERS verification for the dwelling, then compliance may be demonstrated using either one of the applicable ENV-20 variants (a, b).</p> <p>The same variant does not need to be used in all dwellings in a sample group, thus any combination of ENV-20 variants can be used to qualify to be in the same sample group for envelope leakage credit features.</p> |

| Document Number    | Document Type                             | Document Description                                                                                                                                        | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| C2R/CF3R-ENV-21-H  | Certificate of Installation/ Verification | Quality Insulation Installation (QII) Air Infiltration Sealing<br><br>Framing Stage                                                                         | <b>QII Feature</b><br><br>When the CF1R indicates requirement for <u>HERS-field</u> verification of QII, then compliance shall be demonstrated using one <u>HERS-field</u> "Tested" ENV-21 in one of the dwellings in the group, and one <u>HERS-field</u> "Tested" ENV-22 either in the same dwelling or otherwise in another dwelling in same sample group.<br><br>Refer to group sampling procedures in RA2.6.3.5 that allow <u>HERS-field</u> verification "testing" of QII features prior to closing the sample group. |
| CF2R/CF3R-ENV-22-H | Certificate of Installation/ Verification | Quality Insulation Installation (QII) Insulation Stage                                                                                                      | See rules for the ENV-21.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| CF3R-EXC-20-H      | Certificate of Verification               | <u>HERS-field</u> <u>ECC</u> Verification for Existing Conditions for performance compliance for alterations. Required as prerequisite to registration of a | sampling n/a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Document Number     | Document Type                             | Document Description                                                                                                                                                                                                                                                                                                                                                                         | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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|                     |                                           | CF1R-PRF for an altered dwelling.                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| CF2R/CF3R-MCH-20a-H | Certificate of Installation/ Verification | <p>Duct Leakage Measurement</p> <ul style="list-style-type: none"> <li>- New System;</li> <li>- <u>Low Leakage Ducts in Conditioned Space (LLDCS) Compliance Credit;</u></li> <li>- <u>Low Leakage Air-Handling Units</u></li> <li>- <u>Altered (Existing) System (not applicable to sampling)</u></li> <li>- <u>Sealing of All Accessible Leaks (not applicable to sampling)</u></li> </ul> | <p><b>Duct Leakage Feature</b></p> <p>When the MCH-01 indicates a MCH-20 is required for <del>HERS</del> <u>field</u> verification of duct leakage for a ducted indoor unit in the dwelling, <del>qualification for</del> inclusion in a sample group <u>depends on the measurement protocol indicated on the certificate of compliance may be demonstrated using one of MCH-20 any one of the applicable variants (a, b, c, d, e).</u> The same <u>"Duct Leakage Measurement" verification protocol variant does not need must to be used on all indoor units on all systems in all dwellings in a sample group, thus any combination of MCH-20 variants can be used to qualify the dwelling to be in the same sample group. Dwellings containing multiple split systems that have both ducted and</u></p> |

| Document Number | Document Type | Document Description | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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|                 |               |                      | <p><del>ductless indoor units connected to a common outdoor unit may be included in the same sample group along with systems that are entirely ducted or entirely ductless; and any combination of applicable MCH-20 variants may be used to verify those ducted indoor units. Sampling is not allowed for altered systems.</del></p> <p><del>Additionally, if MCH-20, Section C requires use of a smoke test for verification of ducts in a garage space in the dwelling, then the systems in the dwelling that require duct leakage verification must be verified by a ECC-Rater; thus, that dwelling shall not be allowed to comply as a "not tested" dwelling in a sample group, and the dwelling shall not be allowed to be used to represent the "tested" dwelling for duct leakage compliance.</del></p> |

| Document Number                | Document Type                                       | Document Description                                                                          | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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|                                |                                                     |                                                                                               | for the sample group.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <del>CF2R/CF3R-MCH-20b-H</del> | <del>Certificate of Installation/Verification</del> | <del>Duct Leakage Measurement Low Leakage Ducts in Conditioned Space Compliance Credit;</del> | <del>See rules for the MCH-20a.</del>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <del>CF2R/CF3R-MCH-20c-H</del> | <del>Certificate of Installation/Verification</del> | <del>Duct Leakage Measurement Low Leakage Air Handling Units</del>                            | <del>See rules for the MCH-20a.</del>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <del>CF3R-MCH-20d-H</del>      | <del>Certificate of Installation/Verification</del> | <del>Duct Leakage Measurement Altered (Existing) System</del>                                 | <p><del>Sampling is not allowed for altered systems.</del></p> <p><del>See rules for the MCH-20a.</del></p> <p>Additionally, if MCH-20d Section C requires use of a smoke test for verification of ducts in a garage space in the dwelling, then the systems in the dwelling that require duct leakage verification must be verified by a HERS-raterECC-Rater; thus, that dwelling shall not be allowed to comply as a "not tested" dwelling in a sample group, and the dwelling shall not be</p> |



| Document Number     | Document Type                                | Document Description                                     | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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|                     |                                              |                                                          | <del>allowed to be used to represent the "tested" dwelling for duct leakage compliance for the sample group.</del>                                                                                                                                                                                                                                                                                                                                                                                                                 |
| CF2R/CF3R-MCH-20e-H | Certificate of Installation/<br>Verification | Duct Leakage-Measurement Sealing of All Accessible Leaks | <p><u>Sampling is not allowed for altered systems.</u></p> <p>See rules for the MCH-20a.</p> <p>Additionally, If MCH-20e (smoke test) is used for compliance in the dwelling, then the systems in the dwelling that require duct leakage verification must be verified by a HERS-rater; thus, that dwelling shall not be allowed to comply as a "not tested" dwelling in a sample group, and the dwelling shall not be allowed to be used to represent the "tested" dwelling for duct leakage compliance for the sample group.</p> |

| Document Number    | Document Type                             | Document Description       | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| CF2R/CF3R-MCH-21-H | Certificate of Installation/ Verification | Duct Location Verification | <p><b>Duct Location, Surface Area, and R-value Features</b></p> <p>When the MCH-01 indicates requirement for MCH-21 for <b>HERSFIELD</b> field verification for duct location in the dwelling, qualification for inclusion in a sample group may be demonstrated using MCH-21 regardless of which duct location verification protocol is reported on the MCH-21. The same duct location verification protocol does not need to be used in all dwellings in a sample group, thus any combination of MCH-21 verifications <u>given by RA3.1.4.1</u> can be used to qualify to be in the same sample group, <u>unless the verification is part of the LLDCS testing (RA3.1.4.3.8).</u></p> |

| Document Number                | Document Type                                        | Document Description                                                                                                                                                                                                                                                                                                                                                                                                             | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| CF2R/CF3R-MCH-22a-H            | Certificate of Installation/ Verification            | <p>Forced Air System Fan Efficacy (Watt/cfm)</p> <ul style="list-style-type: none"> <li>- Newly Installed Non-Zoned or Zoned Multi-Speed Compressor Systems</li> <li>- <u>Newly Installed Zoned Single Speed Compressor Systems Newly</u></li> <li>- <u>Installed Non-Zoned or Zoned Multi-Speed Compressor Systems with CFVCS</u></li> <li>- <u>Newly Installed Zoned Single Speed Compressor Systems with CFVCS</u></li> </ul> | <p><b>Fan Efficacy Feature</b></p> <p>When the MCH-01 indicates requirement for MCH-22 for <u>HERSFIELD</u> field verification for fan efficacy in the dwelling, qualification for inclusion in a sample group may be demonstrated using <u>any</u> one of <u>MCH-22</u> <del>the applicable variants (a, b, c, d)</del>, or by use of a MCH- 28 when indicated on the MCH-01. The same <u>"Forced Air System Fan Efficacy (Watt/cfm)" verification protocol variant</u> does not need to be used in all dwellings in a sample group, thus any combination of MCH-22 <del>variants</del> or MCH-28 alternatives can be used to qualify to be in the same sample group for the fan efficacy feature.</p> |
| <del>CF2R/CF3R-MCH-22b-H</del> | <del>Certificate of Installation/ Verification</del> | <del>Forced Air System Fan Efficacy (Watt/cfm) Newly Installed Zoned Single Speed Compressor Systems</del>                                                                                                                                                                                                                                                                                                                       | <del>See rules for the MCH-22a.</del>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Document Number                | Document Type                                        | Document Description                                                                                                                                                                                                                                                                                                                                                                        | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <del>CF2R/CF3R-MCH-22c-H</del> | <del>Certificate of Installation/ Verification</del> | <del>Forced Air System Fan Efficacy (Watt/cfm) Newly Installed Non-Zoned or Zoned Multi-Speed Compressor Systems with CFVCS</del>                                                                                                                                                                                                                                                           | <del>See rules for the MCH-22a.</del>                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <del>CF2R/CF3R-MCH-22d-H</del> | <del>Certificate of Installation/ Verification</del> | <del>Forced Air System Fan Efficacy (Watt/cfm) Newly Installed Zoned Single-Speed Compressor Systems with CFVCS</del>                                                                                                                                                                                                                                                                       | <del>See rules for the MCH-22a.</del>                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| CF2R/CF3R-MCH-23a-H            | Certificate of Installation/ Verification            | <p>Forced Air System Airflow Rate (cfm/ton)</p> <p><u>- Single Zone Systems or Zonally Controlled Systems with All Zones Calling</u></p> <p><u>- Zonally Controlled Systems in Every Zonal Control Mode</u></p> <p><u>- Alternative Compliance (best that I can do)</u></p> <p><u>- Measurement Only (CFM) Single Zone Systems or Zonally Controlled Systems with All Zones Calling</u></p> | <p><b>Airflow Rate Feature</b></p> <p>When the MCH-01 indicates requirement for MCH-23 for <u>HERSfield</u> verification for system airflow rate in the dwelling, qualification for inclusion in a sample group may be demonstrated using <u>any one of one of MCH-23 the applicable variants (a, b, c, e, f),</u> or by use of a MCH-28 when indicated on the MCH-01. The same <u>"Forced Air System Airflow Rate (cfm/ton)" verification protocol MCH-23 variant</u></p> |

| Document Number | Document Type | Document Description                                                                                                                                                   | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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|                 |               | <p>- <u>Newly Installed Non- Zoned or Zoned Multi-Speed Compressor with CFVCS</u></p> <p>- <u>Newly Installed Zoned Single Speed Compressor Systems with CFVCS</u></p> | <p>does not need to be used in all dwellings in a sample group, thus any combination of MCH-23 <del>variants</del> or MCH-28 alternatives can be used to qualify to be in the same sample group for the airflow rate verification feature.</p> <p><u>Sampling is not allowed for altered systems.</u></p> <p><del>Additionally, when "Alternative Compliance" is used for compliance, the MCH-23-</del><br/> <del>documented dwelling shall be verified by a ECC Rater (cannot comply as "not tested" dwelling in a sample group); additionally the MCH-23e-</del><br/> <del>documented dwelling cannot be used to represent the "tested" dwelling for airflow rate compliance for the sample group.</del></p> <p><u>Furthermore, a</u></p> |

| Document Number                | Document Type                                       | Document Description                                                                                           | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                 |
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|                                |                                                     |                                                                                                                | <u>"Measurement Only (CFM) Single Zone Systems or Zonally Controlled Systems with All Zones Calling" shall not be used for determining dwelling qualification for field Sample groups.</u>                                                                                                                                                                                                                         |
| <del>CF2R/CF3R-MCH-23b-H</del> | <del>Certificate of Installation/Verification</del> | <del>Forced Air System Airflow Rate (cfm/ton)<br/>Zonally Controlled Systems in Every Zonal Control Mode</del> | <del>See rules for the MCH-23a.</del>                                                                                                                                                                                                                                                                                                                                                                              |
| <del>CF2R/CF3R-MCH-23c-H</del> | <del>Certificate of Installation/Verification</del> | <del>Forced Air System Airflow Rate<br/>Alternative Compliance (best that I can do)</del>                      | <del>Sampling is not allowed for altered systems.</del><br><br><del>See rules for the MCH-23a.</del><br><br><del>Additionally, when MCH-23c is used for compliance, the MCH-23-</del><br><del>documented dwelling shall be verified by a HERS Rater (cannot comply as "not tested" dwelling in a sample group),</del><br><del>additionally the MCH-23c-</del><br><del>documented dwelling cannot be used to-</del> |



| Document Number                | Document Type                                       | Document Description                                                                                                                      | Single Family Group Sampling Rules                                                                                                                                          |
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|                                |                                                     |                                                                                                                                           | represent the "tested" dwelling for airflow-rate compliance for the sample group.                                                                                           |
| <del>CF2R/CF3R-MCH-23d-H</del> | <del>Certificate of Installation/Verification</del> | <del>Forced Air System Airflow Rate Measurement Only (CFM) Single Zone Systems or Zonally Controlled Systems with All Zones Calling</del> | See rules for the MCH-23a.<br><br>Additionally, a MCH-23d shall not be used for determining dwelling qualification for HERSfield Sample groups.                             |
| <del>CF2R/CF3R-MCH-23e-H</del> | <del>Certificate of Installation/Verification</del> | <del>Forced Air System Airflow Rate Newly Installed Non-Zoned or Zoned Multi-Speed Compressor with CFVCS</del>                            | See rules for the MCH-23a.                                                                                                                                                  |
| <del>CF2R/CF3R-MCH-23f-H</del> | <del>Certificate of Installation/Verification</del> | <del>Forced Air System Airflow Rate Newly Installed Zoned Single-Speed Compressor Systems with CFVCS</del>                                | See rules for the MCH-23a.                                                                                                                                                  |
| CF2R/CF3R-MCH-24a-H            | Certificate of Installation/Verification            | Enclosure Air Leakage Worksheet<br><br>_Single-Point Test with Manual Meter_ <u>or Automatic Meter.</u><br><br>_ <u>Multipoint Test</u>   | MCH-24 does not document an applicable sampling feature by itself (MCH-24 is used for completing some MCH-27 docs). A MCH-24 doc shall not be used for determining dwelling |

| Document Number                | Document Type                                       | Document Description                                                              | Single Family Group Sampling Rules                                                                                                                                                                       |
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|                                |                                                     |                                                                                   | qualification for <del>HERS</del> -ECC Sample groups. <u>If a grouped dwelling unit complies using a sampled MCH-27 that necessitated a MCH-24, the sampled dwelling unit does not require a MCH-24.</u> |
| <del>CF2R/CF3R MCH-24b-H</del> | <del>Certificate of Installation/Verification</del> | <del>Enclosure Air Leakage Worksheet Single Point Test with Automatic Meter</del> | <del>See rules for the MCH-24a,</del>                                                                                                                                                                    |

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Text is subject to internal review and revision, and that it is not official guidance until approved by the Commission

| Document Number     | Document Type                             | Document Description                                                                                                                                                                                                                         | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| CF2R/CF3R-MCH-25a-H | Certificate of Installation/ Verification | Refrigerant Charge Verification<br><br>- Superheat Method (Standard Charge Procedure)<br>- Subcooling Method (Standard Charge Procedure)<br>- Weigh-in Charging Procedure<br>- Packaged System Manufacturer Refrigerant Charge Certification | <b>Refrigerant Charge Feature</b><br><br>When the MCH-01 indicates requirement for MCH-25 for <del>HERS</del> field verification for Refrigerant Charge verification for the dwelling, qualification for inclusion in a sample group may be demonstrated using <u>one of MCH-25 any one of the applicable variants (a, b, c, d, e, f).</u> The same "Refrigerant Charge Verification" protocol <del>variant</del> does not need to be used in all dwellings in a sample group, thus any combination of MCH-25 <del>variants</del> can be used to qualify to be in the same sample group.<br><u>When the MCH-25 (Weigh-in Charging Procedure) is used for installation compliance, the MCH-25-documented dwelling system shall be verified by a ECC-Rater (cannot comply as "not tested")</u> |

| Document Number                | Document Type                                       | Document Description                                                                     | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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|                                |                                                     |                                                                                          | <p><u>dwelling in a sample group).</u></p> <p><u>Additionally, when the installer's MCH-25 (Packaged System Manufacturer Refrigerant Charge Certification) is used by the installer, there is no required ECC-Rater verification for the system. These systems are eligible to be included in a sample group for Refrigerant Charge verification features, but these systems cannot be used to represent the "tested" dwelling system for Refrigerant Charge compliance for the sample group.</u></p> |
| <del>CF2R/CF3R-MCH-25b-H</del> | <del>Certificate of Installation/Verification</del> | <del>Refrigerant Charge-Verification-Subcooling Method-(Standard Charge Procedure)</del> | <del>See rules for the MCH-25a.</del>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <del>CF2R/CF3R-MCH-25c-H</del> | <del>Certificate of Installation/Verification</del> | <del>Refrigerant Charge-Verification Weigh-in-Charging-Procedure</del>                   | <p><del>See rules for the MCH-25a.</del></p> <p><del>Additionally, when CF2R-MCH-25c is used for installation compliance, the MCH-25-documented dwelling-</del></p>                                                                                                                                                                                                                                                                                                                                   |

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|                                |                                                       |                                                                                                                                               | <del>shall be verified by a HERS Rater</del> <del>ECC Rater</del><br>(cannot comply as "not tested" dwelling in a sample group), additionally the installer's MCH-25c documented dwelling cannot be used to represent the "tested" dwelling for Refrigerant Charge compliance for the sample group. |
| <del>CF2R/CF3R-MCH-25d-H</del> | <del>Certificate of Installation/- Verification</del> | <del>Refrigerant Charge Verification—Fault Indicator Display (FID)<br/>(embedded in the CF2R-MCH-25a,b,c)<br/>(standalone CF3R-MCH-25d)</del> | <del>See rules for the MCH-25a.</del>                                                                                                                                                                                                                                                               |
| <del>CF2R-MCH-25f-E</del>      | <del>Certificate of Installation</del>                | <del>Refrigerant Charge Verification—</del>                                                                                                   | <del>See rules for the MCH-25a.</del>                                                                                                                                                                                                                                                               |

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|                 |               | <del>Packaged System-<br/>Manufacturer-<br/>Refrigerant Charge-<br/>Certification<br/>(CF2R only for the<br/>"f" variant)</del> | <del>Additionally, when<br/>MCH-25f is used by<br/>the installer, there is<br/>no required HERSECC-<br/>Rater verification for<br/>the system. These<br/>systems are eligible to<br/>be included in a<br/>sample group for<br/>Refrigerant Charge-<br/>verification features,<br/>but these systems<br/>cannot be used to<br/>represent the "tested"<br/>dwelling for<br/>Refrigerant Charge-<br/>compliance for the<br/>sample group.</del> |

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| Document Number    | Document Type                             | Document Description                                   | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| CF2R/CF3R-MCH-26-H | Certificate of Installation/ Verification | Rated Space Conditioning System Equipment Verification | <p><b>Rated Space Conditioning System Feature</b></p> <p>When the MCH-01 indicates requirement for MCH-26 for <b>HERSfield</b> verification of space conditioning systems in the dwelling, qualification for inclusion in a sample group may be demonstrated using MCH-26 regardless of which rating verification protocol is reported on the MCH-26. The same space conditioning system equipment verification protocol does not need to be used in all dwellings in a sample group, thus any combination of _MCH-26_ verifications can be used to qualify to be in the same sample group.</p> |

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| CF2R/CF3R-MCH-27a-H            | Certificate of Installation/ Verification           | <p>IAQ and Mechanical Ventilation</p> <p>Attached/Detached Ventilation</p> <p><u>Non-Dwelling Unit</u></p> | <p><b>IAQ Mechanical Ventilation Features</b></p> <p>When MCH-27 for <u>HERSfield</u> verification for Mechanical Ventilation Airflow Rate is required for the dwelling, qualification for inclusion in a sample group may be demonstrated using <u>any one of the applicable variants (a, c, d) one of MCH-27.</u> The same <u>"IAQ and Mechanical Ventilation"</u> <u>variant verification protocol</u> does not need to be used in all dwellings in a sample group, thus any combination of MCH-27 variants can be used to qualify to be in the same sample group.</p> |
| <del>CF2R/CF3R-MCH-27c-H</del> | <del>Certificate of Installation/Verification</del> | <del>IAQ and Mechanical Ventilation<br/>Scheduled and Real-Time Control</del>                              | <del>See rules for the MCH-27a</del>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <del>CF2R-MCH-27d-H</del>      | <del>Certificate of Installation</del>              | <del>IAQ and Mechanical Ventilation<br/><br/>Non-Dwelling Unit</del>                                       | <del>See rules for the MCH-27a</del>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Document Number    | Document Type                             | Document Description                                                    | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                   |
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| CF2R/CF3R-MCH-28-H | Certificate of Installation/ Verification | Return Duct And Filter Grille Design According to Tables 150.0-B or C   | <p><b>Return Duct Design Alternative to Airflow Rate and Fan Efficacy Verification</b></p> <p>If specified on MCH-01, a MCH-28 shall be used as an alternative to compliance with airflow rate (MCH-23) and fan efficacy (MCH-22) <u>HERSECC</u> features, thus the system qualifies for inclusion in a sample group for airflow rate and Fan Efficacy Features.</p> |
| CF2R/CF3R-MCH-29-H | Certificate of Installation/ Verification | Supply Duct Surface Area and R-Value; Buried Ducts; Deeply Buried Ducts | <p><b>Duct Design Features</b></p> <p>If the MCH-01 specifies requirement for MCH-29 for <u>HERSFIELD</u>field verification for Duct Surface Area and R-Value and Buried or Deeply Buried Ducts Features in the dwelling, then qualification for inclusion in a sample group may be demonstrated using MCH-29 regardless of which verification</p>                   |

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|                    |                                           |                                                                   | protocol is reported on the MCH-29. The same duct verification protocol does not need to be used in all dwellings in a sample group, thus any combination of MCH-29 verifications can be used to qualify to be in the same sample group.                                                                                                                                                                                                                         |
| CF2R/CF3R-MCH-30-H | Certificate of Installation/ Verification | Central Fan Ventilation Cooling Systems (CFVCS) compliance credit | <b>CFVCS Feature</b><br><br>If the MCH-01 specifies requirement for MCH-30 for verification of central fan ventilation cooling systems (CFVCS) in the dwelling, then qualification for inclusion in a sample group may be demonstrated using MCH-30 regardless of which type of CFVCS is documented on the MCH-30. The same CFVCS verification protocol do not need to be used on all systems in all dwellings in a sample group, thus any combination of MCH-30 |

| Document Number                | Document Type                                        | Document Description                                                                                                                                            | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                |                                                      |                                                                                                                                                                 | <p>verifications can be used to qualify to be in the same sample group.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| CF2R/CF3R-MCH-31a-H            | Certificate of Installation/ Verification            | <p><del>HERS</del>-<u>ECC</u> verified WHF</p> <p>- with individual CFM and Watts collection</p> <p>- <u>with total CFM and individual Watts collection</u></p> | <p><b>Whole House Fan (WHF) Airflow and Fan Efficacy</b></p> <p>If MCH-31 for <u>HERSFIELD</u>field verification for WHF verification is required for the dwelling, then qualification for inclusion in a sample group may be demonstrated using <u>one of MCH-31 any</u> <del>one of the applicable variants (a, b, c, d)</del>. The same "<u>ECC verified WHF</u>" <u>verification protocol</u> <del>variant</del> does not need to be used in all dwellings in a sample group, thus any combination of MCH-31 <del>variants</del> can be used to qualify to be in the same sample group</p> |
| <del>CF2R/CF3R-MCH-31b-H</del> | <del>Certificate of Installation/ Verification</del> | <del>HERS-verified WHF with total CFM and individual Watts collection</del>                                                                                     | <del>See rules for the MCH-31a.</del>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| CF2R/CF3R-MCH-32-H             | Certificate of Installation/ Verification            | Local Mechanical Exhaust                                                                                                                                        | <p><b>Kitchen Range Hood Verification</b></p> <p>If <u>HERS</u>field</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Document Number    | Document Type                             | Document Description   | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------|-------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    |                                           |                        | verification of the kitchen range hood is required for the dwelling, then the MCH-32 may be used as qualification for inclusion in a _sample group.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| CF2R/CF3R-MCH-33-H | Certificate of Installation/ Verification | VCHP Compliance Credit | <p><b><u>VCHP</u></b><br/> <b><u>FeatureQualification</u></b></p> <p>If the MCH-01 indicates requirement for MCH-33 for <u>HERSfield</u> _verification of variable capacity heat pump (VCHP) system in the dwelling, then qualification for inclusion in a sample group may be demonstrated using a MCH-33 regardless of which VCHP system type (ducted, ductless, ducted + ductless, non-continuous fan, continuous fan) is documented on the MCH-33. The same combination of MCH-33 compliance protocols <u>de</u><u>does</u> not need to be used for all MCH-33-documented</p> |



| Document Number    | Document Type                             | Document Description                                                              | Single Family Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|-------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    |                                           |                                                                                   | systems in all dwelling units in a sample group, thus any combination of MCH-33_ verifications can be_ used to qualify to be in the same sample group with other MCH-33- documented_ systems.                                                                                                                                                                                                                                                                |
| CF2R/CF3R-PLB-22-H | Certificate of Installation/ Verification | <del>HERS_Single</del> Dwelling Unit Domestic Hot Water (DHW) System Distribution | When the CF1R indicates <del>HERS</del> field verification for a single-family dwelling DHW feature, qualification for inclusion in a sample group may be demonstrated using PLB-22 regardless of which verification protocol is reported on the PLB-22. The same DHW verification protocol does not need to be used in all dwellings in a sample group, thus any combination of PLB-22 verifications can be used to qualify to be in the same sample_group. |

Source: California Energy Commission

Table 7-2: Group Sampling Rules Applicable to 2025 Low-Rise Multifamily  
Compliance Documents ~~and HERS Features~~

| Document Number    | Document Type                                | Document Description | Low-rise Multifamily Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|----------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LMCI/LMCV-ENV-21-H | Certificate of Installation/<br>Verification | QII Framing Stage    | <p>When the CF1R indicates requirement for <b>HERSfield</b> verification of QII, then compliance shall be demonstrated using one <b>HERS-ECC</b> "Tested" ENV-21 in one of the dwellings in the group, and one <b>HERS-ECC</b> "Tested" ENV- 22 either in the same dwelling or otherwise in another dwelling in same sample group.</p> <p>Refer to group sampling procedures in RA2.6.3.5 that allow <b>HERSfield</b> verification "testing" of QII features prior to closing the sample group.</p> |
| LMCI/LMCV-ENV-22-H | Certificate of Installation/<br>Verification | QII Insulation Stage | See rules for the ENV-21.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Document Number                | Document Type                                        | Document Description                                                                                                                                                                                                                                                                                                                                  | Low-rise Multifamily Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LMCI/LMCV-MCH-20a-H            | Certificate of Installation/ Verification            | <p>Duct Leakage <u>Measurement Test</u></p> <p><del>- New Construction</del></p> <p>- <u>New System;</u></p> <p>- <u>Low Leakage Ducts in Conditioned Space Compliance Credit;</u></p> <p>- <u>Low Leakage Air-Handling Units</u></p> <p><del>- <u>Altered (Existing) System</u></del></p> <p><del>- <u>Sealing of All Accessible Leaks</u></del></p> | <p><b>Duct Leakage Feature</b></p> <p>When the MCH-01 indicates a MCH-20 is required for <u>HERS field</u> verification of duct leakage for a ducted indoor unit in the dwelling, <del>qualification for</del> inclusion in a sample group <u>depends on the measurement protocol indicated on the certificate of compliance.</u> <del>may be demonstrated using any one of the applicable variants (a, b, c, d, e).</del> The same variant <del>does not need to be used on all indoor units on all systems in all dwellings in a sample group, thus any combination of MCH-20 variants can be used to qualify the dwelling to be in the same sample group.</del></p> |
| <del>LMCI/LMCV-MCH-20b-H</del> | <del>Certificate of Installation/ Verification</del> | <del>Duct Leakage-LLDCS</del>                                                                                                                                                                                                                                                                                                                         | <del>See rules for the MCH-20a.</del>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Document Number                | Document Type                                        | Document Description                                   | Low-rise Multifamily Group Sampling Rules                                                                                                                                                                                                                                                                                         |
|--------------------------------|------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <del>LMCI/LMCV-MCH-20c-H</del> | <del>Certificate of Installation/ Verification</del> | <del>Duct Leakage-LLAHU</del>                          | <del>See rules for the MCH-20a.</del>                                                                                                                                                                                                                                                                                             |
| <del>LMCI/LMCV-MCH-20d-H</del> | <del>Certificate of Installation/ Verification</del> | <del>Duct Leakage Test- Existing Const</del>           | <p><del>See rules for the MCH-20a.</del></p> <p><del>Additionally, if MCH-20d-Section C requires use of a smoke test for verification of ducts in a garage space, the dwelling unit is prohibited from compliance as "not tested" in the sample group nor can it be used to represent the "tested" system in the group.</del></p> |
| <del>LMCI/LMCV-MCH-20e-H</del> | <del>Certificate of Installation/ Verification</del> | <del>Duct Leakage Test- Sealing Accessible Leaks</del> | <p><del>See rules for the MCH-20a.</del></p> <p><del>Additionally, If MCH-20e (smoke test) is used for compliance, the dwelling unit is prohibited from compliance as "not tested" in the sample group nor can it be used to represent the "tested" system in the group.</del></p>                                                |
| LMCI/LMCV-MCH-21-H             | Certificate of Installation/ Verification            | Duct Location <u>Verification</u>                      | <b>Duct Location, <u>Surface Area,</u> and <u>R-value Features</u></b>                                                                                                                                                                                                                                                            |

| Document Number                 | Document Type                            | Document Description                                                                                                                                          | Low-rise Multifamily Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 |                                          |                                                                                                                                                               | <p>When the MCH-01 indicates requirement for MCH-21 for <del>HERS</del><u>field</u> verification for duct location in the dwelling, qualification for inclusion in a sample group may be demonstrated using MCH-21 regardless of which duct location verification protocol is reported on the MCH-21. The same duct location verification protocol does not need to be used in all dwellings in a sample group, thus any combination of MCH-21 verifications <u>given by RA3.1.4.1</u> can be used to qualify to be in the same sample group, <u>unless the verification is part of the LLDCS testing (RA3.1.4.3.8).</u></p> |
| LMCI/LMCV-MCH-22 <del>a</del> H | Certificate of Installation/Verification | <del>Fan Efficacy All Zones Calling Only</del><br><u>Forced Air System Fan Efficacy (Watt/cfm)</u><br><u>- Newly Installed Non-Zoned or Zoned Multi-Speed</u> | <p><b>Fan Efficacy Feature</b></p> <p>When the MCH-01 indicates requirement for MCH-22 for <del>HERS</del><u>field</u></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Document Number                | Document Type                                       | Document Description                                                                                                                                                                   | Low-rise Multifamily Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                |                                                     | <u>Compressor Systems</u><br><u>- Newly Installed Zoned Single Speed Compressor Systems Newly</u>                                                                                      | verification for fan efficacy in the dwelling, qualification for inclusion in a sample group may be demonstrated using <u>one of MCH-22</u> <del>any one of the applicable variants (a, b, c, d)</del> , or by use of a MCH-28 when indicated on the MCH-01. The same <del>variant</del> <u>"Forced Air System Fan Efficacy (Watt/cfm)" verification protocol</u> does not need to be used in all dwellings in a sample group, thus any combination of MCH-22 <del>variants</del> or MCH-28 alternatives can be used to qualify to be in the same sample group for the fan efficacy feature. |
| <del>LMCI/LMCV-MCH-22b-H</del> | <del>Certificate of Installation/Verification</del> | <del>Fan Efficacy Every Zonal Control Mode</del>                                                                                                                                       | <del>See rules for the MCH-22a.</del>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| LMCI/LMCV-MCH-23a-H            | Certificate of Installation/Verification            | <del>Airflow Rate All Zones Calling Only</del><br><u>Forced Air System Airflow Rate (cfm/ton)</u><br><u>- Single Zone Systems or Zonally Controlled Systems with All Zones Calling</u> | <b>Airflow Rate Feature</b><br>When the MCH-01 indicates requirement for MCH-23 for <del>HERS</del> <u>field</u> verification for system airflow                                                                                                                                                                                                                                                                                                                                                                                                                                             |



| Document Number | Document Type | Document Description                                                                                                                                                                                                                                                                              | Low-rise Multifamily Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 |               | <ul style="list-style-type: none"> <li>- <u>Zonally Controlled Systems in Every Zonal Control Mode</u></li> <li>- <u>Alternative Compliance (best-that-I-can-do)</u></li> <li>- <u>Measurement Only (CFM) Single Zone Systems or Zonally Controlled Systems with All Zones Calling</u></li> </ul> | <p>rate in the dwelling, qualification for inclusion in a sample group may be demonstrated using <u>one of MCH-23</u> <del>any</del> <del>one of the applicable variants (a, b, c, e, f)</del>, or by use of a MCH-28 when indicated on the MCH-01. The same MCH-23 <del>variant</del> <u>"Forced Air System Airflow Rate (cfm/ton)" verification protocol</u> <del>-does not need to be used in all dwellings in a sample group, thus any combination of MCH-23 variants or MCH-28 alternatives can be used to qualify to be in the same sample group for the airflow</del></p> |

| Document Number | Document Type | Document Description | Low-rise Multifamily Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|---------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 |               |                      | <p>rate verification feature.</p> <p><u>Sampling is not allowed for altered systems.</u></p> <p>Additionally, when <u>"Alternative Compliance"</u> is used for compliance, the MCH-23-</p> <p>documented dwelling shall be verified by a ECC Rater (cannot comply as "not tested" dwelling in a sample group), additionally the MCH-23c-</p> <p>documented dwelling cannot be used to represent the "tested" dwelling for airflow rate compliance for the sample group.</p> <p>Furthermore, "Measurement Only (CFM) Single Zone Systems or Zonally Controlled Systems with All Zones Calling" shall not be used for determining dwelling qualification for field Sample</p> |

| Document Number                | Document Type                                       | Document Description                                            | Low-rise Multifamily Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------|-----------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                |                                                     |                                                                 | groups.                                                                                                                                                                                                                                                                                                                                                                                               |
| <del>LMCI/LMCV-MCH-23b-H</del> | <del>Certificate of Installation/Verification</del> | <del>Airflow Rate Every Zonal Control Mode</del>                | <del>See rules for the MCH-23a.</del>                                                                                                                                                                                                                                                                                                                                                                 |
| <del>LMCI/LMCV-MCH-23c-H</del> | <del>Certificate of Installation/Verification</del> | <del>Airflow Rate Best That I Can Do</del>                      | <p><del>See rules for the MCH-23a.</del></p> <p>Additionally, when MCH-23c is used for compliance, the MCH-23 documented dwelling shall be verified by a HERS Rater <u>ECC Rater</u> (cannot comply as "not tested" dwelling in a sample group); additionally the MCH-23c documented dwelling cannot be used to represent the "tested" dwelling for airflow rate compliance for the sample group.</p> |
| <del>LMCI/LMCV-MCH-23d-H</del> | <del>Certificate of Installation/Verification</del> | <del>Airflow Rate—Measurement Only—All Zones Calling Only</del> | <p><del>See rules for the MCH-23a.</del></p> <p>Additionally, a MCH-23d shall not be used for determining dwelling qualification for HERS Sample groups.</p>                                                                                                                                                                                                                                          |

| Document Number                | Document Type                                       | Document Description                                                                                                                                                                                                              | Low-rise Multifamily Group Sampling Rules                                                                                                                                                                                                                                                      |
|--------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <del>LMCI/LMCV-MCH-23e-H</del> | <del>Certificate of Installation/Verification</del> | <del>Airflow Rate All Zones Calling Only with CFVCS</del>                                                                                                                                                                         | <del>See rules for the MCH-23a.</del>                                                                                                                                                                                                                                                          |
| <del>LMCI/LMCV-MCH-23f-H</del> | <del>Certificate of Installation/Verification</del> | <del>Airflow Rate Every Zonal Control Mode with CFVCS</del>                                                                                                                                                                       | <del>See rules for the MCH-23a.</del>                                                                                                                                                                                                                                                          |
| LMCI/LMCV-MCH-24a-H            | Certificate of Installation/Verification            | Enclosure Air Leakage <u>Worksheet Test</u><br>- Single Point Test- Manual Meter <u>or</u> <u>Automatic Meter</u> ,<br>- <u>Multipoint Test</u>                                                                                   | <u>Compartment leakage testing compliance may be demonstrated using any valid testing protocol.</u><br><del>Sampling n/a on the basis that the document is a worksheet for the MCH-27. The MCH-24 can only apply to the dwelling that was measured, not to other dwellings in the group.</del> |
| <del>LMCI/LMCV-MCH-24b-H</del> | <del>Certificate of Installation/Verification</del> | <del>Enclosure Air Leakage Worksheet Single Point Test Automatic Meter</del>                                                                                                                                                      | <del>See rules for the MCH-24a.</del>                                                                                                                                                                                                                                                          |
| LMCI/LMCV-MCH-25a-H            | Certificate of Installation/Verification            | <u>Refrigerant Charge Verification</u><br><br>- <u>Superheat Method (Standard Charge Procedure)</u><br>- <u>Subcooling Method (Standard Charge Procedure)</u><br>- <u>Weigh-in Charging Procedure</u><br>- <u>Packaged System</u> | <b>Refrigerant Charge Feature</b><br><br><u>Sampling is only allowed for newly constructed buildings.</u><br><br>When the MCH-01 indicates requirement for MCH-25 for                                                                                                                          |

| Document Number | Document Type | Document Description                                 | Low-rise Multifamily Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|---------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 |               | <u>Manufacturer Refrigerant Charge Certification</u> | <p><u>HERSfield</u> verification for Refrigerant Charge verification for the dwelling, qualification for inclusion in a sample group may be demonstrated using <del>any one of the applicable variants (a, b, c, d, e, f)</del> <u>one of MCH-25</u>. The same <del>variant</del> <u>"Refrigerant Charge" Verification protocol</u> does not need to be used in all dwellings in a sample group, thus any combination of MCH-25 variants can be used to qualify to be in the same sample group.</p> <p><u>When MCH-25 (Weigh-in Charging Procedure) is used for installation compliance, the MCH-25-documented dwelling shall be verified by a ECC-Rater (cannot comply as "not tested" dwelling in a sample group).</u></p> <p><u>Additionally, when installer's MCH-25 (Packaged System Manufacturer</u></p> |

| Document Number                | Document Type                                       | Document Description                               | Low-rise Multifamily Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------|-----------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                |                                                     |                                                    | <u>Refrigerant Charge Certification</u> ) is used by the installer, there is no required <u>ECC-Rater</u> verification for the system. These systems are eligible to be included in a sample group for <u>Refrigerant Charge</u> verification features, but these systems cannot be used to represent the "tested" dwelling for <u>Refrigerant Charge</u> compliance for the sample group. |
| <del>LMCI/LMCV-MCH-25b-H</del> | <del>Certificate of Installation/Verification</del> | <del>Refrigerant Charge Subcooling</del>           | <del>See rules for the MCH-25a.</del>                                                                                                                                                                                                                                                                                                                                                      |
| <del>LMCI/LMCV-MCH-25c-H</del> | <del>Certificate of Installation/Verification</del> | <del>Refrigerant Charge Weigh-in Observation</del> | <p><del>See rules for the MCH-25a.</del></p> <p>Additionally, when CF2R-MCH-25c is used for installation compliance, the MCH-25-documented dwelling shall be verified by a HERS-Rater <del>ECC-Rater</del> (cannot comply as "not tested" dwelling in a sample group), additionally the installer's MCH-25c-documented dwelling cannot be used to</p>                                      |



| Document Number                | Document Type                                        | Document Description                                              | Low-rise Multifamily Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------|------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                |                                                      |                                                                   | <del>represent the "tested" dwelling for Refrigerant Charge compliance for the sample group.</del>                                                                                                                                                                                                                                                                                                                           |
| <del>LMCI/LMCV-MCH-25e-H</del> | <del>Certificate of Installation/ Verification</del> | <del>Refrigerant Charge- Winter Setup</del>                       | <del>See rules for the MCH-25a.</del>                                                                                                                                                                                                                                                                                                                                                                                        |
| <del>LMCI/LMCV-MCH-25f-E</del> | <del>Certificate of Installation/ Verification</del> | <del>Refrigerant Charge- Packaged System- Manufacturer Cert</del> | <p><del>See rules for the MCH-25a.</del></p> <p><del>Additionally, when MCH-25f is used by the installer, there is no required <u>HERSfield</u> verification for the system. These systems are eligible to be included in a sample group for Refrigerant Charge verification features, but these systems cannot be used to represent the "tested" dwelling for Refrigerant Charge compliance for the sample group.</del></p> |
| LMCI/LMCV-MCH-26-H             | Certificate of Installation/ Verification            | Rated System Verification                                         | <p><b>Rated Space Conditioning System Feature</b></p> <p>When the MCH-01 indicates requirement for MCH-26 for <u>HERSfield</u> verification of space</p>                                                                                                                                                                                                                                                                     |

| Document Number                  | Document Type                            | Document Description                 | Low-rise Multifamily Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------|------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  |                                          |                                      | conditioning systems in the dwelling, qualification for inclusion in a sample group may be demonstrated using MCH-26 regardless of which rating verification protocol is reported on the MCH-26. The same space conditioning system equipment verification protocol does not need to be used in all dwellings in a sample group, thus any combination of MCH-26 verifications can be used to qualify to be in the same sample group. |
| LMCI/LMCV-MCH-27 <sup>b</sup> -H | Certificate of Installation/Verification | Dwelling Unit Mechanical Ventilation | <p>When the MCH-01 requires verification of the dwelling unit ventilation, the MCH-27 may be used to sample in groups having the same features. Dwellings with HRVs may be grouped with dwellings having ERVs or HRVs, for example.</p> <p>However, a dwelling having balanced ventilation without recovery may not be grouped with a</p>                                                                                            |

| Document Number                | Document Type                                       | Document Description                                    | Low-rise Multifamily Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------|-----------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                |                                                     |                                                         | <p>dwelling that features recovery.</p> <p>Additionally, dwellings having CFI and CFVS systems cannot be grouped with dwellings without these systems.</p>                                                                                                                                                                                                                                                                                                                                                                     |
| <del>LMCI/LMCV-MCH-27b-H</del> | <del>Certificate of Installation/Verification</del> | <del>Dwelling Unit Mechanical Ventilation</del>         | <p><del>When the MCH-01 requires verification of the dwelling unit ventilation, the MCH-27 may be used to sample in groups having the same features. Dwellings with HRVs may be grouped with dwellings having ERVs or HRVs, for example.</del></p> <p><del>However, a dwelling having balanced ventilation without recovery may not be grouped with a dwelling that features recovery.</del></p> <p><del>Additionally, dwellings having CFI and CFVS systems cannot be grouped with dwellings without these systems.</del></p> |
| LMCI/LMCV-MCH-28-H             | Certificate of Installation/Verification            | Return Duct and Filter Grille Design- Table 150.0B or C | <p><b>Return Duct Design Alternative to Airflow Rate and Fan Efficacy Verification</b></p> <p>If specified on MCH-</p>                                                                                                                                                                                                                                                                                                                                                                                                         |

| Document Number    | Document Type                             | Document Description                  | Low-rise Multifamily Group Sampling Rules                                                                                                                                                                                                       |
|--------------------|-------------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    |                                           |                                       | 01, a MCH-28 shall be used as an alternative to compliance with airflow rate (MCH- 23) and fan efficacy (MCH-22) <b>HERSECC</b> features, thus the system qualifies for inclusion in a sample group for airflow rate and Fan Efficacy Features. |
| LMCI/LMCV-MCH-29-H | Certificate of Installation/ Verification | Supply Duct Surface Area/Buried Ducts | Sampling n/a                                                                                                                                                                                                                                    |
| LMCI/LMCV-MCH-32-H | Certificate of Installation/ Verification | Kitchen Ventilation                   | <b>Kitchen Range Hood Verification</b><br>If <b>HERSfield</b> verification of the kitchen range hood is required for the dwelling, then the MCH-32 may be used as qualification for inclusion in a sample group.                                |
| LMCI/LMCV-MCH-33-H | Certificate of Installation/ Verification | VCHP Compliance Credit                | <b><u>VCHP Feature</u></b><br><br>If the MCH-01 indicates requirement for MCH-33 for <b>HERSfield</b> verification of variable capacity heat pump (VCHP) system in the dwelling, then qualification for                                         |

| Document Number    | Document Type                             | Document Description                                              | Low-rise Multifamily Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------|-------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    |                                           |                                                                   | inclusion in a sample group may be demonstrated using a MCH-33 regardless of which VCHP system type (ducted, ductless, ducted + ductless, non- continuous fan, continuous fan) is documented on the MCH-33. The same combination of MCH-33 compliance protocols do not need to be used for all MCH-33- documented systems in all dwelling units in a sample group, thus any combination of MCH-33 verifications can be used to qualify to be in the same sample group with other MCH-33- documented systems. |
| LMCI/LMCV-PLB-21-H | Certificate of Installation/ Verification | <del>HERS</del> Multifamily Central Hot Water System Distribution | <b>Multifamily DHW Feature</b><br><br>If <del>HERS</del> -field verification for multifamily central DHW is required, then the PLB-21 may be used as qualification for inclusion in a sample group.                                                                                                                                                                                                                                                                                                          |

| Document Number    | Document Type                             | Document Description                                               | Low-rise Multifamily Group Sampling Rules                                                                                                                                                      |
|--------------------|-------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LMCI/LMCV-PLB-22-H | Certificate of Installation/ Verification | <del>HERS</del> Single Dwelling Unit Hot Water System Distribution | <b>Multifamily DHW Feature</b><br>If <del>HERS</del> field verification for multifamily central DHW is required, then the PLB-22 may be used as qualification for inclusion in a sample group. |

Source: California Energy Commission

**DRAFT**

Text is subject to internal review and revision, and that it is not official guidance until approved by the Commission



Table 7-3: Group Sampling Rules Applicable to 2025 Nonresidential (Including High-Rise Multifamily) Compliance Documents and HERSECC Features

| Document Number | Document Type               | Document Description                                                                                                                                                                                                                                         | Nonresidential Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NRCV-MCH-04a-H  | Certificate of Verification | <p>Nonresidential Duct Leakage Measurement</p> <ul style="list-style-type: none"> <li>- New System</li> <li>- <u>Low Leakage Air-Handling Units</u></li> <li>- <u>Altered (Existing) System</u></li> <li>- <u>Sealing of All Accessible Leaks</u></li> </ul> | <p><b>Nonresidential Duct Leakage Feature</b></p> <p>When the NRCC requires <del>HERS</del>-field verification of duct leakage for system compliance, qualification for inclusion in a sample group may be demonstrated using <u>one of NRCV-MCH-04</u><del>any one of the applicable variants (a, c, d, e).</del> The same <del>variant</del> <u>"Nonresidential Duct Leakage Measurement" verification protocol</u> does not need to be used in all buildings in a sample group, thus any combination of NRCV-MCH-04 variants can be used to qualify to be in the same sample group.</p> <p>-When NRCV-MCH-04e(<u>Sealing of All Accessible Leaks</u>) is used for compliance, the NRCV-MCH-04-</p> |

| Document Number           | Document Type                          | Document Description                                                               | Nonresidential Group Sampling Rules                                                                                                                                                                                                                                                                          |
|---------------------------|----------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           |                                        |                                                                                    | documented system shall be verified by a ECC-Rater (cannot comply as a "not tested" system in a sample group), additionally the NRCV-MCH-04e- ( <u>Sealing of All Accessible Leaks</u> ) documented system cannot be used to represent the "tested" system for duct leakage compliance for the sample group. |
| <del>NRCV-MCH-04c-H</del> | <del>Certificate of Verification</del> | <del>Nonresidential Duct Leakage Measurement Low Leakage Air Handling Units</del>  | <del>See rules for the MCH-04a</del>                                                                                                                                                                                                                                                                         |
| <del>NRCV-MCH-04d-H</del> | <del>Certificate of Verification</del> | <del>Nonresidential Duct Leakage Measurement Altered (Existing) System</del>       | <del>See rules for the MCH-04a</del>                                                                                                                                                                                                                                                                         |
| <del>NRCV-MCH-04e-H</del> | <del>Certificate of Verification</del> | <del>Nonresidential Duct Leakage Measurement Sealing of All Accessible Leaks</del> | <del>When NRCV-MCH-04e is used for compliance, the NRCV-MCH-04 documented system shall be verified by a HERS-Rater <u>ECC-Rater</u> (cannot comply as a "not tested" system in a sample group), additionally the</del>                                                                                       |

| Document Number           | Document Type                          | Document Description                                                                                                                | Nonresidential Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           |                                        |                                                                                                                                     | <del>NRCV-MCH-04e-</del><br><del>documented system-</del><br><del>cannot be used to-</del><br><del>represent the-</del><br><del>"tested" system for-</del><br><del>duct leakage-</del><br><del>compliance for the-</del><br><del>sample group.</del>                                                                                                                                                                                                                                                            |
| NRCV-MCH-24a-H            | Certificate of Verification            | Enclosure Air Leakage Test<br><br><del>Worksheet</del> — Single Point Test with Manual Meter_ or Automatic Meter, — Multipoint Test | <u>Compartment leakage testing</u><br><u>compliance may be demonstrated using any valid testing protocol.</u><br><del>MCH-24 does not document an-</del><br><del>applicable sampling-</del><br><del>feature by itself-</del><br><del>(MCH-24 is used for-</del><br><del>completing some-</del><br><del>MCH-27 does).</del> A<br><del>MCH-24 doc shall not</del><br><del>be used for-</del><br><del>determining dwelling-</del><br><del>qualification for-</del><br><del>HERS Sample</del><br><del>groups.</del> |
| <del>NRCV-MCH-24b-H</del> | <del>Certificate of Verification</del> | <del>Enclosure Air Leakage Worksheet—Single Point Test with Automatic Meter</del>                                                   | <del>See rules for the MCH—24a</del>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| NRCV-MCH-27b-H            | Certificate of Verification            | IAQ and Mechanical Ventilation – High Rise Residential                                                                              | <b>IAQ Mechanical Ventilation Feature</b><br><br>When MCH-27 for <u>HERS field</u>                                                                                                                                                                                                                                                                                                                                                                                                                              |

| Document Number | Document Type                          | Document Description                                                                               | Nonresidential Group Sampling Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------|----------------------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 |                                        |                                                                                                    | <p>verification for Mechanical Ventilation Airflow Rate is required for the dwelling, qualification for inclusion in a sample group may be demonstrated using <u>one of MCH-27</u><del>any</del> <del>one of the applicable variants (b, c).</del> The same <del>variant</del> <u>"IAQ and Mechanical Ventilation – High Rise Residential"</u> does not need to be used in all dwellings in a sample group, thus any combination of MCH-27 <del>variants</del> can be used to qualify to be in the same sample group.</p> |
| NRCV-MCH-27c-H  | <del>Certificate of Verification</del> | <del>IAQ and Mechanical Ventilation – High Rise Residential Scheduled and Real-Time Controls</del> | <del>See rules for the MCH-27b</del>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| NRCV-MCH-32-H   | Certificate of Verification            | Local Mechanical Exhaust                                                                           | <p><b>Kitchen Range Hood Verification</b></p> <p>If <del>HERS</del> <u>field</u> verification of the kitchen range hood is required for the dwelling, then the MCH-32 may be</p>                                                                                                                                                                                                                                                                                                                                          |

| Document Number | Document Type               | Document Description                                                                                                                                                                 | Nonresidential Group Sampling Rules                                                                                                                                                                                                                   |
|-----------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 |                             |                                                                                                                                                                                      | used as qualification for inclusion in a sample group.                                                                                                                                                                                                |
| NRCV-PLB-21-H   | Certificate of Verification | Nonresidential<br><b>HERSECC</b><br>- High Rise Multifamily Central Hot Water System Distribution Multiple Recirculation Loop Design for DHW Systems Serving Multiple Dwelling Units | <b>Nonresidential Multifamily Central DHW Feature</b><br>When the NRCC indicates requirement for <b>HERSfield</b> verification for multifamily central DHW is required, then the PLB-21 may be used as qualification for inclusion in a sample group. |
| NRCV-PLB-22-H   | Certificate of Verification | Nonresidential<br><b>HERSECC</b><br>- High Rise Single Dwelling Unit Hot Water System Distribution                                                                                   | <b>Nonresidential Multifamily Dwelling DHW Feature</b><br>When the NRCC indicates requirement for <b>HERSfield</b> verification for multifamily DHW in the dwelling, then the PLB-22 may be used as qualification for inclusion in a sample group.    |

Source: California Energy Commission

# Appendix A:

## Compliance Document Design Files: Graphical Layouts, User Instructions, Data Field Definitions, and Calculations

Note: For information regarding implementation of the compliance document designs, refer to the current version of the files maintained in the applicable CEC document design file repository at [GitHub.com](https://github.com).

### Residential Single Family

#### ~~CF1R~~2022-

~~2025~~2025-CF1R-ADD-01-E-PrescriptiveAdditionsBuilding.docx

~~2022~~2025-CF1R-ALT-01-E-PrescriptiveAlterationsBuilding.docx

~~2022~~2025-CF1R-ALT-02-E-PrescriptiveAlterationsHVAC.docx

~~2022~~2025-CF1R-ENV-02-E-AreaWeightedAverageWorkSheet.docx

~~2022~~2025-CF1R-ENV-03-E-ShgcWorkSheet.docx

~~2022~~2025-CF1R-ENV-04-E-CoolRoofAndSRIWorksheet.docx

~~2022~~2025-CF1R-ENV-05-E-AlternativeDefaultFenestrationProcedure.docx

~~2022~~2025-CF1R-ENV-06-E-InteriorExteriorInsulationWorksheet.docx

~~2022~~2025-CF1R-NCB-01-E-PrescriptiveNewlyConstructedBuilding.docx

#### ~~CF2R~~2022-

~~2022~~2025-CF2R-ELC-01-E-ElectricReady

~~2022~~2025-CF2R-ENV-01-E-FenestrationInstallation.docx

~~2022~~2025-CF2R-ENV-03-E-InsulationInstallation.docx

~~2022~~2025-CF2R-ENV-04-E-Roofing-RadiantBarrier.docx

~~2022~~2025-CF2R-ENV-20a-E-EnclosureAirLeakageTest-SinglePointTest-Manual-Meter.docx

~~2022~~2025-CF2R-ENV-20b-E-EnclosureAirLeakage-SinglePointTest-Automatic-Meter.docx

~~2022~~2025-CF2R-ENV-21-H-QII-FramingStage.docx

~~2022~~2025-CF2R-ENV-22-H-QII-InsulationStage.docx

~~2022~~2025-CF2R-LTG-01-E-Lighting-SingleFamilyDwellings.docx

~~2022~~2025-CF2R-MCH-01a-E-SpaceConditioningSystem-Performance.docx

~~2022~~2025-CF2R-MCH-01b-E-SpaceConditioningSystem-PrescriptiveAlterations.docx

~~2022~~2025-CF2R-MCH-01c-E-SpaceConditioningSystem-PrescriptiveNCB.docx

~~2022~~2025-CF2R-MCH-01d-E-SpaceConditioningSystem-Performance-E+A+A.docx

~~2022~~2025-CF2R-MCH-02-E-WholeHouseFan.docx

~~2022~~2025-CF2R-MCH-04-E-EvaporativeCoolers.docx

~~2022~~2025-CF2R-MCH-20a-H-DuctLeakageTest-NewConst.docx

~~2022~~2025-CF2R-MCH-21-H-DuctLocation.docx

~~2022~~2025-CF2R-MCH-22a-H-FanEfficacy-AllZonesCallingOnly.docx



~~2022~~2025-CF2R-MCH-23a-H-AirflowRate-~~AllZonesCallingOnly~~.docx  
~~2022~~2025-CF2R-MCH-24a-H-EnclosureAirLeakageWorksheet-~~SinglePointTest-  
Manual Meter~~.docx  
~~2022~~2025-CF2R-MCH-25a-H-RefrigerantCharge-~~Superheat~~.docx  
  
~~2022~~2025-CF2R-MCH-26-H-RatedSystemVerification.docx  
~~2022~~2025-CF3R-MCH-27a-H-  
IndoorAirQualityandMechanicalVentilation~~ContinuousMechVent-  
FanVentRateMethod~~.docx  
~~2022~~2025-CF2R-MCH-28-H-ReturnDuctAndFilterGrilleDesign-Table150.0-BorC.docx  
~~2022~~2025-CF2R-MCH-29-H-SupplyDuctSurfaceAreaBuriedDucts.docx  
~~2022~~2025-CF2R-MCH-30-H-VentilationCooling.docx  
~~2022~~2025-CF2R-MCH-31a-H-WholeHouseFan~~HERS-AirflowandWattsperWHF~~.docx  
~~2022~~2025-CF2R-MCH-32-H-KitchenVentilation.docx  
~~2022~~2025-CF2R-MCH-33-H-VchpComplianceCredit.docx  
~~2022~~2025-CF2R-MCH-34-E-Pre-Cooling.docx  
~~2022~~2025-CF2R-PLB-02-E-SingleDwellingUnitHotWaterSystemDistribution  
~~2022~~2025-CF2R-PLB-03-E-PoolAndSpaHeatingSystems.docx  
~~2022~~2025-CF2R-PLB-22-H-~~HERS~~-  
SingleDwellingUnitHotWaterSystemDistribution.docx  
~~2022~~2025-CF2R-PVB-01-E-PV Systems.docx  
~~2022~~2025-CF2R-PVB-02-E-BatteryStorageSystems.docx  
~~2022~~2025-CF2R-SRA-01-E-SolarReadiness-NewConstruction.docx  
~~2022~~2025-CF2R-SRA-02-E-MinimumSolarZoneAreaWorksheet.docx  
~~2022~~2025-CF2R-STH-01-SolarWaterHeatingSystems.docx  
**CF3R**  
~~2022~~2025-CF3R-ENV-20a-H-EnclosureAirLeakage~~Test-SinglePointTest-  
ManualMeter~~.docx  
~~2022~~2025-CF3R-ENV-21-H-QII-FramingStage.docx  
~~2022~~2025-CF3R-ENV-22-H-QII-InsulationStage.docx  
~~2022~~2025-CF3R-EXC-20-H-VerificationOfExistingConditionsForAlterations.docx  
~~2022~~2025-CF3R-MCH-20a-H-DuctLeakageTest-~~NewConst~~.docx  
~~2022~~2025-CF3R-MCH-21-H-DuctLocation.docx  
~~2022~~2025-CF3R-MCH-22a-H-FanEfficacy-~~AllZonesCallingOnly~~.docx  
~~2022~~2025-CF3R-MCH-23a-H-AirflowRate-~~AllZonesCallingOnly~~.docx  
~~2022~~2025-CF3R-MCH-24a-EnclosureAirLeakageWorksheet-~~SinglePointTest-Manual  
Meter~~.docx  
~~2022~~2025-CF3R-MCH-25a-H-RefrigerantCharge-~~Superheat~~.docx  
~~2022~~2025-CF3R-MCH-26-H-RatedSystemVerification.docx  
~~2022~~2025-CF3R-MCH-27a-H- Indoor Air Quality and Mechanical  
Ventilation ~~ContinuousMechVent-FanVentRateMethod~~.docx  
~~2022~~2025-CF3R-MCH-28-H-ReturnDuctAndFilterGrilleDesign-Table150.0-BorC.docx  
~~2022~~2025-CF3R-MCH-29-H-SupplyDuctSurfaceAreaBuriedDucts.docx  
~~2022~~2025-CF3R-MCH-30-H-VentilationCooling.docx  
~~2022~~2025-CF3R-MCH-31a-H-WholeHouseFan~~HERS-AirflowandWattsperWHF~~.docx  
~~2022~~2025-CF3R-MCH-32-H-KitchenVentilation.docx  
~~2022~~2025-CF3R-MCH-33-H-VchpComplianceCredit.docx

~~2022-2025~~-CF3R-PLB-22-H-SingleDwellingUnitHotWaterSystemDistribution.docx

~~2022-2025~~-CF2R-MCH-20b-H-DuctLeakage-LLDCS.docx

~~2022-2025~~-CF2R-MCH-20c-H-DuctLeakage-LLAHU.docx

~~2022-2025~~-CF2R-MCH-20d-H-DuctLeakageTest-ExistingConst.docx

~~2022-2025~~-CF2R-MCH-20e-H-DuctleakageTest-SealingAccesibleLeaks.docx

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Text is subject to internal review and revision, and that it  
is not official guidance until approved by the Commission

**Low-Rise Multifamily****LMCC**

~~2022~~ 2025-LMCC-CXR-01-E-Commissioning.xlsx  
~~2022~~ 2025-LMCC-ELC-01-E-ElectricalPowerDistributionandElectricReady.xlsx  
~~2022~~ 2025-LMCC-ENV-01-E-Envelope.xlsx  
~~2022~~ 2025-LMCC-LTI-01-E-Indoor Lighting.xlsx  
~~2022~~ 2025-LMCC-LTO-01-E-Outdoor Lighting.xlsx  
~~2022~~ 2025-LMCC-LTS-01-E-Sign Lighting.xlsx  
~~2022~~ 2025-LMCC-MCH-01-E-Mechanical.xlsx  
2025-LMCC-MCH-02-E-PrescriptiveAlterationsHVAC.xlsx  
~~2022~~ 2025-LMCC-PLB-01-E-Water Heating.xlsx  
~~2022~~ 2025-LMCC-PRC-01-E-Process Systems.xlsx  
~~2022~~ 2025-LMCC-SAB-01-E-SolarReady,PVandBattery.xlsx

**LMCI**

~~2022~~ 2025-LMCI-ELC-01-E-ElectricReady  
~~2022~~ 2025-LMCI-ENV-21-H-QII Framing Stage.docx  
~~2022~~ 2025-LMCI-ENV-22-H-QII Insulation Stage.docx  
~~2022~~ 2025-LMCI-MCH-01a-E-SpaceConditioningSystem-Performance.docx  
~~2022~~ 2025-LMCI-MCH-01b-E-SpaceConditioningSystem-  
 PrescriptiveAlterations.docx  
~~2022~~ 2025-LMCI-MCH-01c-E-SpaceConditioningSystem-PrescriptiveNCB.docx  
~~2022~~ 2025-LMCI-MCH-01d-E-SpaceConditioningSystem-Performance-E+A+A.docx  
~~2022~~ 2025-LMCI-MCH-20-H-DuctLeakageTest.docx  
~~2022~~ 2025-LMCI-MCH-21-H-Duct Location.docx  
~~2022~~ 2025-LMCI-MCH-22-H-FanEfficacy.docx  
~~2022~~ 2025-LMCI-MCH-23-H-AirflowRate.docx  
~~2022~~ 2025-LMCI-MCH-24-H-EnclosureAirLeakageTestWorksheet.docx  
~~2022~~ 2025-LMCI-MCH-25-H-RefrigerantCharge-Superheat.docx  
~~2022~~ 2025-LMCI-MCH-26-H-RatedSystemVerification.docx  
~~2022~~ 2025-LMCI-MCH-27-H-IndoorAirQualityandMechanicalVentilation.docx  
~~2022~~ 2025-LMCI-MCH-28-H-ReturnDuctAndFilterGrilleDesign-Table150.0-BorC.docx  
~~2022~~ 2025-LMCI-MCH-29-H-SupplyDuctSurfaceAreaBuriedDucts.docx  
~~2022~~ 2025-LMCI-MCH-32-H-LocalMechanicalExhaustKitchenVentilation.docx  
~~2022~~ 2025-LMCI-MCH-33-H-VchpComplianceCredit.docx  
~~2022~~ 2025-LMCI-PLB-01-E-MultifamilyCentralHotWaterSystemDistribution.docx  
~~2022~~ 2025-LMCI-PLB-02-E-SingleDwellingUnitHotWaterSystemDistribution.docx  
2025-LMCI-PLB-03-E-PoolAndSpaHeatingSystems.docx  
~~2022~~ 2025-LMCI-PLB-21-H-MultifamilyCentralHotWaterSystemDistribution.docx  
~~2022~~ 2025-LMCI-PLB-22-H-SingleDwellingUnitHotWaterSystemDistribution.docx

**LMCV**

~~2022~~ 2025-LMCV-ENV-21-H-QII Framing Stage.docx  
~~2022~~ 2025-LMCV-ENV-22-H-QII Insulation Stage.docx  
~~2022~~ 2025-LMCV-EXC-20-H- VerificationOfExistingConditionsForAlterations.docx  
~~2022~~ 2025-LMCV-MCH-20-H-DuctLeakageTest.docx

~~2022-2025~~ LMCV-MCH-21-H-DuctLocation.docx  
~~2022-2025~~ LMCV-MCH-22-H-FanEfficacy.docx  
~~2022-2025~~ LMCV-MCH-23-H-AirflowRate.docx  
~~2022-2025~~ LMCV-MCH-24-H-EnclosureAirLeakageWorksheet.docx  
~~2022-2025~~ LMCV-MCH-25a-H-RefrigerantCharge.docx  
~~2022-2025~~ LMCV-MCH-26-H-RatedSystemVerification.docx  
~~2022-2025~~ LMCV-MCH-27-H-IndoorAirQualityandMechanicalVentilation.docx  
~~2022-2025~~ LMCV-MCH-28-H-ReturnDuctAndFilterGrilleDesign-Table150.0-BorC.docx  
~~2022-2025~~ LMCV-MCH-29-H-SupplyDuctSurfaceAreaBuriedDucts.docx  
~~2022-2025~~ LMCV-MCH-32-H-LocalMechanicalExhaust~~KitchenVentilation~~.docx  
~~2022-2025~~ LMCV-MCH-33-H-VchpComplianceCredit.docx  
~~2022-2025~~ LMCV-PLB-21-H-MultifamilyCentralHotWaterSystemDistribution.docx  
~~2022-2025~~ LMCV-PLB-22-H-SingleDwellingUnitHotWaterSystemDistribution.docx

## Nonresidential and High-Rise Multifamily

### NRCV

~~2022-2025~~ NRCV-MCH-04-H-DuctLeakageTest.docx  
2025-NRCV-MCH-04-H-DuctLeakageTest.docx  
~~2022-2025~~ NRCV-MCH-04c-H-DuctLeakage-LLAHU.docx  
~~2022-2025~~ NRCV-MCH-04d-H-DuctLeakageTest-ExistingConst.docx  
~~2022-2025~~ NRCV-MCH-04e-H-DuctleakageTest-SealingAccesibleLeaks.docx  
~~2022-2025~~ NRCV-MCH-24-H-EnclosureAirLeakageWorksheet.docx  
~~2022-2025~~ NRCV-MCH-24b-H-EnclosureAirLeakageWorksheet-SinglePointTest-  
~~2022-2025~~ NRCV-MCH-27b-H- IndoorAirQualityandMechanicalVentilation-  
HighriseResidential.docx  
~~2022-2025~~ NRCV-MCH-32-H-LocalMechanicalExhaust.docx  
~~2022-2025~~ NRCV-PLB-21-H-MultifamilyCentralHotWaterSystemDistribution.docx  
~~2022-2025~~ NRCV-PLB-22-H-IndividualDwellingUnitHotWaterSystemDistribution.docx

An example of the single family compliance document layout, the contents of the file named ~~2022-~~2025-CF2R-MCH-01b-SpaceConditioningSystem-PrescriptiveAlterations.docx is available at the following URL.

<https://github.com/california-energy-commission/Compliance-Documents-Mockups/tree/main/2025/Drafts/Compliance%20Documents/CF2R>

<https://github.com/california-energy-commission/Compliance-Documents-Mockups/tree/main/2022/Revision%20Batch/V2022.1.000/CF2R>

As an example of the low-rise multifamily compliance document layout, the contents of the file named ~~2022-~~2025-LMCC-ENV-E-Envelope.xlsx is available at the following URL.

<https://github.com/california-energy-commission/Compliance-Documents-Mockups/tree/main/2025/Drafts/Compliance%20Documents/LMCC>

<https://github.com/california-energy-commission/Compliance-Documents-Mockups/tree/main/2022/Drafts/Compliance%20Documents/LMCC>

As an example of the nonresidential compliance document layout, the contents of the file named ~~2022-~~2025-NRCV-MCH-04a-DuctLeakageTest-~~NewConst~~.docx is available at the following URL.

<https://github.com/california-energy-commission/Compliance-Documents-Mockups/tree/main/2025/Drafts/Compliance%20Documents/NRCV>

<https://github.com/california-energy-commission/Compliance-Documents-Mockups/tree/main/2022/Revision%20Batch/V2022.1.000/NRCV>

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Text is subject to internal review and revision, and that it is not official guidance until approved by the Commission



## Appendix B

### Compliance Document XML Schema Files (XSD)

Note: For information for implementation of the compliance document schemas, refer to the current version of the files maintained in the applicable CEC XSD repository at [GitHub.com](https://github.com).

#### Single Family

CF1RADD01E.xsd

CF1RALT01E.xsd

CF1RALT02E.xsd

CF1RENV02E.xsd

CF1RENV03E.xsd

CF1RENV04E.xsd

CF1RENV05E.xsd

CF1RENV06E.xsd

CF1RNCB01E.xsd

~~CF1RPLB01E.xsd~~

CF2RELC01E.xsd

CF2RENV01E.xsd

CF2RENV03E.xsd

CF2RENV04E.xsd

CF2RENV20aH.xsd

~~CF2RENV20bH.xsd~~

CF2RENV21H.xsd

CF2RENV22H.xsd

CF2RLTG01E.xsd

CF2RMCH01aE.xsd

CF2RMCH01bE.xsd

CF2RMCH01cE.xsd

CF2RMCH01dE.xsd

CF2RMCH02E.xsd

CF2RMCH04E.xsd

CF2RMCH20aH.xsd

~~CF2RMCH20bH.xsd~~

~~CF2RMCH20cH.xsd~~

~~CF2RMCH20dH.xsd~~

~~CF2RMCH20eH.xsd~~

CF2RMCH21H.xsd

CF2RMCH22aH.xsd

~~CF2RMCH22bH.xsd~~

~~CF2RMCH22cH.xsd~~



~~CF2RMCH22dH.xsd~~  
CF2RMCH23aH.xsd  
~~CF2RMCH23bH.xsd~~  
~~CF2RMCH23cH.xsd~~  
~~CF2RMCH23dH.xsd~~  
~~CF2RMCH23eH.xsd~~  
~~CF2RMCH23fH.xsd~~  
CF2RMCH24aH.xsd  
~~CF2RMCH24bH.xsd~~  
CF2RMCH25aH.xsd  
~~CF2RMCH25bH.xsd~~  
~~CF2RMCH25cH.xsd~~  
~~CF2RMCH25fE.xsd~~  
CF2RMCH26H.xsd  
CF2RMCH27aH.xsd  
~~CF2RMCH27dE.xsd~~  
CF2RMCH28H.xsd  
CF2RMCH29H.xsd  
CF2RMCH30H.xsd  
CF2RMCH31aH.xsd  
~~CF2RMCH31bH.xsd~~  
CF2RMCH32H.Hxsd  
CF2RMCH33H.xsd  
CF2RMCH34E.xsd  
CF2RPLB02E.xsd  
CF2RPLB03E.xsd  
CF2RPLB22H.xsd  
CF2RPVB01E.xsd  
CF2RPVB02E.xsd  
CF2RSRA01E.xsd  
CF2RSRA02E.xsd  
CF2RSTH01E.xsd  
CF3RENV20aH.xsd  
~~CF3RENV20bH.xsd~~  
CF3RENV21H.xsd  
CF3RENV22H.xsd  
CF3REXC20H.xsd  
CF3RMCH20aH.xsd  
~~CF3RMCH20bH.xsd~~  
~~CF3RMCH20cH.xsd~~  
~~CF3RMCH20dH.xsd~~  
~~CF3RMCH20eH.xsd~~  
CF3RMCH21H.xsd  
CF3RMCH22aH.xsd

~~CF3RMCH22bH.xsd~~  
~~CF3RMCH22cH.xsd~~  
~~CF3RMCH22dH.xsd~~  
CF3RMCH23aH.xsd  
~~CF3RMCH23bH.xsd~~  
~~CF3RMCH23cH.xsd~~  
~~CF3RMCH23dH.xsd~~  
~~CF3RMCH23eH.xsd~~  
~~CF3RMCH23fH.xsd~~  
CF3RMCH24aH.xsd  
~~CF3RMCH24bH.xsd~~  
CF3RMCH25aH.xsd  
~~CF3RMCH25bH.xsd~~  
~~CF3RMCH25cH.xsd~~  
~~CF3RMCH25dH.xsd~~  
CF3RMCH26H.xsd  
CF3RMCH27aH.xsd  
CF3RMCH28H.xsd  
CF3RMCH29H.xsd  
CF3RMCH30H.xsd  
CF3RMCH31aH.xsd  
~~CF3RMCH31bH.xsd~~  
CF3RMCH32H.xsd  
CF3RMCH33H.xsd  
CF3RPLB22H.xsd  
CF3RNoTestH.xsd  
DataTypes.xsd  
ResBuilding.xsd  
ResCommon.xsd  
ResCompliance.xsd  
ResEnvelope.xsd  
ResHvac.xsd  
ResLighting.xsd

DRAFT

Text is subject to internal review and revision, and that it is not official guidance until approved by the Commission

**Low-Rise Multifamily**

LMCCCXR01E.xsd

LMCCELC01E.xsd

LMCCENV01E.xsd

LMCCLTI01E.xsd

LMCCLTO01E.xsd

LMCCLTS01E.xsd

LMCCMCH01E.xsd

LMCCMCH02E.xsd

LMCCPLB01E.xsd

LMCCSAB01E.xsd

LMCCPRC01E.xsd

LMCIELC01E.xsd

LMCIENV21H.xsd

LMCIENV22H.xsd

LMCIMCH01aE.xsd

LMCIMCH01bE.xsd

LMCIMCH01cE.xsd

LMCIMCH01dE.xsd

LMCIMCH20aH.xsd

~~LMCIMCH20bH.xsd~~~~LMCIMCH20cH.xsd~~~~LMCIMCH20dH.xsd~~~~LMCIMCH20eH.xsd~~

LMCIMCH21H.xsd

LMCIMCH22aH.xsd

~~LMCIMCH22bH.xsd~~

LMCIMCH23aH.xsd

~~LMCIMCH23bH.xsd~~~~LMCIMCH23cH.xsd~~~~LMCIMCH23dH.xsd~~

LMCIMCH24aH.xsd

~~LMCIMCH24bH.xsd~~

LMCIMCH25aH.xsd

~~LMCIMCH25bH.xsd~~~~LMCIMCH25cH.xsd~~~~LMCIMCH25dH.xsd~~~~LMCIMCH25fE.xsd~~

LMCIMCH26H.xsd

LMCIMCH27bH.xsd

LMCIMCH28H.xsd

LMCIMCH29H.xsd

LMCIMCH32.Hxsd

LMCIMCH33H.xsd

LMCIPLB01E.xsd  
LMCIPLB02E.xsd  
LMCIPLB03E.xsd  
LMCIPLB21E.xsd  
LMCIPLB22H.xsd  
LMCVENV21H.xsd  
LMCVENV22H.xsd  
LMCVEXC20H.xsd  
LMCVMCH20aH.xsd  
~~LMCVMCH20bH.xsd~~  
~~LMCVMCH20cH.xsd~~  
~~LMCVMCH20dH.xsd~~  
~~LMCVMCH20eH.xsd~~  
LMCVMCH21H.xsd  
LMCVMCH22aH.xsd  
~~LMCVMCH22bH.xsd~~  
LMCVMCH23aH.xsd  
~~LMCVMCH23bH.xsd~~  
~~LMCVMCH23cH.xsd~~  
~~LMCVMCH23dH.xsd~~  
LMCVMCH24aH.xsd  
~~LMCVMCH24bH.xsd~~  
LMCVMCH25aH.xsd  
~~LMCVMCH25bH.xsd~~  
~~LMCVMCH25cH.xsd~~  
~~LMCVMCH25dH.xsd~~  
LMCVMCH26H.xsd  
LMCVMCH27bH.xsd  
LMCVMCH28H.xsd  
LMCVMCH29H.xsd  
LMCVMCH32H.xsd  
LMCVMCH33H.xsd  
LMCVPLB21H.xsd  
LMCVPLB22H.xsd  
DataTypes.xsd  
ResBuilding.xsd  
ResCommon.xsd  
ResCompliance.xsd  
ResEnvelope.xsd  
ResHvac.xsd  
ResLighting.xsd

**Nonresidential and High-Rise Multifamily**

NRCVMCH04aH.xsd  
~~NRCVMCH04cH.xsd~~

~~NRCVMCH04dH.xsd~~  
~~NRCVMCH04eH.xsd~~  
NRCVMCH24aH.xsd  
~~NRCVMCH24bH.xsd~~  
NRCVMCH27bH.xsd  
NRCVMCH32H.xsd  
NRCVPLB21H.xsd  
NRCVPLB22H.xsd  
DataTypes.xsd  
ResBuilding.xsd  
ResCommon.xsd  
ResCompliance.xsd  
ResEnvelope.xsd  
ResHvac.xsd  
ResLighting.xsd

An example of the single-family compliance document schemas, the contents of the file named CF2RMCH01bE.xsd is available at the following URL.

<https://github.com/california-energy-commission/2025-SF-Documents-Schema/tree/main/deployed/CF2R>

~~<https://github.com/california-energy-commission/2022-HERS-Documents-Schema/tree/main/deployed/CF2R>~~

An example of the low-rise multifamily compliance document schemas, the contents of the file named LMCCENVE.xsd is available at the following URL.

<https://github.com/california-energy-commission/2025-Prescriptive-Residential-LM-Schema/tree/main/deployed/LMCC>

As an example of the nonresidential compliance document schemas, the contents of the file named NRCVMCH04aH.xsd are available at the following URL.

<https://github.com/california-energy-commission/2025-SF-Documents-Schema/tree/main/deployed/NRCV>

~~<https://github.com/california-energy-commission/2022-HERS-Documents-Schema/tree/main/deployed/NRCV>~~